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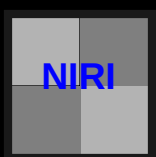
# Near IR Imager (NIRI)—Contents

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niri\_h ty\_TOC / 07-00 / 2001-07-02

This report contains a collection of tables of contents for the documentation for the Gemini Near Infra-Red Imager (NIRI).









|       |                                                                           |    |
|-------|---------------------------------------------------------------------------|----|
| 1     | Introduction .....                                                        | 1  |
| 1.1   | Purpose .....                                                             | 1  |
| 1.2   | Definitions, Acronyms and Abbreviations .....                             | 1  |
| 1.3   | Stylistic Conventions.....                                                | 2  |
| 1.4   | References.....                                                           | 2  |
| 2     | Release Notes .....                                                       | 2  |
| 2.1   | The 1.00 release (2000-06-27) .....                                       | 2  |
| 2.2   | The 1.00 alpha 6 release (1999-11-11) .....                               | 2  |
| 2.3   | The 1.00 alpha 5 release (1999-10-23) .....                               | 3  |
| 2.4   | Release V003, Releases 1.00 alpha 1 - 1.00 alpha 4.....                   | 3  |
| 2.5   | Release V002.....                                                         | 3  |
| 2.6   | Release V001.....                                                         | 3  |
| 3     | Hardware configuration .....                                              | 3  |
| 3.1   | VME Boards .....                                                          | 3  |
| 3.2   | Omega Temperature controllers .....                                       | 4  |
| 3.3   | MV167 Real time clock .....                                               | 4  |
| 4     | Prerequisites .....                                                       | 4  |
| 4.1   | VxWorks routing bug .....                                                 | 5  |
| 4.2   | Bug fix for e2sr (pre-GEM5).....                                          | 5  |
| 5     | Building, Loading, and Using the NIRI Low-Level Engineering Software..... | 6  |
| 5.1   | Obtaining the Software .....                                              | 6  |
| 5.2   | Unpacking the Software .....                                              | 7  |
| 5.3   | Configuring a standard EPICS system .....                                 | 7  |
| 5.4   | Configuration.....                                                        | 7  |
| 5.4.1 | Initial Setup .....                                                       | 8  |
| 5.4.2 | Environment subsystem .....                                               | 8  |
| 5.5   | Building the software .....                                               | 8  |
| 5.6   | Making dependency list.....                                               | 9  |
| 5.7   | Loading the NIRI CC and OIWFS control software or simulator .....         | 9  |
| 5.8   | Running the diagnostic software.....                                      | 9  |
| 5.9   | Cleaning and rebuilding .....                                             | 9  |
| 6     | Building, Loading, and Using the CICS/NIRI/OIWFS software .....           | 10 |
| 6.1   | Obtaining the Software .....                                              | 10 |
| 6.2   | Unpacking the Software .....                                              | 10 |
| 6.3   | Configuring a standard EPICS system .....                                 | 10 |
| 6.4   | Configuration.....                                                        | 10 |
| 6.5   | Building the system.....                                                  | 13 |
| 6.6   | Making dependency list.....                                               | 13 |
| 6.7   | Loading the NIRI/CICS OIWFS/CICS control software or simulator .....      | 13 |
| 6.8   | Running the user interface.....                                           | 13 |
| 6.9   | Cleaning and rebuilding .....                                             | 14 |
| 7     | Multiple copies of the software .....                                     | 14 |
| 8     | Limitations of the simulation .....                                       | 14 |
| 9     | Unfinished tasks .....                                                    | 15 |
| 9.1   | WCS .....                                                                 | 15 |
| 9.2   | Networking and communications .....                                       | 15 |
| 9.3   | wfsBeam .....                                                             | 15 |

---

---

|      |                                                                |    |
|------|----------------------------------------------------------------|----|
| 10   | Known bugs and limitations . . . . .                           | 15 |
| 10.1 | Problems with the simulation . . . . .                         | 16 |
| 10.2 | Problems with edd / dm . . . . .                               | 16 |
| 10.3 | Stability problems . . . . .                                   | 17 |
| 10.4 | Compiler Problems, Code, and Coding Standards . . . . .        | 17 |
| 10.5 | Mechanical problems . . . . .                                  | 17 |
| 10.6 | Datuming and diagnostic commands . . . . .                     | 18 |
| 10.7 | Calibration tools . . . . .                                    | 18 |
| 10.8 | Other Problems . . . . .                                       | 19 |
| 11   | Possible Enhancements . . . . .                                | 19 |
| 12   | Design Decisions (“It’s not a bug, it’s a feature!”) . . . . . | 20 |

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**niri\_h ty\_009.fm    Near IR Imager (NIRI) and Wavefront Sensor (OIWFS)—Definitions, Abbreviations  
and References**

---

|       |                                      |   |
|-------|--------------------------------------|---|
| 1     | Introduction . . . . .               | 1 |
| 1.1   | Purpose of the document . . . . .    | 1 |
| 1.2   | Scope . . . . .                      | 1 |
| 1.3   | Abbreviations and Acronyms . . . . . | 1 |
| 1.3.1 | Stylistic Conventions . . . . .      | 2 |
| 1.4   | Definitions . . . . .                | 2 |
| 1.5   | References . . . . .                 | 2 |





|     |                                                   |   |
|-----|---------------------------------------------------|---|
| 1   | Introduction . . . . .                            | 1 |
| 1.1 | Purpose of the document . . . . .                 | 1 |
| 1.2 | Scope . . . . .                                   | 1 |
| 1.3 | Definitions, Acronyms and Abbreviations . . . . . | 1 |
| 1.4 | Stylistic Conventions . . . . .                   | 2 |
| 1.5 | References . . . . .                              | 2 |
| 2   | Implementation . . . . .                          | 2 |



|       |                                                                               |    |
|-------|-------------------------------------------------------------------------------|----|
| 1     | Introduction . . . . .                                                        | 1  |
| 1.1   | Purpose of the document . . . . .                                             | 1  |
| 1.2   | Scope . . . . .                                                               | 2  |
| 1.3   | Definitions, Acronyms and Abbreviations . . . . .                             | 2  |
| 1.4   | Stylistic Conventions . . . . .                                               | 2  |
| 1.5   | References . . . . .                                                          | 2  |
| 2     | Implementation . . . . .                                                      | 2  |
| 2.1   | Overview . . . . .                                                            | 2  |
| 2.2   | Static Storage . . . . .                                                      | 4  |
| 2.2.1 | Static Look-Up Tables . . . . .                                               | 4  |
| 3     | Stepper Motor / Hall-Effect Sensor Control . . . . .                          | 4  |
| 3.0.1 | Notes on Figure 3 . . . . .                                                   | 4  |
| 3.1   | Hall-Effect Sensors and the Home Position . . . . .                           | 5  |
| 3.1.1 | A discrete positioning mechanism (4-position filter wheel) . . . . .          | 5  |
| 3.1.2 | Continuous-positioning mechanisms (OIWFS focusing stage) . . . . .            | 10 |
| 3.1.3 | Other continuous-positioning mechanisms (science focus and gimbals) . . . . . | 12 |
| 3.1.4 | Two position mechanisms (the pupil viewer) . . . . .                          | 14 |
| 3.2   | Peak Finding Algorithm . . . . .                                              | 16 |
| 3.3   | Stepper-motor errors . . . . .                                                | 16 |
| 3.4   | Backup sensors . . . . .                                                      | 16 |
| 3.5   | Hard Limits and Limit Switches . . . . .                                      | 16 |
| 3.5.1 | The focusing stages . . . . .                                                 | 16 |
| 3.5.2 | The Pupil Viewer . . . . .                                                    | 16 |
| 3.5.3 | The OIWFS Gimbal Mirror . . . . .                                             | 17 |
| 3.6   | Simultaneous motor motion . . . . .                                           | 17 |
| 3.7   | Stepper Motor Control Module Failure . . . . .                                | 17 |
| 3.8   | Diagnose . . . . .                                                            | 17 |
| 3.9   | Redatum . . . . .                                                             | 18 |
| 4     | Environment Control . . . . .                                                 | 18 |
| 4.1   | Cooling Motor . . . . .                                                       | 18 |
| 4.2   | Temperature regulation . . . . .                                              | 18 |
| 4.3   | Heating Resistors . . . . .                                                   | 18 |
| 4.4   | Temperature Sensors . . . . .                                                 | 18 |
| 5     | Special Calibration Programs . . . . .                                        | 19 |
| 6     | The Gemini Interlock System . . . . .                                         | 19 |
| 7     | Special Epics Records . . . . .                                               | 19 |
| 7.1   | Modified records . . . . .                                                    | 19 |
| 7.1.1 | Motor record (OMS VME8, VME44 device/driver) . . . . .                        | 19 |
| 7.2   | Custom Records, Device Support, and Subroutine Records . . . . .              | 19 |



|       |                                                         |   |
|-------|---------------------------------------------------------|---|
| 1     | Introduction .....                                      | 1 |
| 1.1   | Purpose of the document .....                           | 1 |
| 2     | Overview .....                                          | 1 |
| 2.1   | The ENG directory .....                                 | 2 |
| 2.1.1 | The CC Database .....                                   | 2 |
| 2.1.2 | The WFS Database .....                                  | 2 |
| 2.1.3 | The GBL Database .....                                  | 3 |
| 2.1.4 | The ENV Database .....                                  | 3 |
| 3     | The Mech Symbol and Diagram .....                       | 3 |
| 3.0.1 | The Mech Symbol .....                                   | 3 |
| 3.1   | External interfaces (the NIRI/CICS interface) .....     | 4 |
| 4     | The HallStep Record and Symbol .....                    | 6 |
| 4.1   | Record processing .....                                 | 9 |
| 4.1.1 | When an operation code is processed: .....              | 9 |
| 4.1.2 | When the completion task is given a new operation ..... | 9 |



|     |                                                                                                                                                                                              |    |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 1   | Introduction .....                                                                                                                                                                           | 1  |
| 1.1 | Purpose .....                                                                                                                                                                                | 1  |
| 1.2 | Definitions, Acronyms and Abbreviations .....                                                                                                                                                | 1  |
| 1.3 | Stylistic Conventions.....                                                                                                                                                                   | 1  |
| 1.4 | References.....                                                                                                                                                                              | 1  |
| 2   | When is it necessary to recalibrate a mechanism? .....                                                                                                                                       | 2  |
| 2.1 | What is the difference between normal calibration and low-level calibration.....                                                                                                             | 2  |
| 2.2 | When is normal calibration necessary?.....                                                                                                                                                   | 2  |
| 2.3 | When is low-level calibration necessary?.....                                                                                                                                                | 2  |
| 2.4 | When should the mechanism not be calibrated?.....                                                                                                                                            | 2  |
| 3   | Running the tools .....                                                                                                                                                                      | 2  |
| 3.1 | The Engineering User Interface .....                                                                                                                                                         | 2  |
| 4   | Scan Tool .....                                                                                                                                                                              | 2  |
| 4.1 | Starting the “Scan Tool” .....                                                                                                                                                               | 3  |
| 4.2 | Using The “Scan Tool” .....                                                                                                                                                                  | 3  |
| 5   | The “Cycle Tool” .....                                                                                                                                                                       | 4  |
| 5.1 | Starting the “Cycle Tool” .....                                                                                                                                                              | 4  |
| 5.2 | Using the “Cycle Tool” .....                                                                                                                                                                 | 5  |
| 5.3 | Normal Calibration (Phase 1) .....                                                                                                                                                           | 6  |
| 5.4 | Expected Values (Phase 1) .....                                                                                                                                                              | 6  |
| 5.5 | Configuration file.....                                                                                                                                                                      | 8  |
| 6   | The “Position Tool” .....                                                                                                                                                                    | 8  |
| 6.1 | Starting the “Position Tool” .....                                                                                                                                                           | 8  |
| 6.2 | Using The “Position Tool” .....                                                                                                                                                              | 8  |
| 6.3 | Warning: If the “Cycle Tool” is run while the mechanism is not working correctly, and if the values are written to disk, then the automatic verification of a move command will be suspect.9 |    |
| 6.4 | Calibration Procedure (Phase 2) .....                                                                                                                                                        | 10 |
| 6.5 | Expected Values (Phase 2) .....                                                                                                                                                              | 10 |
| 6.6 | Configuration file.....                                                                                                                                                                      | 11 |
| 7   | Low Level Calibration Procedure.....                                                                                                                                                         | 11 |
| 7.1 | The Wheels and Wheel-like Mechanisms, the Pupil Viewer, and the Window Cover11                                                                                                               |    |
| 7.2 | The Focus Stages.....                                                                                                                                                                        | 11 |
| 7.3 | The Gimbal Mirror .....                                                                                                                                                                      | 12 |
| 8   | Warnings.....                                                                                                                                                                                | 12 |





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**niri\_h ty\_007.fm    Near IR Imager (NIRI)—The Engineering User Interface**

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|       |                                                      |   |
|-------|------------------------------------------------------|---|
| 1     | Introduction .....                                   | 1 |
| 1.1   | Purpose .....                                        | 1 |
| 1.2   | Definitions, Acronyms and Abbreviations .....        | 2 |
| 1.3   | Stylistic Conventions.....                           | 2 |
| 1.4   | References.....                                      | 2 |
| 2     | Warnings and General Information .....               | 2 |
| 2.1   | Warnings (Very Important!) .....                     | 2 |
| 2.2   | A Few Notes About Edd/Dm Interfaces .....            | 3 |
| 3     | General User Commands .....                          | 3 |
| 3.1   | Starting the User Interface.....                     | 3 |
| 3.2   | The Introduction Screen .....                        | 3 |
| 3.3   | The Components Controller Main Screen.....           | 3 |
| 3.4   | The On-Instrument Wavefront Sensor Main Screen ..... | 3 |
| 4     | Restarting After the IOC Has Been Rebooted .....     | 3 |
| 4.1   | Wait for initialization to complete.....             | 3 |
| 4.2   | Restart screens.....                                 | 3 |
| 4.3   | Look for RED .....                                   | 3 |
| 4.3.1 | Set the Temperature Interlock .....                  | 3 |
| 4.3.2 | Start the Closed Cycle Coolers (if needed) .....     | 3 |
| 4.4   | Look for YELLOW.....                                 | 3 |
| 4.4.1 | Datum Mechanisms .....                               | 3 |
| 5     | Advanced User commands .....                         | 3 |
| 5.1   | Starting the Low-Level Engineering Interface .....   | 4 |
| 5.2   | The Main Engineering Screens .....                   | 4 |
| 5.3   | The Global Parameters Screen .....                   | 6 |
| 5.4   | The Engineering Interlocks Screen .....              | 6 |
| 5.5   | The Low Level Mechanism Control Screen .....         | 7 |
| 5.6   | The Parameter Screen .....                           | 8 |
| 5.7   | The Temperature Control Screen .....                 | 9 |

**niri\_h ty\_003.fm    “Near IR Imager (NIRI) to Observatory Control System”**

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|           |                                               |   |
|-----------|-----------------------------------------------|---|
| Chapter 1 | Introduction .....                            | 3 |
| 1–1.0     | Description .....                             | 3 |
| 1–1.1     | Purpose .....                                 | 3 |
| 1–1.2     | Scope .....                                   | 3 |
| 1–2.0     | Related Documents and Drawings .....          | 4 |
| 1–2.1     | References.....                               | 4 |
| 1–2.2     | Definitions, Acronyms and Abbreviations ..... | 4 |
| 1–2.3     | Stylistic Conventions.....                    | 4 |
| Chapter 2 | Details of the NIRI to OCS Interface.....     | 5 |
| 2–1.0     | Introduction .....                            | 5 |

---

---

|         |                                                   |    |
|---------|---------------------------------------------------|----|
| 2-2.0   | Physical System Interfaces                        | 5  |
| 2-2.1   | Mechanical Interface                              | 5  |
| 2-2.2   | Optical Interface                                 | 5  |
| 2-2.3   | Electronic Interface                              | 5  |
| 2-2.4   | Mass/Balance                                      | 6  |
| 2-2.5   | Thermal Interface                                 | 6  |
| 2-3.0   | Software / Control Function Interface             | 6  |
| 2-3.1   | Overview                                          | 6  |
| 2-3.1.1 | Communications infrastructure                     | 6  |
| 2-3.1.2 | Software architecture                             | 6  |
| 2-3.2   | Behavior                                          | 6  |
| 2-3.3   | Implementation                                    | 6  |
| 2-3.3.1 | Command (CAD/CAR) records                         | 6  |
| 2-3.3.2 | Status Information Records                        | 12 |
| 2-3.3.3 | Position names                                    | 26 |
| 2-3.4   | Detailed Command Description                      | 27 |
| 2-3.4.1 | Sequence Commands                                 | 27 |
| 2-3.4.2 | General Engineering Commands                      | 30 |
| 2-3.4.3 | NIRI Standard Mechanisms                          | 31 |
| 2-3.5   | Detailed Status information                       | 34 |
| 2-3.5.1 | General records                                   | 34 |
| 2-3.5.2 | Component Status Records                          | 34 |
| 2-3.5.3 | Component Health Records                          | 36 |
| 2-3.5.4 | Temperature Monitoring and Control Status Records | 37 |
| 2-3.5.5 | Temperature Monitoring and Control—Health Records | 38 |
| 2-3.6   | Alarm conditions                                  | 38 |
| 2-3.7   | Debugging and Maintenance                         | 39 |
| 2-3.7.1 | Debugging Modes                                   | 39 |
| 2-3.7.2 | Calibration                                       | 39 |
| 2-3.8   | Simulation                                        | 39 |
| 2-4.0   | Safety Issues                                     | 39 |

---

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**niri\_h ty\_013.fm    Near IR Imager (NIRI) — Component Position Names**

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|     |                                                                              |   |
|-----|------------------------------------------------------------------------------|---|
| 1   | Introduction .....                                                           | 1 |
| 1.1 | Purpose .....                                                                | 1 |
| 1.2 | Definitions, Acronyms and Abbreviations .....                                | 1 |
| 1.3 | Stylistic Conventions.....                                                   | 1 |
| 1.4 | References.....                                                              | 1 |
| 2   | Tables .....                                                                 | 2 |
| 2.1 | Filter Wheel 1 .....                                                         | 2 |
| 2.2 | Filter Wheel 2 .....                                                         | 2 |
| 2.3 | Filter Wheel 3 .....                                                         | 3 |
| 2.4 | Table 3 describes the named positions for filter wheel 3Filter Wheel 3 ..... | 3 |

**niri\_h ty\_008.fm    “A&G to NIRI/GNIRS On Instrument Wavefront Sensors”**

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|           |                                                        |   |
|-----------|--------------------------------------------------------|---|
| Chapter 1 | Introduction .....                                     | 3 |
| 1–1.0     | Description .....                                      | 3 |
| 1–1.1     | Purpose .....                                          | 3 |
| 1–1.2     | Scope .....                                            | 3 |
| 1–2.0     | Related Documents and Drawings .....                   | 4 |
| 1–2.1     | References.....                                        | 4 |
| 1–2.2     | Definitions, Acronyms and Abbreviations .....          | 4 |
| 1–2.3     | Stylistic Conventions.....                             | 4 |
| Chapter 2 | Details of the A&G to NIRI/GNIRS OIWFS Interface ..... | 5 |
| 2–1.0     | Introduction .....                                     | 5 |
| 2–2.0     | Physical System Interfaces .....                       | 5 |
| 2–2.1     | Mechanical Interface .....                             | 5 |
| 2–2.2     | Optical Interface .....                                | 5 |
| 2–2.3     | Electronic Interface .....                             | 5 |
| 2–2.3.1   | NIRI .....                                             | 5 |
| 2–2.3.2   | GNIRS / Standalone OIWFS .....                         | 5 |
| 2–2.4     | Mass/Balance .....                                     | 6 |
| 2–2.5     | Thermal Interface .....                                | 6 |

---

|         |                                               |    |
|---------|-----------------------------------------------|----|
| 2-3.0   | Software/Control Function Interface . . . . . | 6  |
| 2-3.1   | Overview . . . . .                            | 6  |
| 2-3.1.1 | Communications infrastructure . . . . .       | 6  |
| 2-3.1.2 | Software architecture . . . . .               | 6  |
| 2-3.2   | Behavior . . . . .                            | 6  |
| 2-3.3   | Implementation . . . . .                      | 6  |
| 2-3.3.1 | Command (CAD/CAR) records . . . . .           | 7  |
| 2-3.3.2 | Status Information Records . . . . .          | 9  |
| 2-3.3.3 | Position names . . . . .                      | 16 |
| 2-3.4   | Detailed Command Description . . . . .        | 17 |
| 2-3.4.1 | Sequence Commands . . . . .                   | 17 |
| 2-3.4.2 | General Engineering Commands . . . . .        | 18 |
| 2-3.4.3 | OIWFS Standard Mechanisms . . . . .           | 18 |
| 2-3.5   | Detailed Status information . . . . .         | 21 |
| 2-3.5.1 | General records . . . . .                     | 21 |
| 2-3.5.2 | Component Status Records . . . . .            | 21 |
| 2-3.5.3 | Component Health Records . . . . .            | 23 |
| 2-3.6   | Alarm conditions . . . . .                    | 24 |
| 2-3.7   | Debugging and Maintenance . . . . .           | 25 |
| 2-3.7.1 | Debugging Modes . . . . .                     | 25 |
| 2-3.7.2 | Calibration . . . . .                         | 25 |
| 2-3.8   | Simulation . . . . .                          | 25 |
| 2-4.0   | Safety Issues . . . . .                       | 25 |

---

---

|                         |                                                                         |
|-------------------------|-------------------------------------------------------------------------|
| <b>niri_h ty_014.fm</b> | <b>Near IR Imager (NIRI) and OIWFS — OIWFS Component Position Names</b> |
|-------------------------|-------------------------------------------------------------------------|

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|     |                                                   |   |
|-----|---------------------------------------------------|---|
| 1   | Introduction . . . . .                            | 1 |
| 1.1 | Purpose . . . . .                                 | 1 |
| 1.2 | Definitions, Acronyms and Abbreviations . . . . . | 1 |
| 1.3 | Stylistic Conventions. . . . .                    | 1 |
| 1.4 | References. . . . .                               | 2 |
| 2   | Tables . . . . .                                  | 2 |
| 2.1 | Filter Wheel. . . . .                             | 2 |

|                         |                                                                          |
|-------------------------|--------------------------------------------------------------------------|
| <b>niri_h ty_005.fm</b> | <b>“NIRI/GNIRS Science Instrument to On Instrument Wavefront Sensor”</b> |
|-------------------------|--------------------------------------------------------------------------|

---

|           |                                                                      |    |
|-----------|----------------------------------------------------------------------|----|
| Chapter 1 | Introduction . . . . .                                               | 3  |
| 1–1.0     | Description . . . . .                                                | 3  |
| 1–1.1     | Purpose . . . . .                                                    | 3  |
| 1–1.2     | Scope . . . . .                                                      | 3  |
| 1–1.3     | Differences between NIRI and GNIRS. . . . .                          | 3  |
| 1–2.0     | Related Documents and Drawings . . . . .                             | 3  |
| 1–2.1     | References. . . . .                                                  | 3  |
| 1–2.2     | Definitions, Acronyms and Abbreviations . . . . .                    | 4  |
| 1–2.3     | Stylistic Conventions. . . . .                                       | 4  |
| Chapter 2 | Details of the NIRI/GNIRS to OIWFS Interface . . . . .               | 5  |
| 2–1.0     | Introduction . . . . .                                               | 5  |
| 2–2.0     | Physical System Interfaces . . . . .                                 | 5  |
| 2–2.1     | Mechanical Interface . . . . .                                       | 5  |
| 2–2.2     | Optical Interface . . . . .                                          | 5  |
| 2–2.3     | Electronic Interface . . . . .                                       | 5  |
| 2–2.4     | Mass/Balance . . . . .                                               | 5  |
| 2–2.5     | Thermal Interface . . . . .                                          | 5  |
| 2–3.0     | Software/Control Function Interface (wavefront sensor). . . . .      | 5  |
| 2–3.1     | Overview . . . . .                                                   | 5  |
| 2–3.1.1   | Communications infrastructure . . . . .                              | 5  |
| 2–3.2     | Implementation . . . . .                                             | 5  |
| 2–3.2.1   | Command (CAD/CAR) records. . . . .                                   | 5  |
| 2–3.2.2   | Status Information Records . . . . .                                 | 8  |
| 2–3.2.3   | Detailed SIR record description. . . . .                             | 9  |
| 2–4.0     | Software/Control Function Interface (Components Controller). . . . . | 10 |
| 2–4.1     | Overview . . . . .                                                   | 10 |
| 2–4.1.1   | Communications Infrastructure . . . . .                              | 10 |
| 2–4.2     | Implementation . . . . .                                             | 10 |
| 2–4.2.1   | Monitoring an external record. . . . .                               | 10 |
| 2–4.2.2   | Status Information Records . . . . .                                 | 11 |
| 2–5.0     | Safety Issues. . . . .                                               | 11 |
| 2–6.0     | Activity Completion . . . . .                                        | 11 |



|     |                                                   |   |
|-----|---------------------------------------------------|---|
| 1   | Introduction . . . . .                            | 1 |
| 1.1 | Purpose of the document . . . . .                 | 1 |
| 1.2 | Scope . . . . .                                   | 1 |
| 1.3 | Definitions, Acronyms and Abbreviations . . . . . | 1 |
| 1.4 | Stylistic Conventions . . . . .                   | 2 |
| 1.5 | References . . . . .                              | 2 |
| 2   | Caveats . . . . .                                 | 2 |
| 3   | History . . . . .                                 | 2 |
| 3.1 | Early attempts . . . . .                          | 2 |
| 3.2 | The first working system . . . . .                | 4 |
| 3.3 | Integration with the CICS . . . . .               | 5 |
| 3.4 | The interlock system . . . . .                    | 5 |
| 3.5 | The NIRI temperature control system . . . . .     | 6 |
| 3.6 | The NIRI diagnostic functions . . . . .           | 7 |
| 3.7 | The NIRI verification functions . . . . .         | 7 |
| 4   | CICS . . . . .                                    | 7 |
| 4.1 | Successful elements of the CICS . . . . .         | 7 |
| 4.2 | Problems with the CICS . . . . .                  | 7 |
| 5   | The NIRI software . . . . .                       | 7 |
| 5.1 | Successful elements of NIRI . . . . .             | 7 |
| 5.2 | Problems with NIRI . . . . .                      | 8 |
| 6   | Lessons learned . . . . .                         | 8 |
| 7   | Suggestions . . . . .                             | 8 |
| 8   | Conclusion . . . . .                              | 8 |





|     |                                    |   |
|-----|------------------------------------|---|
| 1   | Introduction . . . . .             | 1 |
| 1.1 | Purpose of the document . . . . .  | 1 |
| 2   | The Motor Driver Modules . . . . . | 2 |



|     |                                               |     |
|-----|-----------------------------------------------|-----|
| 1   | Introduction .....                            | i   |
| 1.1 | Purpose .....                                 | i   |
| 1.2 | Definitions, Acronyms and Abbreviations ..... | ii  |
| 1.3 | Stylistic Conventions.....                    | ii  |
| 1.4 | References.....                               | ii  |
| 2   | Precautions .....                             | ii  |
| 2.1 | Temperature.....                              | ii  |
| 2.2 | Motor Heating.....                            | ii  |
| 2.3 | Wear.....                                     | ii  |
| 3   | Procedure for Testing Mechanisms .....        | ii  |
| 3.1 | Preparation .....                             | iii |
| 4   | Filter Wheel .....                            | iii |
| 4.1 | Focus Stage .....                             | iv  |
| 4.2 | Gimbal.....                                   | iv  |



|     |                                               |   |
|-----|-----------------------------------------------|---|
| 1   | Introduction .....                            | 1 |
| 1.1 | Definitions, Acronyms and Abbreviations ..... | 1 |
| 1.2 | Stylistic Conventions.....                    | 1 |
| 1.3 | References.....                               | 1 |
| 2   | Filter Wheel .....                            | 2 |
| 3   | Focus Stage.....                              | 2 |
| 4   | Gimbal Mirror .....                           | 3 |



# Gemini Near Infra-Red Imager and Wavefront Sensor



# Near IR Imager (NIRI) and On-Instrument Wavefront Sensor (OIWFS)—Software Installation and Release Notes

**Hubert Yamada and Steven Beard**

niri\_h ty\_006 / 1.00-00 / 2001-06-27

This document describes the procedure for building and installing the NIRI/OIWFS software, the NIRI engineering software, and the Near Infra-Red OIWFS software, release 1.00.

## 1 Introduction

### 1.1 Purpose

This document describes the procedure for building and installing the NIRI CC and OIWFS software. It also describes version-specific features of this release.

This package provides all of the software that is necessary to control NIRI or the OIWFS according to gemini specifications. The higher level CAD/CAR/APPLY interface is provided by niri\_1.00 which is based on the CICS. The CICS and the higher-level interface software was written by Steven Beard.

The low-level engineering package is part of the software distribution. It provides the functions that are necessary to operate the NIRI and OIWFS mechanisms. It may also be used as a stand-alone package which provides all the tools that are necessary to interface to the NIRI CC and OIWFS hardware, to verify that the hardware is working correctly, and to calibrate the hardware. When used as a standalone package, it does not satisfy all of the gemini interface requirements.

The intended audience of this document is:

- the NIRI reviewers;
- the developers of the instruments which will be using the NIRI OIWFS.

### 1.2 Definitions, Acronyms and Abbreviations

See “Near IR Imager (NIRI)—Definitions, Abbreviations, and References”.

### **1.3 Stylistic Conventions**

References to documents in Section 1.5, “References,” on page 2 of “Near IR Imager (NIRI)—Definitions, Abbreviations, and References” are given like this; [1].

### **1.4 References**

See “Near IR Imager (NIRI)—Definitions, Abbreviations, and References”.

---

## **2 Release Notes**

---

This is NIRI software 1.00, which is intended to be the final software release of the University of Hawaii software.

The following features are missing or broken and are expected to be fixed by the Gemini telescope developers.

- Some hardware subsystems (window cover, air ionizer) are not supported;
- The World Coordinate System is not supported;
- Communications with the detector controller need further work;
- Health problems cannot all be traced to alarms;

### **2.1 The 1.00 release (2000-06-27)**

Currently under development.

- Feature Freeze;
- WCS code now added (though WCS support still not present);
- A&G Interface now conforms fully to the ICD’s;
- Some of the A&G interface and time-handling software was changed, to conform to the A&G system, as implemented;
- The documentation has been finalized.

### **2.2 The 1.00 alpha 6 release (1999-11-11)**

The alpha 6 release includes the following changes:

- Fixed many small problems with the Engineering User interface;
- Fixed some problems with the OMS driver which occurred when the requested step count was too large;
- Added start-up scripts for OIWFS only mode;
- The “INVALID” position warning and the move-verification errors are now triggered by the same tolerances;
- The A&G interface now conforms better to the ICD’s;



- The directory structure has been significantly rearranged, to be more consistent;
  - The CICSICC directory is renamed to CC;
  - The CICSWFS directory has been renamed to WFS;
  - The gnaac-interface directory is now a complete EPICS-application tree (the DC directory);
  - The sequencer functions have been moved into an IS directory;
  - The GNAAC simulator directory has been removed;
- The startup scripts have been rewritten to be easier to modify;
- WfsBeam is now implemented;
- The directory structure and configuration files are now finalized.

### 2.3 The 1.00 alpha 5 release (1999-10-23)

The alpha 5 release includes the following changes:

- Restructured CAD files to make the logic clearer;
- The A&G interface is now functional.

### 2.4 Release V003, Releases 1.00 alpha 1 - 1.00 alpha 4

These were not publicly released. They involved extensive restructuring of the CICS-derived code to make it easier to modify. They also included changes to support communications with the detector controller; better support for the Hall effect sensors; temperature control; interlocks; temperature-dependent records; support for the full range of NIRI mechanisms; and many other smaller changes.

### 2.5 Release V002

Mostly bug fixes and changes in the way that configuration is handled. There is now a sequencer control screen.

### 2.6 Release V001

This is the initial release and was not widely distributed.

---

## 3 Hardware configuration

---

### 3.1 VME Boards

All boards should be set as specified in [30], except as described below.

The address of the Oregon Microsystems boards must be set to 0xFC80 (first board) and 0xFC90 (second board). The address of the XYCOM 490 board must be set to 0xE000. (See your Oregon Microsystems and Xycom documentation). Interrupt vector 0xE0 must be available for use by the Xycom board.

### 3.2 Omega Temperature controllers

The Omega temperature controllers should be set to 1200 baud, using the front panel commands.

### 3.3 MV167 Real time clock

Unfortunately, some of the EPICS device support (e.g., the steppermotor record) assumes a 60 Hz clock, which can cause occasional, intermittent problems; since the NIRI/OIWFS software no longer requires the steppermotor record, it may no longer be necessary to set the real time clock, but it is recommended as a precaution

The following instructions will fix the problem. Warning: These instructions are only used for the MV167, which is the gemini standard CPU for instrument systems. *Do not follow these instructions on any other system!* At the VxWorks prompt:

```
m 0xffffc00fe
```

You will see 0xffffc00fe:, followed by a 4 digit hexadecimal number, followed by a dash. Record it for later reference. Enter:

```
21ee
```

You will see 0xffffc0100, followed by a four digit hexadecimal number, followed by a dash. Enter a period:

```
.
```

To undo these changes, repeat the procedure, except that you must enter the number that you recorded earlier, instead of 21ee.

---

## 4 Prerequisites

---

This software was developed under Solaris 2.5.1 and epics3.12.2GEM5 / VxWorks 5.2. I do not guarantee that it will work correctly with any other operating system release or EPICS release. At this time no customization to the operating system is necessary. Minor changes to EPICS may be necessary, as described below.

In order to run the CICS/NIRI interface to the GNAAC detector controller, a current version of the GNAAC detector controller software must have already been installed and running on a separate IOC. (The GNAAC simulator, is no longer included with the software, because it is very out of date.)

In order to build and to run the CICS/NIRI software, the Gemini slalib, Gemini astlib, and Gemini timelib are all required.<sup>1</sup>

Also required:

- gunzip.

---

1. These packages are available via ftp from <ftp://ftp.gemini.edu/pub/epics/gemLibs.tar.gz>

#### 4.1 VxWorks routing bug

There is a bug in VxWorks that sometimes causes communications problems between the instrument sequencer and the detector controller software (running on a separate CPU).

Execute a `routeShow` command, from the VxWorks prompt. This will show your routing table. Typically, you will see two entries

| destination                | gateway                      |
|----------------------------|------------------------------|
| 0.0.0.0                    | xxx.xxx.xxx.rrr <sup>a</sup> |
| xxx.xxx.xxx.0 <sup>b</sup> | xxx.xxx.xxx.yyy <sup>c</sup> |

- a. The address of your router.
- b. Your local subnetwork address.
- c. Your host address.

On some systems, you may see the following:

| destination                | gateway                      |
|----------------------------|------------------------------|
| 0.0.0.0                    | xxx.xxx.xxx.rrr <sup>a</sup> |
| xxx.xxx.xxx.0 <sup>b</sup> | xxx.xxx.xxx.yyy <sup>c</sup> |
| xxx.xxx.xxx.0              | xxx.xxx.xxx.rrr              |

- a. The address of your router.
- b. Your local subnetwork address.
- c. Your host address.

If you see this, you should edit your `$local.vws` file, located in `$EPICS/extensions/src/uae/templates/startup`. There will normally be a line in this file of the form:

```
routeAdd "0", "xxx.xxx.xxx.rrr"
```

where `xxx.xxx.xxx.rrr` is the address of your router.

After this line, add the command:

```
routeDelete "xxx.xxx.xxx.0", "xxx.xxx.xxx.rrr"
```

where `xxx.xxx.xxx.0` is the address of your local subnetwork and `xxx.xxx.xxx.rrr` is the address of your router.

If you are unable to modify your epics distribution, you may add this command to your VxWorks startup script instead.

#### 4.2 Bug fix for e2sr (pre-GEM5)

*Important:* There is a bug in the `e2sr` command, which is part of the EPICS distribution (fixed in GEM5). This bug will cause `e2sr` to crash, if a variable is set to an empty value. It is not possible to compile the software, unless the bug is patched.

If an version of EPICS preceding GEM5 is used<sup>2</sup>, the following patch should be applied:

Edit `edifParser()`, in `$EPICS/extensions/src/edif/edifLib.c`, by replacing

```
while (*in != EOS) {  
  
    ...  
  
}
```

with

```
if (in != NULL) {  
    while (*in != EOS) {  
  
        ...  
  
    }  
  
}
```

---

## 5 Building, Loading, and Using the NIRI Low-Level Engineering Software

---

The `niriEng` package is a minimal package which is designed to run the NIRI and OIWFS hardware, and to simulate the hardware. It does not satisfy the Gemini interface requirements. It does provide all functions which are necessary to calibrate and debug the NIRI hardware.

Note: the full NIRI/CICS package, described below in “Building, Loading, and Using the CICS/NIRI/OIWFS software” on page 10, uses the `niriEng` package<sup>3</sup> for its low-level functions. All functions which are provided by the `niriEng` package are available in the CICS/NIRI package; there are three reasons to use the `niriEng` package directly:

- It loads a *lot* faster
- It compiles a *lot* faster
- It uses less memory (probably not a real problem)

### 5.1 Obtaining the Software

Ftp to `gemftp.ifa.hawaii.edu`, and download

```
/pub/gemini_IFA/niri/source/niri_1.00.tar.gz
```

---

2. This is not to imply that releases of EPICS other than GEM5 are supported!

3. The previous release of this package was called `niriCC`

For instruments other than NIRI, there will also be a file:

```
/pub/gemini_IFA/niri/source/XXXX_1.00.gz
```

Where XXXX will be the name of the instrument, e.g., “gnirs.” Get this file in addition to the standard niri file. This file contains hardware parameters which are unique for each mechanism that is built.

## 5.2 Unpacking the Software

Execute the following commands:

```
mkdir niri_1.00
cd niri_1.00
cp (...) /niri_1.00.tar.gz .
gunzip < niri_1.00.tar.gz | tar -xvf -
```

For instrument XXXX (other than niri), after executing the previous commands, execute:

```
cp (...) /niri_1.00_XXXX_00.tar.gz .
gunzip < niri_1.00_XXXX_00.tar.gz | tar -xvf -
```

You only need the ENG subdirectory. The use of the remaining contents of this directory is described below, in “Building, Loading, and Using the CICS/NIRI/OIWFS software” on page 10.

## 5.3 Configuring a standard EPICS system

*Previous releases of this software required several changes to the EPICS distribution. This package now uses a modified version of the motor record, and does not need to replace the stepper motor record.*

## 5.4 Configuration

Create a `niriLogin` file; the file `niriLogin_SAMPLE` is provided as an example. You will need to edit this file to set the path to your installed NIRI software, and to set your prefix (or prefixes).<sup>4</sup>

Warning: The instrument prefix, e.g., `niri:`, must contain at most five characters, including the final colon. The status-and-alarm-database prefix, e.g., `niri:` or `niri::`, must contain at most six characters, including the final colon. The instrument name, e.g., `NIRI`, must contain at most four characters.

*You should not need to change any other files. If you do need to do so, it should be reported as a bug.*

The `csh/tcsh` configuration file “`niriLogin`” sets up various environment variables and your path. You must source `niriLogin` before building or running any of the `niri` software. It might be wise to source “`niriLogin`” in your `.login` or `.cshrc` file:

```
source niriLogin
```

---

4. `NiriLogin` assumes that you are using `csh`, `tcsh`, or some other shell that is compatible with the `csh/tcsh` syntax. If not, you will need to make an equivalent file for your shell, using `niriLogin` as a starting point.

The variables which must be defined in niriLogin are:

**TABLE 1.**

| Variable     | Default value    | Purpose                                                                                                                                                                                   | Change? <sup>a</sup> |
|--------------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| NIRI_ENGDIR  | /gemini/niri/ENG | The path to the NIRI engineering source directory.                                                                                                                                        | Always               |
| NIRI_PREFIX  | niri:            | This prefix is prepended to all records. It should be four or fewer alphanumeric characters, followed by a colon.                                                                         | If needed            |
| NIRI_NAME    | NIRI             | This is used to generate a description of various records. It should be four or fewer alphanumeric characters.                                                                            | If needed            |
| NIRI_TMPSCAN | “Passive”        | This is used by the temperature interlock, which is described in [6], Section 2–4.2.1, “Monitoring an external record.”, on page 10. Set this to a scan rate, e.g., “1 second”.           | If needed            |
| NIRI_TMPVAR  | 0                | This is used by the temperature interlock, which is described in [6], Section 2–4.2.1, “Monitoring an external record.”, on page 10. Set this to a record field, e.g., niri:cc:tmpvar.    | If needed            |
| NIRI_TMPHB   | 0                | This is used by the temperature interlock, which is described in [6], Section 2–4.2.1, “Monitoring an external record.”, on page 10. Set this to a record field, e.g., niri:cc:heartBeat. | If needed            |
| NIRI_TARGETS | mv167            | Which EPICS-supported hardware platform <sup>b</sup> .                                                                                                                                    | Not recommended      |

- a. Items which are marked “Not recommended” will have unexpected results, unless the niri software is modified accordingly. Details are outside the scope of this document.
- b. The mv167 is the only supported architecture at this time.

#### 5.4.1 Initial Setup

In the \$NIRI\_ENGDIR directory, type:

```
./niriSetup
```

*You will get a few warning messages, e.g., that “.cvsignore” does not exist. These are harmless and can be ignored.*

#### 5.4.2 Environment subsystem

The environment simulator is now working. *It is no longer necessary to disable it if the appropriate supporting hardware is not available.*

### 5.5 Building the software

In the \$NIRI\_ENGDIR directory, type

```
gmake
```

This should build the system. The software should compile without warnings, *except*

- All C modules will warn “‘rcsid’ defined but not used.”
- The SNL code compiler (\*.stpp files) generates a lot of warnings about undefined variables.

### **5.6 Making dependency list**

Switch to the source directory

```
cd $NIRI_ENGDIR/src
```

Make the dependency list

```
gmake depends
```

This will insure that all files will be remade if the record configuration files or C header files are changed.

### **5.7 Loading the NIRI CC and OIWFS control software or simulator**

Connect to your IOC and set the start-up script to:

- (...) /bin/mv167/startupSim (simulator),
- (...) /bin/mv167/startup (control software),
- (...) /bin/mv167/startupWfsSim (OIWFS simulator), or
- (...) /bin/mv167/startupWfs (OIWFS control software).

where (...) represents the full, absolute path of the niri engineering directory. Then reboot the IOC. The IOC should reboot without any errors. At this time, you should be running the software that you selected.

### **5.8 Running the diagnostic software**

- Make sure that NIRI\_ENGDIR is defined.
- Make sure that \$NIRI\_ENGDIR/bin/solaris is in your path

Type:

```
niriEng
```

This should start up the niri low-level engineering screen. Note: This screen is designed to control the whole NIRI system. If it is used to control the OIWFS system, non-OIWFS control widgets will be white. Separate screens to control the OIWFS only are not provided.

The low level engineering screens are documented in [5].

### **5.9 Cleaning and rebuilding**

The set up and installation process generates a number of files automatically. To insure that the old files are removed before building the system with a new configuration, type:

```
gmake reset
```

*Do not use*

```
gmake clean
```

because this fails to remove some files which should be removed, and remove some files which should not be removed. To restore the system to its original state, type:

```
gmake resettall.
```

---

## 6 Building, Loading, and Using the CICS/NIRI/OIWFS software

---

The niri package is designed to run the NIRI and OIWFS hardware, and to simulate the hardware. This package is intended to satisfy all Gemini interface requirements for a science instrument. It provides all functions which are necessary to calibrate and debug the NIRI hardware.

Note: the full NIRI/CICS package uses the niriEng package (described above in “Building, Loading, and Using the NIRI Low-Level Engineering Software” on page 6) for its low-level functions.

### 6.1 Obtaining the Software

Ftp to gemftp.ifa.hawaii.edu, and download

```
/pub/gemini_IFA/niri/source/niri_1.00.tar.gz
```

### 6.2 Unpacking the Software

Execute the following commands:

```
mkdir niri_1.00
cd niri_1.00
cp (...)niri_1.00.tar.gz .
gunzip < niri_1.00.tar.gz | tar -xvf -
```

For instrument XXXX (other than niri), after executing the previous commands, execute:

```
cp (...)niri_1.00_XXXX_00.tar.gz .
gunzip < niri_1.00_XXXX.tar.gz | tar -xvf -
```

### 6.3 Configuring a standard EPICS system

*It is no longer necessary to modify your EPICS distribution. This package now uses a modified version of the motor record, and does not need to replace the stepper motor record.*

However, the address of the Oregon Microsystems board must be set to 0xFC80 and the address of the XYCOM 490 board must be set to 0xE000. (See your Oregon Microsystems and Xycom documentation). Note also that interrupt vector 0xE0 must be available for use by the Xycom board.

### 6.4 Configuration

Create a niriLogin file; the file niriLogin\_SAMPLE is provided as an example. You will need to edit this file to set the path to your installed NIRI software, and to set your prefix (or prefixes).<sup>5</sup>

Warning: The instrument prefix and detector controller prefix, e.g., niri:, must contain at most five characters, including the final colon. The status-and-alarm-database prefix and detector-controller status-and-alarm-database prefix, e.g., niri: or niri::, must contain

---

5. NiriLogin assumes that you are using csh, tcsh, or some other shell that is compatible with the csh/tcsh syntax. If not, you will need to make an equivalent file for your shell, using niriLogin as a starting point.



at most five characters, including the final colon. The instrument name, e.g., **NIRI**, should contain at most four characters.

The variables which must be defined in niriLogin are:

---

**TABLE 2.** Variables

| Variable         | Default value          | Purpose                                                                                                                                                                                                                                                                                           | Change <sup>a</sup> |
|------------------|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| NIRI_DIR         | /gemini/niri           | The path to the NIRI software source directory.                                                                                                                                                                                                                                                   | Always              |
| NIRI_DCPKGDIR    | /gemini/gnaac/gnaacSrc | The path to the NIRI detector-controller (GNAAC) source directory.                                                                                                                                                                                                                                | Always              |
| NIRI_SLALIBDIR   | /gemini/slalib         | This is the path to the installed Gemini slalib package.                                                                                                                                                                                                                                          | Always              |
| NIRI_TIMELIBDIR  | /gemini/timelib        | This is the path to the installed Gemini timelib package.                                                                                                                                                                                                                                         | Always              |
| NIRI_ASTLIBDIR   | /gemini/astlib         | This is the path to the installed Gemini astlib package.                                                                                                                                                                                                                                          | Always              |
| NIRI_PREFIX      | niri:                  | This prefix is prepended to all records. It should be four or fewer alphanumeric characters, followed by a colon.                                                                                                                                                                                 | If needed           |
| NIRI_DCPREFIX    | niri:                  | When accessing records in the detector controller, this prefix will be prepended to all records. It should be four or fewer alphanumeric characters, followed by a colon. It should not contain the dc: prefix, which is normally added to detector-controller records.                           | If needed           |
| NIRI_SADPREFIX   | \${NIRI_PREFIX}        | This prefix is prepended to all records in the status and alarm database. It should be four or fewer alphanumeric characters, followed by one or two colons.                                                                                                                                      | If needed           |
| NIRI_DCSADPREFIX | \${NIRI_DCPREFIX}      | When accessing records in the detector controller status and alarm database, this prefix will be prepended to all records. It should be four or fewer alphanumeric characters, followed by a colon. It should not contain the dc: prefix, which is normally added to detector-controller records. | If needed           |
| NIRI_NAME        | NIRI                   | This is used to generate a description of various records. It should be four or fewer alphanumeric characters.                                                                                                                                                                                    | If needed           |
| NIRI_TMPSCAN     | “Passive”              | This is used by the temperature interlock, which is described in [6], Section 2–4.2.1, “Monitoring an external record.”, on page 10. Set this to a scan rate, e.g., “1 second”.                                                                                                                   | If needed           |

**TABLE 2.** Variables

| Variable        | Default value    | Purpose                                                                                                                                                                                                                        | Change <sup>a</sup> |
|-----------------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| NIRI_TMPVAR     | 0                | This is used by the temperature interlock, which is described in [6], Section 2–4.2.1, "Monitoring an external record.", on page 10. Set this to a record field, e.g., niri:cc:tmpvar.                                         | If needed           |
| NIRI_TMPHB      | 0                | This is used by the temperature interlock, which is described in [6], Section 2–4.2.1, "Monitoring an external record.", on page 10. Set this to a record field, e.g., niri:cc:heartBeat.                                      | If needed           |
| NIRI_USEBANCOMM | 0                | If this is set to 1, then the Bancomm clock board is not initialized. This is useful for debugging the system when the timebus is not available <sup>b</sup> .                                                                 | For debugging only  |
| NIRI_USEVXTIME  | 1                | If this is set to 0, then the internal system clock is used, instead of the timebus. This is useful for debugging the system when the timebus is not available <sup>c</sup> .                                                  | For debugging only  |
| NIRI_AGPREFIX   | ag:              | This is the top-level prefix of the A&G system. The default value points to a simulation record, to avoid writing to the real A&G inadvertently. This may be set to aid in \${NIRI_PREFIX} debugging the system <sup>d</sup> . | For debugging only  |
| NIRI_LIBDIR     | \${NIRI_DIR}/LIB | The path to the NIRI shared source directory.                                                                                                                                                                                  | Not recommended     |
| NIRI_ENGDIR     | \${NIRI_DIR}/ENG | The path to the NIRI engineering source directory.                                                                                                                                                                             | Not recommended     |
| NIRI_CCDIR      | \${NIRI_DIR}/CC  | The path to the NIRI public components controller interface directory                                                                                                                                                          | Not recommended     |
| NIRI_WFSDIR     | \${NIRI_DIR}/WFS | The path to the NIRI public wavefront sensor components source directory                                                                                                                                                       | Not recommended     |
| NIRI_WCSDIR     | \${NIRI_DIR}/WCS | The path to the NIRI world coordinate system software directory.                                                                                                                                                               | Not recommended     |
| NIRI_ISDIR      | \${NIRI_DIR}/IS  | The path to the NIRI instrument sequencer source directory                                                                                                                                                                     | Not recommended     |
| NIRI_DCDIR      | \${NIRI_DIR}/DC  | The path to the NIRI to GNAAC interface directory                                                                                                                                                                              | Not recommended     |
| NIRI_TARGETS    | mv167            | Which EPICS-supported hardware platform <sup>e</sup> .                                                                                                                                                                         | Not recommended     |

- Items which are marked “Not recommended” will have unexpected results, unless the niri software is modified accordingly. Items which are marked “For debugging only” should be returned to the default value before running a working system. Details are outside the scope of this document.
- Note that this causes incorrect behavior, because local time is different from Gemini time.
- Note that this causes incorrect behavior, because local time is different from Gemini time.
- Note that this causes incorrect behavior, because the A&G status record will not be updated.
- The mv167 is the only supported architecture at this time.

*You should not need to change any other files. If you do need to do so, it should be reported as a bug.*

The csh/tcsh configuration file “niriLogin” sets up various environment variables and your path. You must source niriLogin before building or running any of the niri software. It would probably be wise to source “niriLogin” in your .login or .cshrc file:

```
source niriLogin
```

Run niriSetup to configure the software for your EPICS system, and copy in some standard EPICS files.

```
./niriSetup
```

## **6.5 Building the system**

Execute the following command:

```
gmake
```

This should build the system. The software should compile without warnings, *except*

- All C modules will warn “‘rcsid’ defined but not used.”
- The SNL code compiler (\*.stpp files) generates a lot of warnings about undefined variables.

## **6.6 Making dependency list**

Switch to the source directory

```
cd $NIRI_DIR/src
```

Make the dependency list

```
gmake depends
```

Repeat with \$NIRI\_LIBDIR/src, \$NIRI\_ENGDIR/src, and \$NIRI\_DCDIR/src.

This will insure that all files will be remade if the record configuration files or C header files are changed.

## **6.7 Loading the NIRI/CICS OIWFS/CICS control software or simulator**

Connect to your IOC and set the start-up script to

- (...) /bin/mv167/startupSim (simulator),
- (...) /bin/mv167/startup (control software),
- (...) /WFS/bin/mv167/startupSim (OIWFS simulator), or
- (...) /WFS/bin/mv167/startup (OIWFS control software),

where (...) represents the full, absolute path to the niri\_1.00 directory. Then reboot the IOC. The IOC should reboot without any errors. At this time, you should be running the software that you selected.

### **6.8 Running the user interface**

The system may be started by running the command:

```
niriStart
```

This will start a top level control screen from which you can start the NIRI CC, WFS, DC, and low-level engineering screens. Note: This screen is designed to control the whole NIRI system. If it is used to control the OIWFS system, non-OIWFS control widgets will be white. Separate screens to control the OIWFS only are not provided.

### **6.9 Cleaning and rebuilding**

The set up and installation process generates a number of files automatically. To insure that the old files are removed before building the system with a new configuration, type:

```
gmake reset
```

*Do not use*

```
gmake clean
```

because this fails to remove some files which should be removed, and remove some files which should not be removed. To restore the system to its original state, type:

```
gmake resetall.
```

---

## **7 Multiple copies of the software**

---

The NIRI/OIWFS software is designed to make it easy for multiple copies of the software to be built and in use simultaneously on separate IOC's. Multiple copies may be kept in synch by using CVS. Each copy should be built in a separate directory and should have its own version of niriLogin, with the program prefix and name defined differently. In order to make it easy to have two copies simultaneously (i.e., a database which is intended for general use, and a version with experimental changes, or a simulator running simultaneously on another IOC, the following features are provided:

- All symbolic links are created by the configuration script, for ease of use with CVS.
- All configuration information is stored in niriLogin, and in files that are automatically generated by the configuration process.
- The only time that it is actually necessary to source niriLogin is before running niriSetup and the user interface scripts (niriEng, niriStart, etc.).
- It is not necessary to source niriLogin before running gmake.

---

## **8 Limitations of the simulation**

---

The simulation is a reasonably accurate representation of the working niri system except:

- Mechanisms are idealized, i.e., all Hall-effect sensors lie on top of each other, there is no noise in the signals, there is no backlash, and the mechanisms cannot fail.
- The Hall effect sensors do not read the same values as the real mechanisms.
- The "FAST" simulation mode is not really very fast for datuming.

- The environment subsystem is a very poor simulation.

---

## **9 Unfinished tasks**

---

This section describes tasks which remain to be done. These tasks, for the most part, are tasks that are fairly easy to do at the telescope, and very difficult to do with a standalone system.

### **9.1 WCS**

**1. Calibration constants for the WCS have not been computed.**

This is easily done empirically, with far less effort than it would take to compute the values.

### **9.2 Networking and communications**

**1. The time bus has not been tested. Time functions have not been tested.**

This requires equipment which is not available to the NIRI developers. The necessary subroutines are in place, and if they function as expected, this should be a simple operation.

**2. The Interlock bus has not been tested.**

NIRI currently does not need to generate an interlock event, nor will it need to respond to an interlock demand. The software and hardware are provided so that it will be possible to interface with the interlock bus if needed. This should be tested.

### **9.3 wfsBeam**

**1. The filter positions which block the light path to the wfs have yet to be entered in the appropriate .lut file.**

This cannot be done until the final arrangements of filters is known.

---

## **10 Known bugs and limitations**

---

This section describes problems with the NIRI software. All bugs listed here are believed to be non-critical or have a reasonable work-around. In general, bugs in the EPICS or GEM EPICS have not been fixed, if there is an adequate work-around, in the interest of portability. The problems here are generally of the following types:

- Non-fatal bugs which have proven difficult to isolate;
- Bugs which would require extensive redesign to repair;
- Bugs which would require significant time to repair, and which do not have a significant effect on the quality of the system;
- Bugs in the underlying EPICS or GEM EPICS system.

### **10.1 Problems with the simulation**

**1. The simulation starts in an invalid location.**

When the system is first rebooted, its initial position is 0, and its initial Hall-effect sensor values are also 0.0. The Hall sensors will be computed correctly after the first motion command.

**2. It is not possible to switch into simulation mode using the sequencer commands.**

This was discussed with gemini, and it was decided that it was too difficult to implement with the existing tools. Simulation mode requires a separate startup script.

**3. The GNAAC simulator is not supported.**

The GNAAC simulator is very out of date and is not supported. The changes which are required to make it work with the NIRI software are probably small. The GNAAC simulator should be able to run on the same IOC as the NIRI software.

**4. FAST Simulation mode is not really “FAST”**

The FAST simulation mode takes a fair amount of time to complete some routine tasks (notably datuming). This could be improved with a small amount of effort, if the time and resources were made available.

**5. VSM Simulation modes is not supported at all**

The requirement for VSM was relaxed in discussions with Gemini. “FAST” mode is probably adequate for most purposes.

### **10.2 Problems with edd / dm**

**1. When the IOC is rebooted, some of the menus in the edd/dm user interface do not get loaded properly. The contents of the menu will be listed as “no menu.” Furthermore, some of the text / numeric displays show incorrect values (usually, obviously meaningless values).**

This is a problem with edd/dm. A work-around is to close (not just iconify) the affected windows, and restart them.

**2. The user interface dies unexpectedly.**

This is a known bug with edd/dm. If you open a window (call it window A), and use a button on window A to open another window (call it window B), then close window A, then the user interface may crash unexpectedly at some time in the future. This is a known problem with edd/dm. There is no known work-around, other than to use caution when closing windows.

**3. The user interface corrupts the color map of an X (including Openwindows) display.**

This is a known bug with edd/dm. There is no known work-around. You will have to restart your windowing system.

**4. The user interface screens cannot be edited with edd or displayed with dm.**

All user interface tools for NIRI have been built using edd2.4 and dm2.4, which uses an incompatible file format. The following commands should be used in place of the edd and dm commands.

```
env EPICS_DISPLAY_PATH=`pwd`/O.solaris:`pwd` edd2.4
env EPICS_DISPLAY_PATH=`pwd`/O.solaris:`pwd` dm2.4
```

**10.3 Stability problems**

**1. The system sometimes stops responding to EPICS channel-access commands (once per month).**

This problem occurs randomly, and only when the cooling motors are in use. This may have been caused by a bug in the motor record, which has now been fixed.

**10.4 Compiler Problems, Code, and Coding Standards**

**1. All C modules warn “‘rcsid’ defined but not used” during compilation.**

The Gemini Programming Standards Document [22] requires “rcsid” to be defined and requires code to be compiled with a high warning level. These two requirements cannot be met simultaneously.

**2. The SNL code compiler (\*.stpp files) generates a lot of warnings about undefined variables.**

These warnings are very difficult to suppress, because many constants are defined in C include files, but the SNL compiler cannot understand most C include files. Suppressing this error would require significant changes to header files, including system files, EPICS files, and Gemini-provided files or would require constants to be defined in more than one place, which is poor programming practice.

**3. Boolean variables are not used consistently. Sometimes TRUE is 0, and sometimes it is any non-zero value. Sometimes FALSE is 1, and sometimes it is 0.**

The EPICS Boolean convention (0 True, 1 False) is the reverse of the C convention (0 False, non-zero True). The EPICS convention is being used, the definitions EPICS\_TRUE and EPICS\_FALSE are always used. When using the C convention definitions are not always used. The engineering subsystem uses the C convention exclusively. All other systems should use the EPICS convention exclusively, except when interfacing to the Engineering subsystem, though there may be a few places where the C convention is used (these need to be corrected when found.)

**10.5 Mechanical problems**

**1. The cooling motors do not stay synchronized.**

There does not seem to be any way to fix this with the current hardware.

2. The software assumes that the number of motor-steps for a full rotation of one of the filter wheels and other wheel-like mechanisms is an integer.

Unfortunately, some of the wheels are a non-integer number of motor steps. This can cause slight positioning problems when the wheels have rotated a large number of turns from their home position. This is not believed to be a serious problem.

It may be necessary to datum the system occasionally or to use the diagnose command to return it to 0.

## **10.6 Datuming and diagnostic commands**

1. The diagnose commands and redatum commands can be very slow for the wheels.

These commands move the wheels back to the zero point, without taking advantage of the coordinate wraparound. This could be improved, but the diagnose and redatum commands are intended for infrequent, engineering use.

2. If a mechanism is put into motion by a low-level engineering command, and if it is in the not-datumed state, then a warning message appears: “Mechanism position lost spontaneously.”

It is not trivial to get rid of this warning message, because of the way that the SNL code is written. The warning message should just be ignored.

3. There is no warning if the pupil viewer is not properly seated.

The pupil viewer is an unusual mechanism. It seats itself by overrunning its limits and forcing itself into a hard stop. Handling this case would require a significant rewrite of the control code. In general, if the mechanism is ever handled manually, it is necessary to move it from one limit to the other or to datum it. (The high-level datum command automatically does a diagnose).

4. The wfsBeam record is of limited accuracy.

It would be difficult and time-consuming to figure out which orientations of the gimbal mirror would permit light to fall on the wave-front sensor with different pick-off mirrors. This and similar effects are completely neglected. The only things which are considered are mirror positioning and filter transparency.

## **10.7 Calibration tools**

1. The scan tool times out a lot, especially for the focussing stage. When it times out, the user interface leaves most of the controls disabled. The interface must be exited and restarted to fix the problem.

The scan tool is used very infrequently, and never by the end users. This really *should* be fixed, but is a low priority.

2. The scan tool, the position tool, and the cycle tool handle don’t handle text input very well.

The version of tcl / tk which is shipped with the GEM5 release is quite old. I do have a newer version of tcl / tk and have ported etwish to that release, but I don’t use it with the



NIRI software, because that would make my utilities incompatible with the unmodified GEM5 release.

### **10.8 Other Problems**

1. The number of frames for the observe command should really be part of the setup command.

The frames parameter is likely to change frequently, and shouldn't really require a complete set up of the detector controller in order to change its value. There could be a separate command, just to set the frame count.

2. Using caget or caput to access elements of type "double" in the HallStep record causes a fault, which renders the record non-functional.

There is probably some EPICS requirement that is not being satisfied by the HallStep record. Oddly enough, this problem only occurs with caget and caput. Edd / dm, cau, pvload, and etwish all have no problem.

3. There is no easy way, short of rebooting, to restore all parameters that can be set using the low-level engineering interface to their initial values. Using the low-level engineering interface, it is easy to put the system into a state in which the usual commands do not function properly, even though the system may appear normal. The low-level engineering interface is capable of putting the system into a state from which rebooting will not recover and can even put the system into a state from which software recovery is not possible.

There are just too many things that can be done with the low-level engineering interface to safeguard against all possibilities. There is no way to avoid this, except by disabling the low-level engineering commands completely. The low-level engineering commands are on screens which have a distinct black title bar with bright yellow letters. Be sure that you know what you are doing, before using the low-level engineering commands!

4. The current position for the guide probe can show "INVALID" when it should show "Undefined."

This is because of a limitation on the number of entries in a lutin/lutout record. In practice, the named positions are not used for the guide probe.

## **11 Possible Enhancements**

---

The following modifications are candidates for release 1.01.

1. Datuming is too slow.

The datuming algorithm is very conservative, in order to work reliably with no modification under all conditions, both at room temperature and cryogenic temperatures.

2. The temperature control system is not as well automated as it should be.

Scheduling will make it impossible to complete the front end to the temperature control system. The existing code is sufficiently usable that simplifying the task is considered a low priority.

3. There are problems with the motor record.

Janet Tvedt discovered several problems with the motor record, and provided patches to fix these problems. These patches have not been incorporated into the hmotor record (NIRI’s customized motor record), and some of them should probably be applied. Currently, the hmotor record is working fairly reliably in the mode in which NIRI requires it, so there seems to be little need to apply these changes. If the hmotor record is used in other projects, however, there may be problems.

4. Utility scripts need to be rewritten.

The utility scripts in “scripts” should be rewritten in another language. They are really too complicated to be shell scripts. In particular, there are a number of global variables which are reused everywhere. It is not a serious problem with the existing scripts, but requires careful attention to which variables are reused.

5. Health problems cannot be traced to alarms

This is a Gemini requirement which is not properly met. In fact, although the health tree is quite extensive, alarms are practically non-existent.

6. Issuing multiple movement commands to a single mechanism can cause problems. In particular, it is not possible to issue a datum command and a motion command at the same time.

There is a race condition, which can occur. It would be possible to add an interlock to keep this from happening, but for now, nothing is done. This should never occur in normal operation anyway; however, it should really be fixed.

---

## 12 Design Decisions (“It’s not a bug, it’s a feature!”)

---

1. Mechanisms do not datum themselves automatically.

Mechanisms that datum themselves automatically can be a real pain to work with during hardware and software testing and integration. Furthermore, the mechanisms retain knowledge of their current positions, even through a reboot (though not a power cycle), which can be a significant time saver, allowing the entire time-consuming datuming procedure to be bypassed. Since NIRI does not appear to need frequent rebooting under normal conditions, this is not a serious problem.

2. The Health state is not set to the WARNING mode if an operation fails.

The CICS sets its health to WARNING if an operation fails. This seemed to be rather unintuitive and was misleading to the people who tested the NIRI software, so it was removed. In general, any failure which can cause an operation to fail will trigger some warning, most commonly, the backlash warning or the not-datumed warning.

**3. If a mechanism times out, the CAR record is not set to the ERR state.**

Timeout is now handed via a warning SIR record, instead of via the CAR record. This is because certain operations can take a very long time under certain conditions. For example, datuming all mechanisms is much more time consuming than might be expected, because all mechanisms cannot be operated simultaneously.



# Gemini Near Infra-Red Imager and Wavefront Sensor



## Near IR Imager (NIRI) and Wavefront Sensor (OIWFS)— Definitions, Abbreviations and References

Hubert Yamada

niri\_h ty\_009 / 05 / 1999-10-23

This report lists all abbreviations and references which are specific to the Gemini “Near Infra-Red Imager” and “On Instrument Wavefront Sensor.”

### 1 Introduction

#### 1.1 Purpose of the document

This is a master list of all abbreviations and acronyms that are used by the software documentation for the Near Infra-Red Imager (NIRI) and On Instrument Wavefront Sensor (OIWFS).

#### 1.2 Scope

The software product to be developed from this document will be the control system for the Near Infra-Red Imager (NIRI) and the On Instrument Wavefront Sensor (OIWFS).

#### 1.3 Abbreviations and Acronyms

*This list is incomplete. For a more complete listing, see the Abbreviations and Acronyms list in [16].*

|       |                                      |
|-------|--------------------------------------|
| A&G   | Acquisition and Guiding              |
| A/D   | Analog to Digital Converter          |
| CICS  | Core Instrument Control System       |
| CC    | Components Controller                |
| DC    | Detector Controller                  |
| GNAAC | Gemini NOAO Aladdin Array Controller |
| GNIRS | Gemini Near Infra-Red Spectrograph   |
| HS    | Hall-effect Sensor                   |
| ICD   | Interface Control Document           |
| IS    | Instrument Sequencer                 |

|        |                                               |
|--------|-----------------------------------------------|
| N/A    | Not Applicable                                |
| NIRI   | Near Infra-Red Imager                         |
| OCS    | Observatory Control System                    |
| OIWFS  | On Instrument Wave-Front Sensor               |
| OMS    | Oregon Microsystems                           |
| TBD    | To Be Determined                              |
| TCS    | Telescope Control System                      |
| WCS    | World Coordinate System                       |
| UH IfA | University of Hawaii, Institute for Astronomy |

### 1.3.1 Stylistic Conventions

References to documents in Section 1.5, “References,” on page 2 are given like this [1].

## 1.4 Definitions

*This list is incomplete. For a more complete listing, see the CICS definitions list in [17].*

**Hall-effect Sensor.** An electronic component which is used to detect and to measure magnetic field strength.

## 1.5 References

- [1] niri\_h ty\_000.fm (obsolete, superseded by [2] and by [3]), “NIRI Optical-Components Controller: Implementation Issues,” Hubert Yamada, Ev Irwin, Tony Young, University of Hawaii, Institute for Astronomy.
- [2] niri\_h ty\_001.fm<sup>1</sup>, “Near IR Imager (NIRI)—Architectural Design Document,” Hubert Yamada, University of Hawaii, Institute for Astronomy.
- [3] niri\_h ty\_002.fm, “Near IR Imager (NIRI)—Detailed Design Document,” Hubert Yamada, University of Hawaii, Institute for Astronomy.
- [4] niri\_h ty\_003.fm, “Near IR Imager (NIRI) to Observatory Control System,” Hubert Yamada, University of Hawaii, Institute for Astronomy.
- [5] niri\_h ty\_004.fm, “Near IR Imager (NIRI)—Calibration and Maintenance Tools,” Hubert Yamada, University of Hawaii, Institute for Astronomy
- [6] niri\_h ty\_005.fm “NIRI Science Instrument to On Instrument Wavefront Sensor”, Hubert Yamada, University of Hawaii, Institute for Astronomy.
- [7] niri\_h ty\_006.fm “Near IR Imager (NIRI)-Software Release Notes”, Hubert Yamada, University of Hawaii, Institute for Astronomy; Steven Beard, Royal Observatory Edinburgh.
- [8] niri\_h ty\_007.fm “Near IR Imager (NIRI)-The Engineering User Interface”, Hubert Yamada, University of Hawaii, Institute for Astronomy.
- [9] niri\_h ty\_008.fm “A&G to NIRI/GNIRS On Instrument Wavefront Sensor”, Hubert Yamada, University of Hawaii, Institute for Astronomy.
- [10] niri\_h ty\_009.fm, “Near IR Imager (NIRI)—Definitions, Abbreviations and References, Hubert Yamada, University of Hawaii, Institute for Astronomy

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1. The NIRI and OIWFS software documentation is currently under development. Working drafts are available by ftp from [ftp://gemftp.ifa.hawaii.edu/gemini\\_IFA/niri](ftp://gemftp.ifa.hawaii.edu/gemini_IFA/niri).

- [11] NIRI Hardware-specification document(s), under preparation, Tony Young
- [12] NIRI Electronics-specification document(s), under preparation, Ev Irwin
- [13] NIRI Wavefront-sensor detector document(s), under preparation, Ev Irwin
- [14] cics\_smb\_041, “CICS—Detailed Design Document,” Steven Beard, Royal Observatory Edinburgh.
- [15] cics\_smb\_022, “CICS—Architectural Design Document,” Steven Beard, Royal Observatory Edinburgh.
- [16] cics\_smb\_013, “Science Instrument to Observatory Control System”, Steven Beard, Royal Observatory Edinburgh.
- [17] cics\_smb\_039, “CICS Project Glossary and References,” Steven Beard, Royal Observatory Edinburgh.
- [18] GNAAC.ICD.193\_195ab.fm, “ICD 1.9.3[ab]—Detector Controller to NIRI/NIRS ICS Interface”, Nick C. Buchholz, NOAO.
- [19] ic16110.doc, “ICD 1.6/1.10 A&G to On-Instrument Wavefront Sensors”, Nick Dillon, Malcolm Stewart, Steven Beard, Brian Leckie.
- [20] ag\_jms\_004, “Control of Wavefront Sensors”, Malcolm Stewart
- [21] SPE-C-G0037, “*Software Design Description*”, Gemini 8m Telescopes Project.
- [22] SPE-I-G0009, “*Software Programming Standards*”, Gemini 8m Telescopes Project.
- [23] “*Instrumentation (Facilities and Science) ICD’s Guidelines*”, Susan Wieland, Gemini 8m Telescopes Project.
- [24] gscg.kkg.009, “*ICD 1a — The System Command Interface*”, Kim Gillies, NOAO.
- [25] gscg.kkg.024, “*ICD 1b — The Baseline Attribute/Value Interface*”, Kim Gillies, NOAO.
- [26] gscg.kkg.010, “*ICD 2 — Systems Status and Alarm Interfaces*”, Gemini 8m Telescopes Project.
- [27] gscg.grp.015, “*ICD 7a — ICS Subsystem Interfaces*”, Steven Beard, Steve Wampler and Chris Mayer, Gemini 8m Telescopes Project.
- [28] gscg.grp.009, “*ICD 7b— TCS Subsystem Interfaces*”, Philip Taylor
- [29] gscg.grp.???, “*ICD 12—Interlock System*”, Peregrin McGehee
- [30] gscg.grp.???, “*ICD 13 — Standard Controller*”, Bret Goodrich & Andrew Johnson, Gemini 8m Telescopes Project.
- [31] gscg.grp.025, “*ICD 16 — The Parameter Definition Format*”, Steve Wampler, Gemini 8m Telescopes Project.
- [32] ocs.kkg.031, “*Sequence Command Specifications*”, Kim Gillies, Shane Walker & Steven Beard, NOAO/Gemini 8m Telescopes Project.
- [33] pg-c-0019, “*An Engineering back-door for Gemini instruments*”, Steve Wampler, Gemini 8m Telescopes Project.
- [34] cics\_smb\_045, “*CICS — Project History Document*”, Steven Beard, Royal Observatory Edinburgh.
- [35] niri\_h ty\_011.fm, “Near IR Imager (NIRI) and On Instrument Wavefront Sensor (OIWFS) —Implementation Notes and Record Reference Manual,” Hubert Yamada, University of Hawaii, Institute for Astronomy.
- [36] niri\_h ty\_013.fm, “Near IR Imager (NIRI)—Position Names”
- [37] niri\_h ty\_014.fm, “Near IR Imager (NIRI) and OIWFS—Position Names”

- [38] wcsForDummies.txt, “Gemini World Coordinate Systems for Dummies,” Steven Beard, Royal Observatory Edinburgh.



# Gemini Near Infra-Red Imager and Wavefront Sensor



# Near IR Imager (NIRI) and On-Instrument Wavefront Sensor (OIWFS)—Architectural Design Document

**Hubert Yamada**

niri\_hyt\_001 / 05 / 1999-10-23

**This report gives a architectural software design for the Gemini “Near Infra-Red Imager and On-Instrument Wavefront Sensor software.”**

---

## 1 Introduction

---

This report gives an architectural software design for the Near Infra-Red Imager (NIRI) and provides description of its software decomposition.

### 1.1 Purpose of the document

This document describes the difference between the NIRI design and the template which is provided by the CICS, described in [15]. It is assumed that the reader of this document is familiar with that document.

The intended audience of this document is:

- the NIRI reviewers;
- the developers of the Gemini Observatory Control System (OCS);
- the developers of the Gemini Acquisition and Guidance units
- the developers of the GNIRS OIWFS components controller and other groups which will be using the design of the NIRI OIWFS.

### 1.2 Scope

The software product to be developed from this document will be control software for the Near Infra-Red Imager (NIRI).

### 1.3 Definitions, Acronyms and Abbreviations

See “Near IR Imager (NIRI)—Definitions, Abbreviations, and References”.

## 1.4 Stylistic Conventions

References to documents in Section 1.5, “References,” on page 2 of “Near IR Imager (NIRI)—Definitions, Abbreviations, and References” are given like this; [1].

## 1.5 References

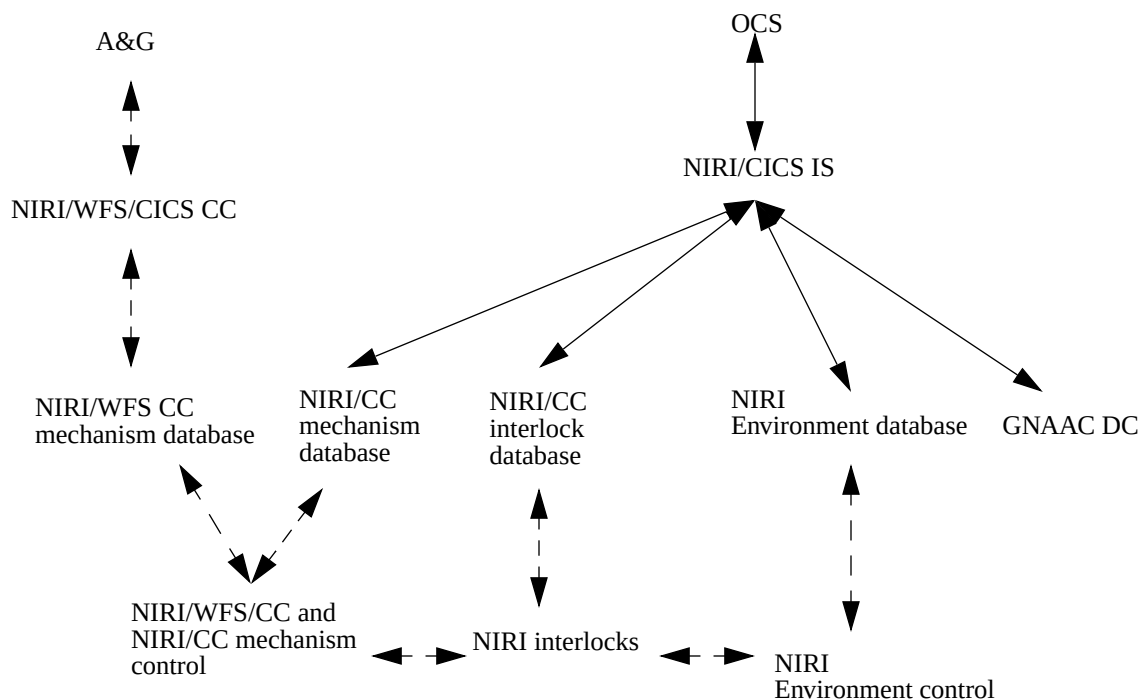
See “Near IR Imager (NIRI)—Definitions, Abbreviations, and References”.

---

## 2 Implementation

---

At this time, the NIRI software is expected to follow the CICS architectural design very closely. The differences will be detailed in a later version of this document.



Solid lines represent commands that are linked by apply records

# Gemini Near Infra-Red Imager and Wavefront Sensor



# Near IR Imager (NIRI) and On-Instrument Wavefront Sensor (OIWFS)—Detailed Design Document

**Hubert Yamada**

niri\_h ty\_002 / 09 / 1999-10-23

**This report gives a detailed software design for the Gemini “Near Infra-Red Imager” and On Instrument Wavefront Sensor.**

---

## 1 Introduction

---

This report presents a the software design for the control software for the Gemini Near Infra-Red Imager.

This report supersedes the preliminary document, niri\_h ty\_000.fm, “Optical-Components Controller: Implementation Issues,” which was released as part of the NIRI PDR documents.

Note that the design of many of the hardware elements is not finalized at this time. Some control algorithms may need to be modified.

### 1.1 Purpose of the document

This report gives a describes the differences between the design of the software for the Near Infra-Red Imager (NIRI) and the prototype instrument which is described in the CICS documents [14]. This document includes a description of the software for both NIRI and the NIRI OIWFS<sup>1</sup> components controllers. At some time, the OIWFS documents will be placed into a separate document.

The intended audience of this document is:

- the NIRI reviewers;
- the developers of the Gemini Observatory Control System (OCS);
- the developers of the Gemini Acquisition and Guidance units;

---

1. For purposes of this document, the terms NIRI OIWFS and OIWFS will be considered synonymous, though the NIRI OIWFS will be used by other systems.

- the developers of the GNIRS OIWFS components controller and other groups which will be using the design of the NIRI OIWFS.

It is assumed that readers of this document are familiar with the CICS prototype instrument system which is described in [14], [15], [16]. Throughout this document, unless otherwise stated, it is assumed that the software design for the NIRI control software is the same as the software design for the CICS prototype instrument

## **1.2 Scope**

The software product to be developed from this document will be the control system for the Near Infra-Red Imager (NIRI).

## **1.3 Definitions, Acronyms and Abbreviations**

See “Near IR Imager (NIRI)—Definitions, Abbreviations, and References”.

## **1.4 Stylistic Conventions**

References to documents in Section 1.5, “References,” on page 2 of “Near IR Imager (NIRI)—Definitions, Abbreviations, and References” are given like this; [1].

## **1.5 References**

See “Near IR Imager (NIRI)—Definitions, Abbreviations, and References”.

---

# **2 Implementation**

---

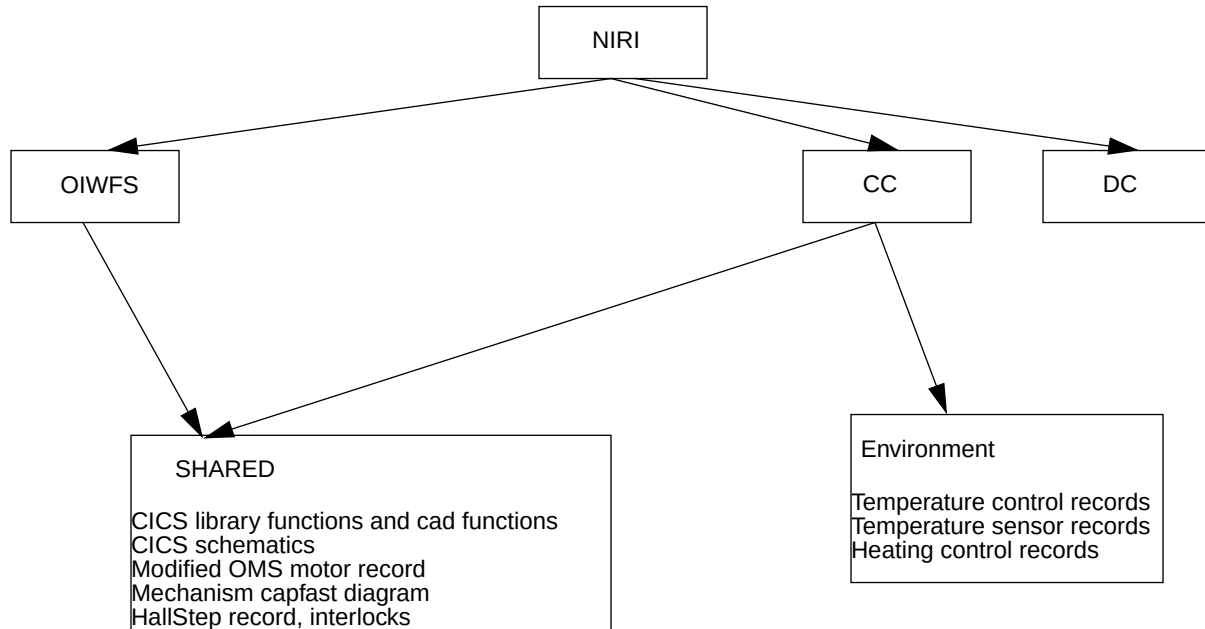
## **2.1 Overview**

The NIRI and NIRI OIWFS control software will be based on the Core Instrument Control System (CICS). Whenever possible, CICS commands will be duplicated. New commands and their corresponding records will be as similar to existing CICS commands as possible. If there is nothing in the CICS which can be adapted to our purposes, a new command will be designed and implemented, using the specifications in the ICD’s (especially ICD 7a and ICD 1b).

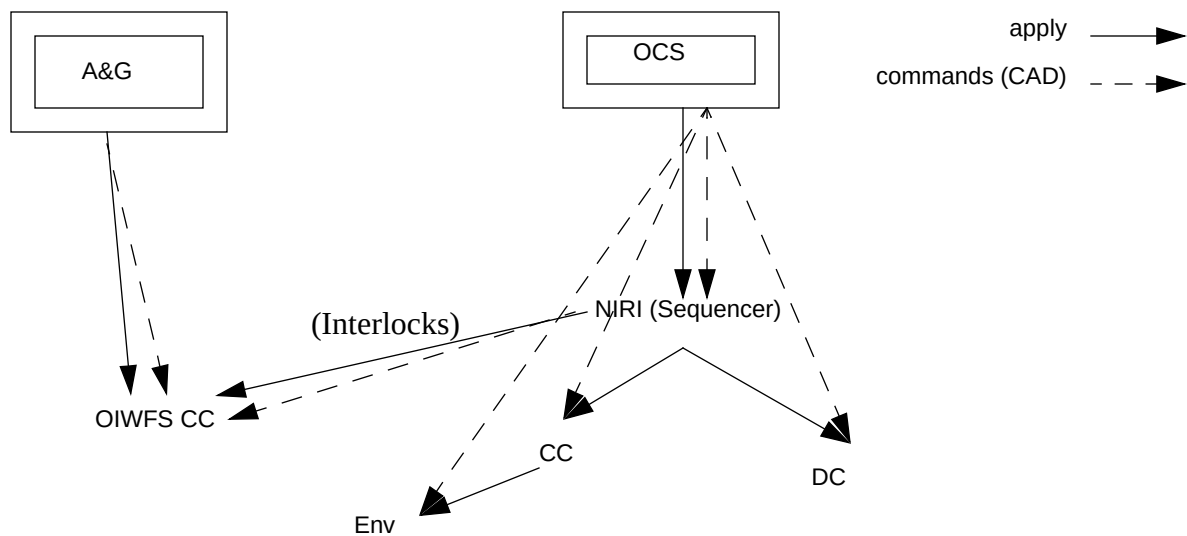
The instrument sequencer (IS) will forward sequence commands to the components controller (CC) and to the detector controller (DC); it will get the World Coordinate System (WCS) information from the TCS; it will modify the WCS information to include instrument-specific information; it will forward the modified coordinates to the DC; and it will set the TCS pointing origin.

The detector controller is a separate work package, and is described in [18].

**FIGURE 1.** Module Dependencies



**FIGURE 2.** Command Flow



## 2.2 Static Storage

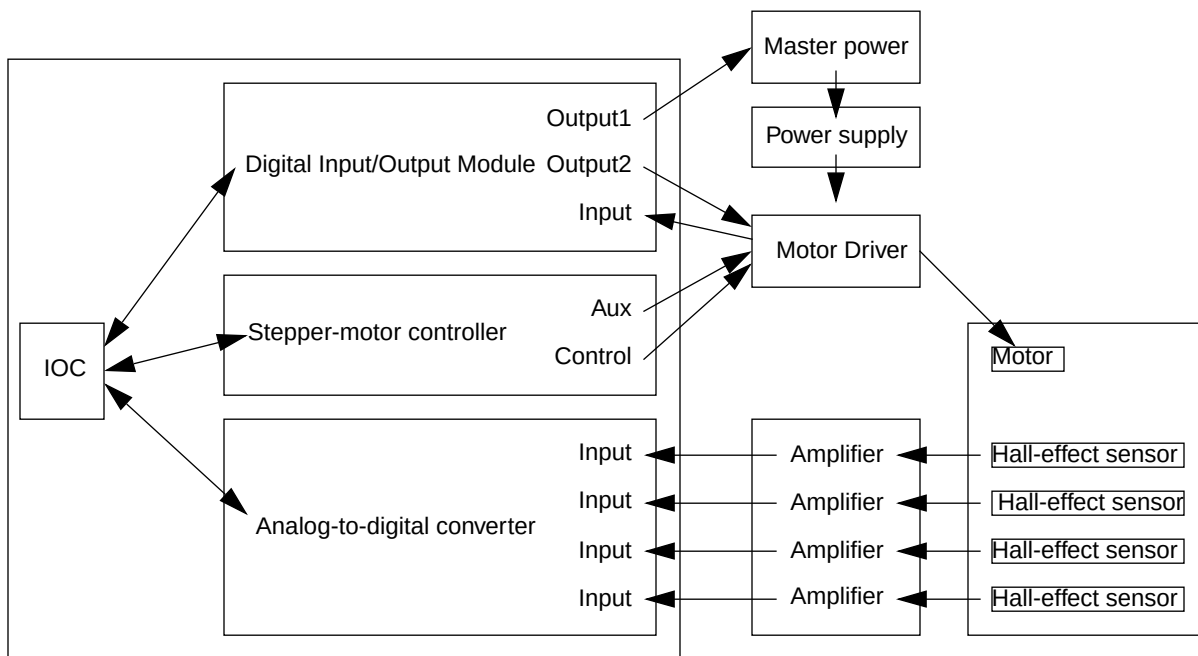
### 2.2.1 Static Look-Up Tables

Upon initialization of the system, calibration information will be loaded into appropriate EPICS process variables using pload. This information will be necessary to calibrate the Hall-effect sensors. The calibration data will be generated by a separate program, which will be described in [5].

## 3 Stepper Motor / Hall-Effect Sensor Control

All mechanisms will be controlled using a combination of stepper motors and Hall-effect sensors.

**FIGURE 3.** Control system overview (single mechanism)



### 3.0.1 Notes on Figure 3

- All inputs in Figure 3 lead in to the same amplifier and A/D boards. This will probably not be true. The inputs may be distributed among several boards, so that the system will remain partially functional if a single board fails.
- Although the master power control is shown in this single-component diagram, this will actually be shared among all mechanical components. The master power will be shut off only when all components are idle.
- The Digital Input/Output, “Output2” and “Input” are used to check the status of the stepper motor driver, and to reset it if there is a fault.

- The Stepper-motor controller Aux output will be used to shut down the power to the motor when it is not in motion.

### **3.1 Hall-Effect Sensors and the Home Position**

The stepper motors are not encoded. Instead, each component will have a magnet at its datum (home) position. A Hall-effect sensor will be used to detect this magnet to locate the datum position. For some components there may also be other magnets giving physical limits and other reference positions. These secondary magnets may be distinguished from the home-position magnet by using the opposite polarity, and may be distinguished from each other by a different peak field strengths at the sensor<sup>2</sup>.

One sensor is designated as the “primary” sensor. This sensor will be connected to the “home” input of the Oregon Microsystems stepper-motor controller, which will speed up the datuming process. If this sensor fails, then the datum position may still be found, but will require a slower search process.

#### **3.1.1 A discrete positioning mechanism (4-position filter wheel)**

Figure 4 describes a simple, 4-position filter wheel. For mechanisms with discrete positions (e.g., filter wheels) there will be reference magnets for each distinct position, as well as the magnets for the home position. For the filter wheel, the reference magnets (and their corresponding sensors) are mounted on two concentric rings. By using a combination of magnets with different field strengths at the sensor, each position may be uniquely encoded. Note that if either sensor fails, the system will continue to operate, though the encoding scheme may no longer be unique.

---

2. Different field strengths at the sensor may be produced either by varying the depth of the magnet or by using magnets with different strengths.

**FIGURE 4.**

A simple mechanism (wheel)

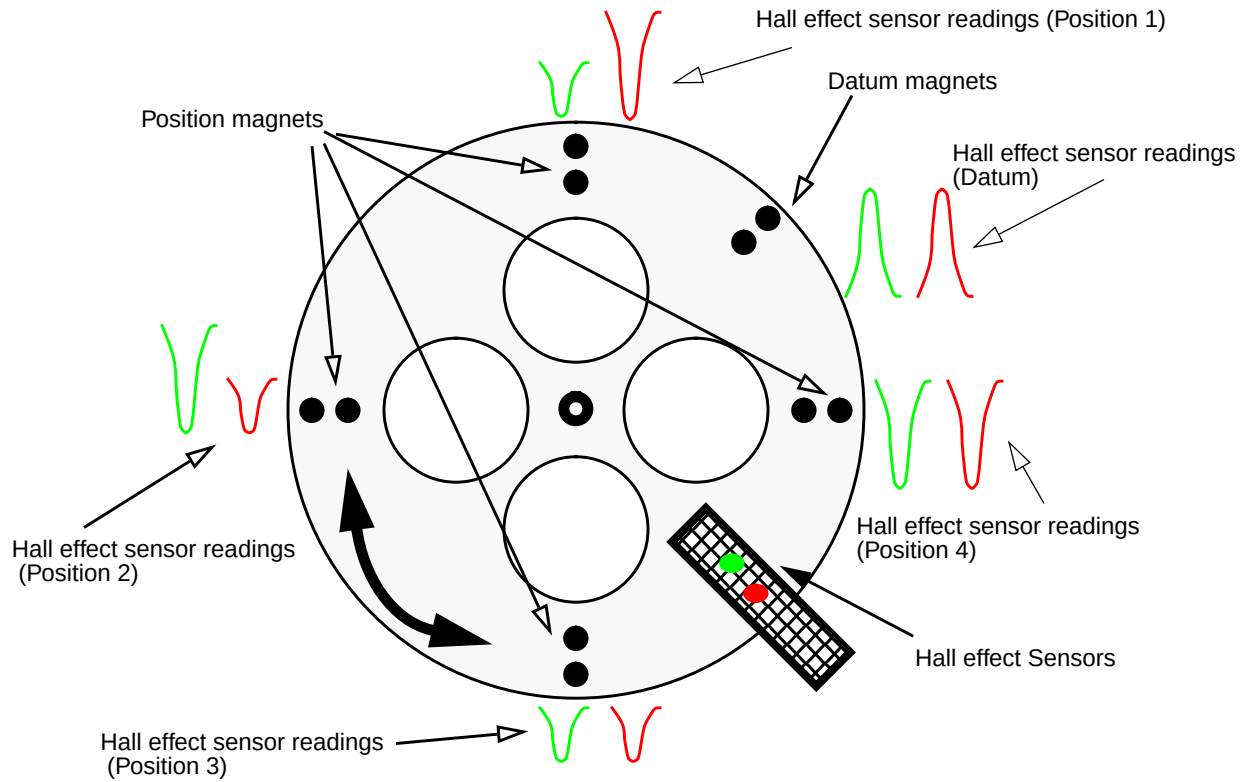


Table 1 describes the sensor values corresponding to Figure 4.

**TABLE 1.**

Sensor Values for Figure 4

| Position | Sensor 1 (V) | Sensor 2 (V) |
|----------|--------------|--------------|
| 1        | -2.0         | -4.0         |
| 2        | -4.0         | -2.0         |
| 3        | -2.0         | -2.0         |
| 4        | -4.0         | -4.0         |
| Datum    | +4.0         | +4.0         |



Table 2 describes typical sensor values for a twelve-position wheel. Note that it is assumed that it is only possible to distinguish four different magnet strengths. In practice, it may be possible to distinguish more values.

---

**TABLE 2.**

Typical Sensor Values (12 position wheel)

| Position | Sensor 1 (V) | Sensor 2 (V) |
|----------|--------------|--------------|
| 1        | -4.0         | -4.0         |
| 2        | -4.0         | -3.0         |
| 3        | -4.0         | -2.0         |
| 4        | -4.0         | -1.0         |
| 5        | -3.0         | -4.0         |
| 6        | -3.0         | -3.0         |
| 7        | -3.0         | -2.0         |
| 8        | -3.0         | -1.0         |
| 9        | -2.0         | -4.0         |
| 10       | -2.0         | -3.0         |
| 11       | -2.0         | -2.0         |
| 12       | -2.0         | -1.0         |
| Datum    | +4.0         | +4.0         |

For smaller mechanisms with a small number of positions, the two-sensor arrangement will be modified to use a single back-to-back sensor pair.

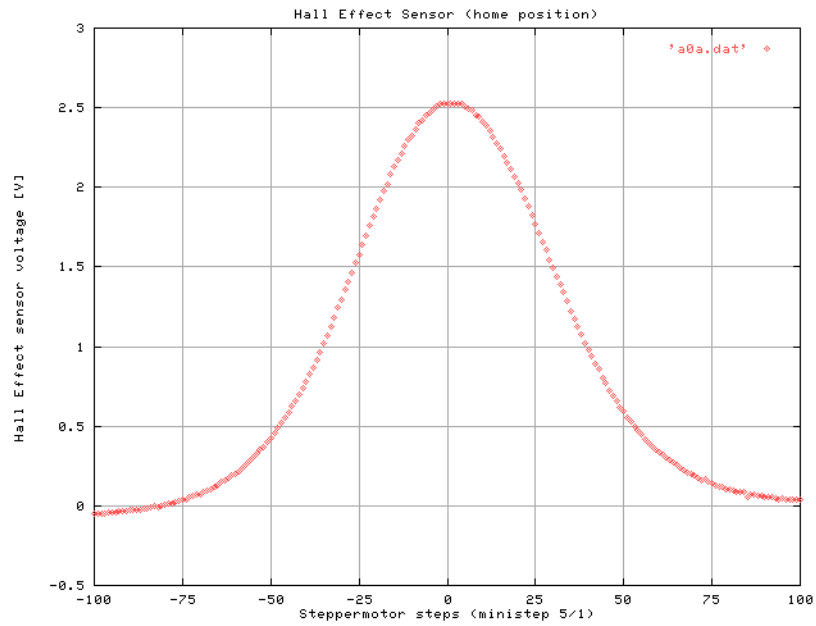
#### 3.1.1.1      Sensor data plots (home magnet)

Figure 5 is a data plot of the Hall-Effect sensor values near a datum magnet. Note that the peak is sharply defined and nearly symmetric, though careful inspection shows a slight asymmetry. The data values become negative in the wings of the peak. This is probably a real effect, due to the reversal of the direction of the magnetic field near a magnet.

This data was taken using a 5/1 mini-step stepper motor controller. In practice, the mini-steps may prove to be useless, because the motor will tend to fall into a detente when power to the motor is removed.

FIGURE 5.

Hall effect sensor readings (Datum magnet)



### 3.1.1.2 Sensor data plots (Confirmation magnet)

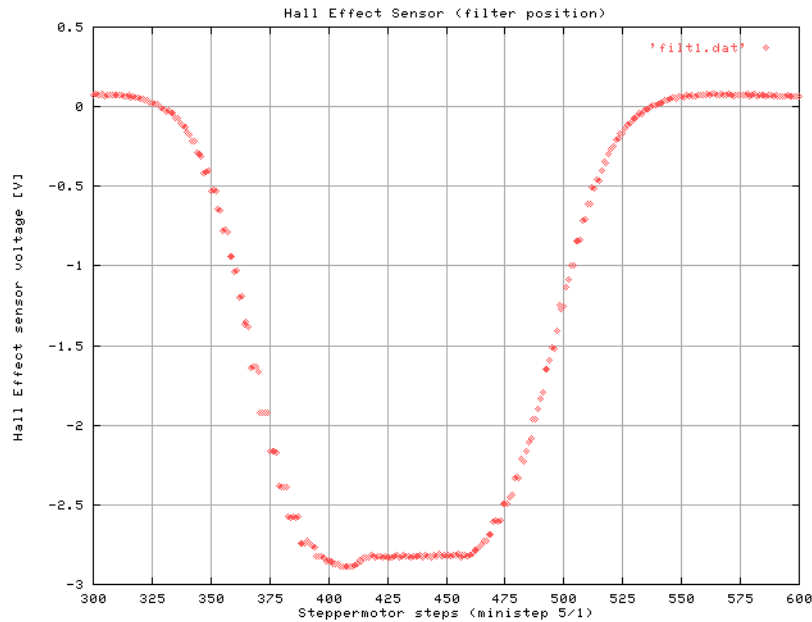
Figure 5 is a data plot of the Hall-Effect sensor values near a datum magnet. Note that the (negative) peak is flattened. This demonstrates the effectiveness of the Geneva mechanism which is used to turn the filter wheel. (The Geneva mechanism causes the motion of the wheel to be non-uniform. The motion of the wheel is slowed in the vicinity of the filters, making the hardware relatively insensitive to positioning errors.) The data values become positive in the wings of the peak. This is probably a real effect, due to the reversal of the direction of the magnetic field near a magnet.

This data was taken using a 5/1 mini-step stepper motor controller. In practice, the mini-steps may prove to be useless, because the motor will tend to fall into a detente when power to the motor is removed.

---

**FIGURE 6.**

**Hall effect sensor readings (Filter magnet)**



### 3.1.2 Continuous-positioning mechanisms (OIWFS focusing stage)

In some cases, a single sensor will be unable to provide adequate resolution. Higher resolution will be provided by using a combination of magnets and sensors on different parts of the mechanism (e.g., for the focusing stage, one magnet/sensor pair on the moving part of the stage, and a second pair on the main gear).

Figure 7 describes an (over)simplified model of a focusing stage, which requires two sensor pairs to define the datum position. The coarse sensor is able to position the stage in the vicinity of the datum, unambiguously, but with poor resolution. The gear sensor is able to determine the position very accurately, but the true position can be incorrect by an integral number of gear rotations.

A dual-magnet linear sensor arrangement is shown. The coarse-positioning or “linear” magnets are placed with opposite polarities, producing two adjacent peaks, one positive, and one negative. Between the peaks, there is a ramp, which is (almost) linear. The stage is designed so that it always operates in the linear region, providing a useful (though not extremely accurate) encoding of the position of the stage. Note that any changes in the sensitivity of the sensor electronics or constant offset can affect this encoding scheme. This is sufficient, because the encoding scheme needs only to position the stage at the datum, to within one turn of the main gear.

**FIGURE 7.** A two-sensor arrangement (OIWFS Focusing stage)

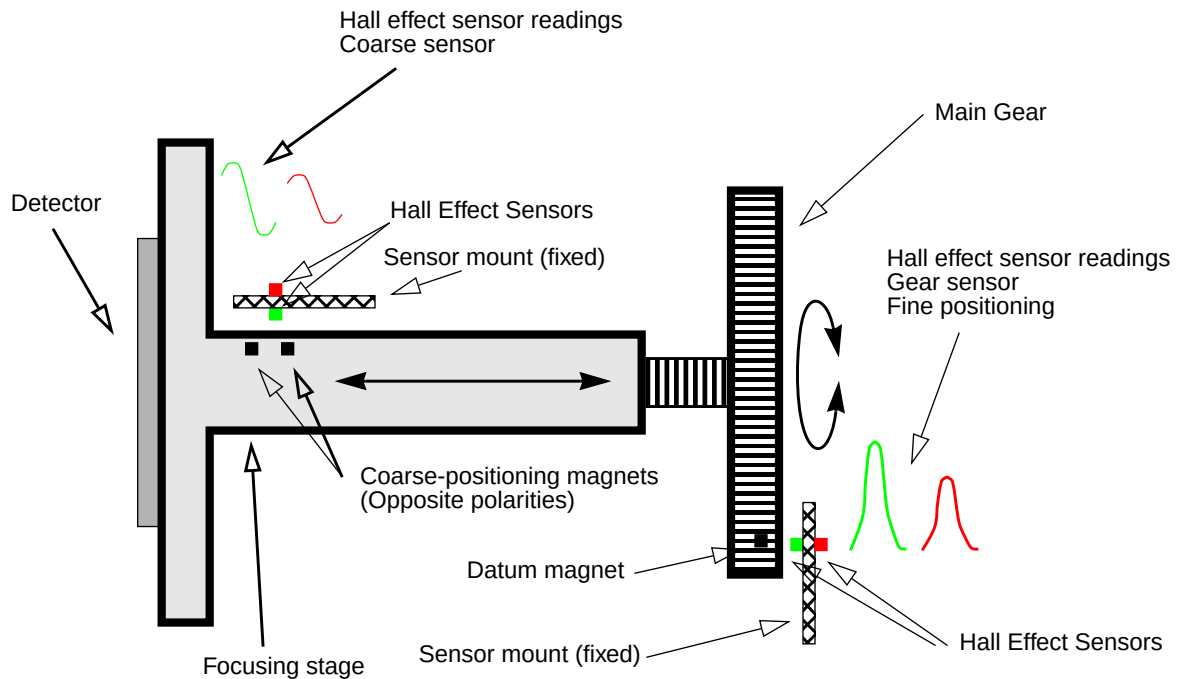
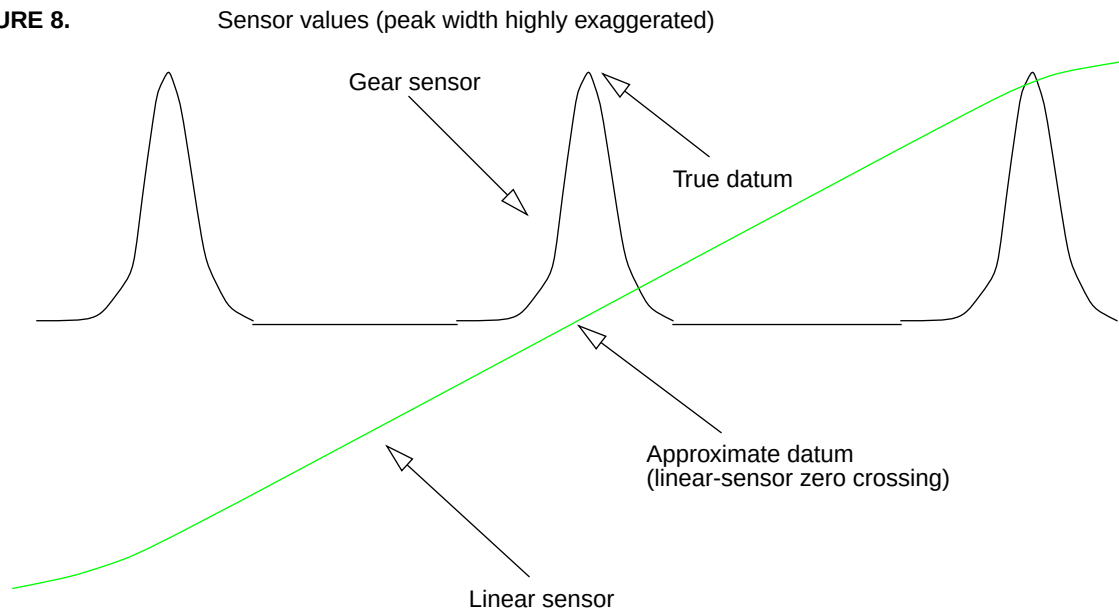


Figure 8 shows the expected reading from the Hall-effect sensors.

**FIGURE 8.**



Unlike the filter wheel, the Hall-effect sensors here are mounted on top of each other. This arrangement is simpler than for the filter wheel, because it requires only a single magnet, not a pair of magnets. Unfortunately, because it is not possible to insure that the sensors are precisely concentric with each other, there is a slight difference in the peak position. This must be compensated for via an offset value, which must be determined empirically. Furthermore, the signal strength for the sensor which is further away from the magnets is noticeably weaker.

One sensor is designated as the “primary” sensor. This sensor will be connected to the “home” input of the Oregon Microsystems stepper motor controller, which will speed up the datuming process. If this sensor fails, then the datum position may still be found, but will require a slower search process.

#### 3.1.2.1 Reference points

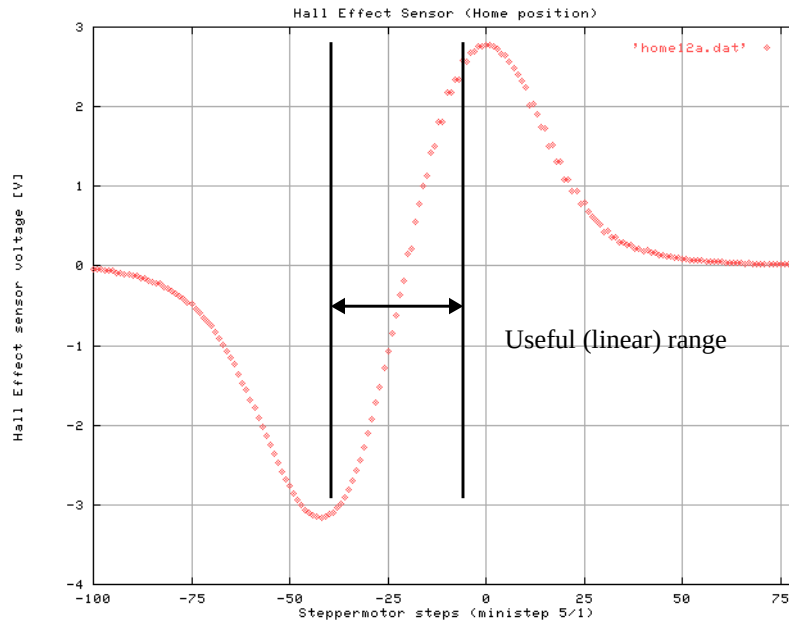
Although there are no predefined locations, a list of (arbitrarily chosen) reference points will be defined. To insure proper operation of the system, a command (“DIAGNOSE”) will be provided to move the mechanism to each reference point, and the expected value of the hall-effect sensors will be compared the actual value.

### 3.1.2.2 Sensor data plots (“linear” arrangement)

Figure 9 shows measured values for the Hall effect sensors in a “linear” arrangement. Note that the data values are almost linear in the region between the peaks. This provides a useful partial encoding scheme; however it is too susceptible to instabilities in the electronics to be used as a precise encoding scheme.

**FIGURE 9.**

Hall effect sensor readings (“linear” magnet arrangement)



### 3.1.3 Other continuous-positioning mechanisms (science focus and gimbals)

Unfortunately, not all of the continuous-positioning mechanisms are identical. The focussing stage for the science detector and the OIWFS gimbal mirrors are similar to the focussing stage for the OIWFS detector, but are not identical.

#### 3.1.3.1 The focussing stage for the science detector

The larger range of travel of the focussing stage for the science detector means that the full range of travel of the stage cannot be uniquely encoded, as it is for the OIWFS focussing stage. To get around this problem, additional magnets are installed on the translation stage to insure that there is a region in which the signal value is a one-to-one function of the position. The ends do not provide a unique signal, but the sign of the signal is sufficient to indicate the location of the linear region.

**FIGURE 10.** A two-sensor arrangement (Science Focusing stage)

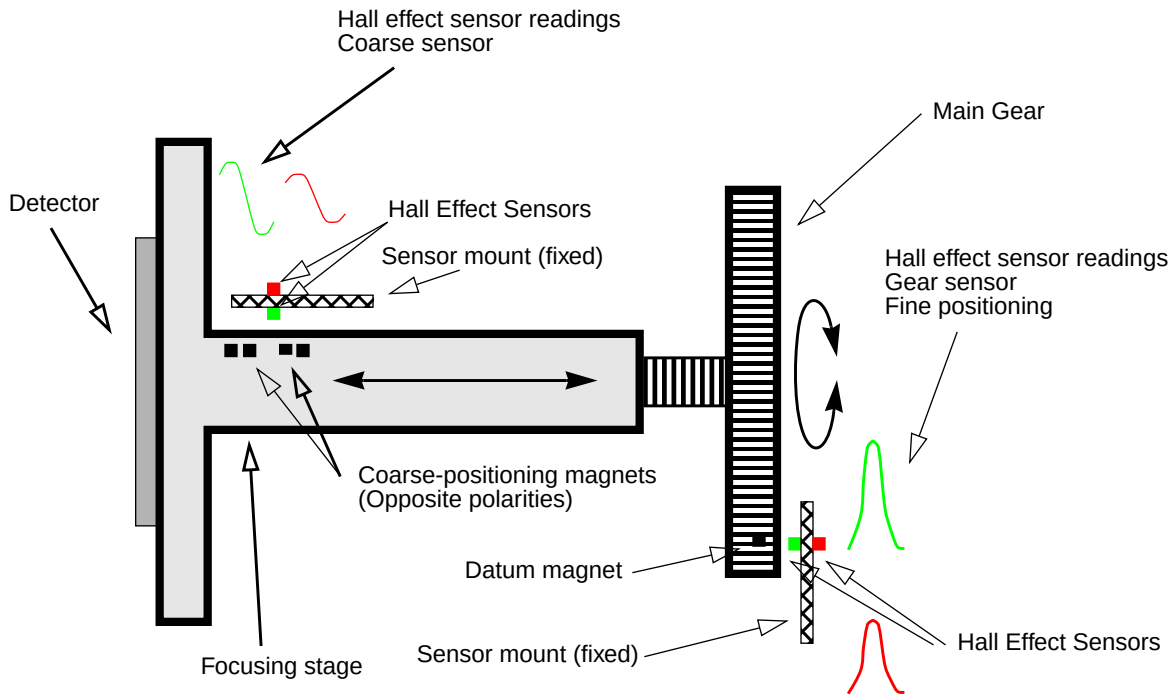
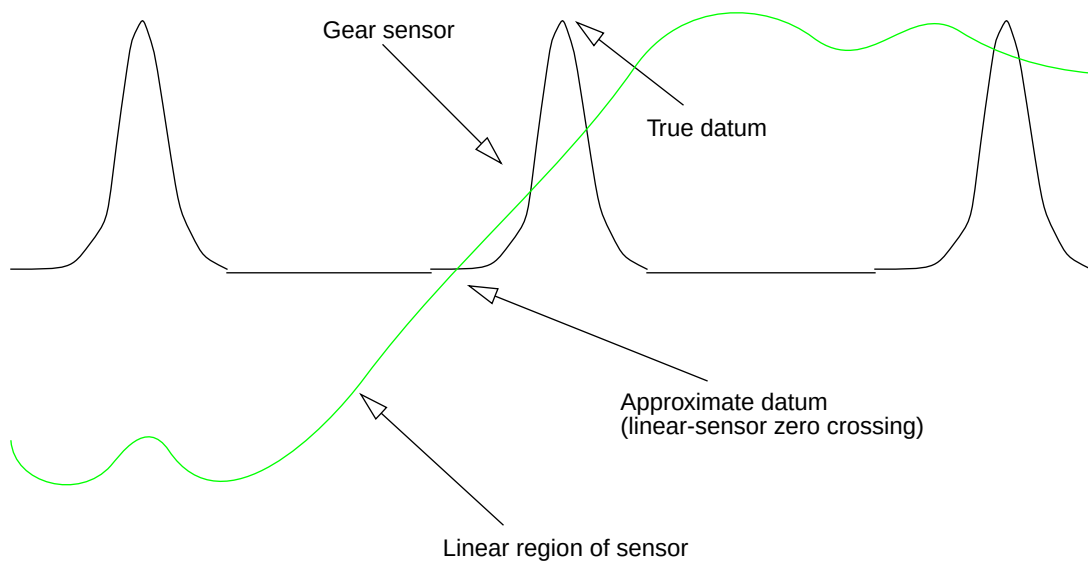


Figure 11 shows the expected reading from the Hall-effect sensors.

**FIGURE 11.** Sensor values (peak width highly exaggerated)



### 3.1.3.2 The OIWFS gimbal mirrors

The larger range of travel of the gimbal mirrors stage and the precise positioning requirements require still another variation for the gimbal mirrors. Instead of a linear positioning sensor, a single magnet is used, and the location of the peak is used to determine the true home position.

**FIGURE 12.** A two-sensor arrangement (gimbal mirror stage)

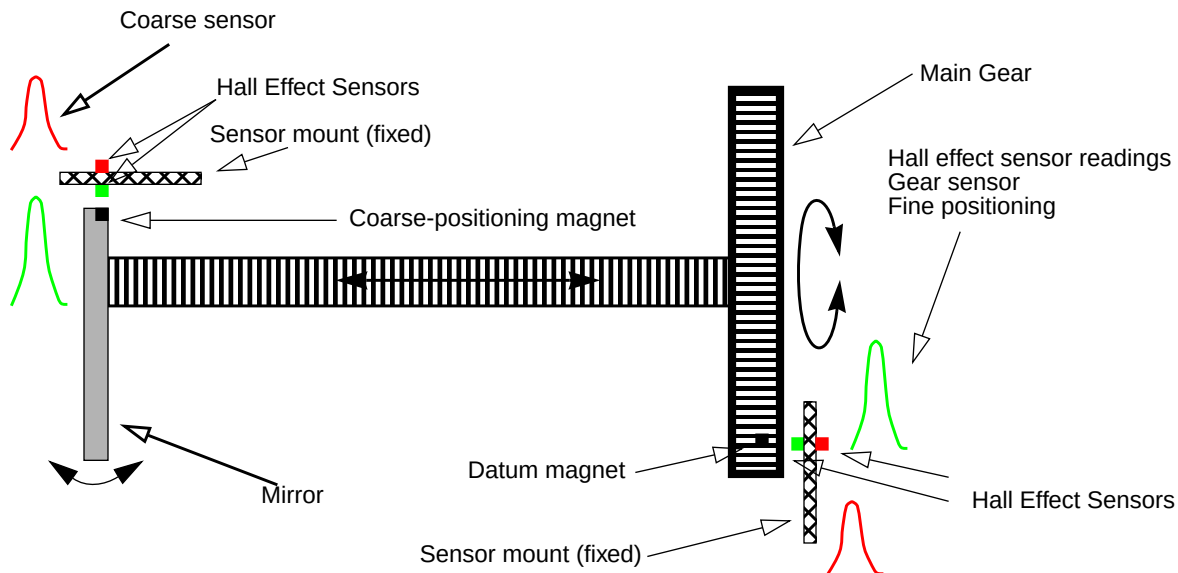
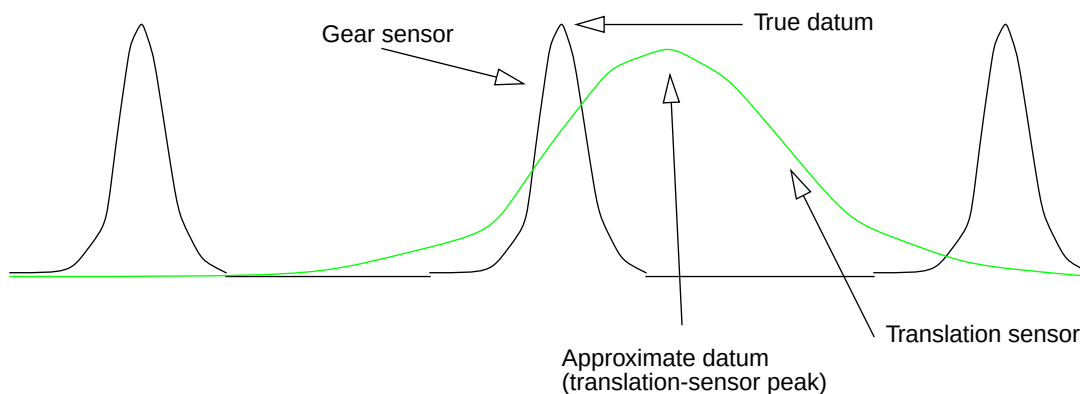


Figure 13 shows the expected reading from the Hall-effect sensors.

**FIGURE 13.** Sensor values (peak width highly exaggerated)



### 3.1.4 Two position mechanisms (the pupil viewer)

The pupil viewer has two defined positions. Each position is located at a hard stop, at an end of travel.



**FIGURE 14.** A one-sensor arrangement (pupil viewer))

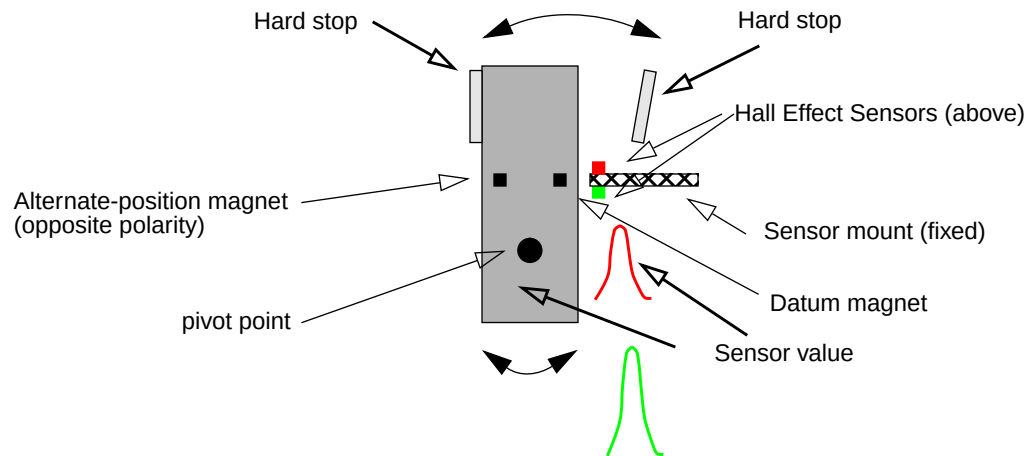
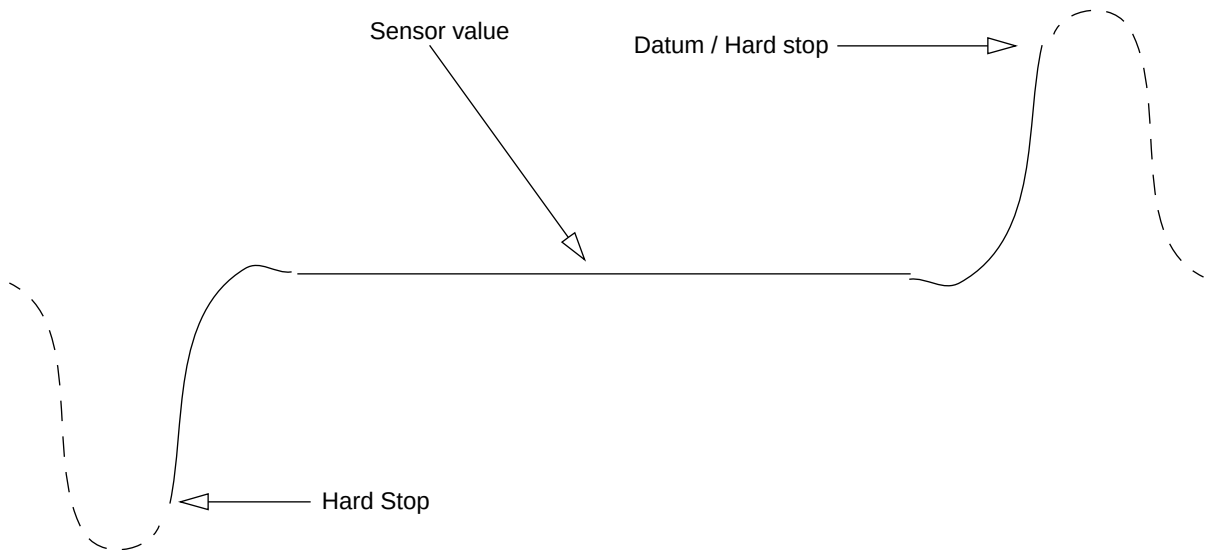


Figure 13 shows the expected reading from the Hall-effect sensors.

**FIGURE 15.** Sensor values (peak width highly exaggerated)



### **3.2 Peak Finding Algorithm<sup>3</sup>**

For the rest of this document, we will refer to “finding the peak value” associated with each magnet/sensor pair, using the following algorithm

Locate the largest sensor value with a brute-force search.  
Sample the sensors values near the maximum sensor value.  
Fit a parabola to the sensor values.  
Use the peak of the parabola as the sensor value.

This symmetric algorithm should be relatively insensitive to slight changes in the gain of the electronics, and to any biases in the electronics.

### **3.3 Stepper-motor errors**

The hardware will be designed so that the stepper motors will never lose counts if all mechanisms are working properly. If any counts are lost, a problem exists which must be detected and corrected. The Hall-effect sensors may be able to detect the problem. If the sensors are able to detect a problem, an ERROR will be generated so that appropriate error recovery may be taken.

### **3.4 Backup sensors**

All of the Hall-effect sensors have backup sensors. The system will be designed to use a primary sensor to datum the corresponding mechanism and to confirm the instrument configuration, and a secondary sensor for confirmation purposes only. Either sensor may be disabled if it fails. If either sensor is disabled, it may not be possible to confirm the configuration of the corresponding mechanism. If the primary sensor fails a slower datuming algorithm will be used. If both sensors are disabled, it will not be possible to confirm the current position nor will it be possible to datum the mechanism. Disabling either sensor will generate a WARNING.

### **3.5 Hard Limits and Limit Switches**

Most of the mechanisms do not have hard limits and can be driven continuously indefinitely. The exceptions are described in the following section.

#### **3.5.1 The focusing stages**

The wavefront-sensor focusing stage and the science-detector focusing stage both have hard stops. Neither mechanism has a limit switch or a passive way of detecting that the mechanism has hit a limit. Both mechanisms are designed so that they will not be damaged by *occasionally* hitting a hard stop. The software is designed with soft limits and it should be impossible to hit a hard stop unless there is a hardware problem. If a hard stop is encountered, it will be necessary to redatum the mechanism.

#### **3.5.2 The Pupil Viewer**

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3. The peak-finding algorithm is implemented within the HallStep record, documented in [35] in Section 4 on page 6.

The pupil viewer is designed so that it always hits a hard stop whenever it hits the limits of its travel.

### **3.5.3 The OIWFS Gimbal Mirror**

The gimbal mirror can be damaged by overrunning its limits. There are limit switches to prevent this from happening. These switches are not intended to be used in normal operation. The software will be designed with soft limits so that the mechanism will never hit a limit.

If a motion command causes the hardware to hit a limit switch, the mechanism will stop and the motion command will complete with an error message. It will then be possible only to back the mechanism out of the limit. It should not be necessary to redatum the mechanism after hitting a limit.

Warning: The limit switches for both axes are tied together. If either axis hit the positive limit, then both axes will stop, and it will only be possible to move the axes in a negative direction. The same holds true for the negative limits. It is possible to reach a state in which it is not possible to move the mechanism: If one axis actuates the positive limit, and the other axis actuates the negative limit, then the mechanism will not be able to move. In this case, it will be necessary to determine which mechanism is in which limit, and to carefully back out of each limit in the proper direction. The procedure for determining which limit switch has been actuated by which axis is described in the hardware reference manual.

## **3.6 Simultaneous motor motion**

It will not be possible to move all motors in the system simultaneously, because of power restrictions. (This will be handled by the lower level software, and should be transparent to the upper level interface.)

Whenever a motor becomes active, a counter will be incremented. When the maximum number of motors is running, new operations will be delayed until the motor count drops below the maximum.

## **3.7 Stepper Motor Control Module Failure**

If the motor draws too much current, the stepper-motor control modules can enter into a fault state. The motor is unable to operate if a fault occurs. This fault generates an immediate ERROR until the stepper-motor control module is reset. A command will be provided to reset the fault, which should be a very infrequent event.

Note: A fault condition also exists if power is not applied to the control module. This is normal. Obviously, it is impossible to reset this fault. It is not possible to distinguish this from the over-current fault condition.

## **3.8 Diagnose**

A DIAGNOSE command will be provided for each component (This should not be confused with the standard TEST and VERIFY commands.) This command will move the hardware and monitor the Hall-effect sensors in order to determine if the hardware is functioning. It should be possible to discover a single, failed, sensor. It will not be possible to distinguish a

double sensor failure from a failed motor. Unlike the TEST command, the DIAGNOSE command *will* move the hardware.

### **3.9 Redatum**

A REDATUM command will be provided for each component. (This should not be confused with the standard TEST and VERIFY commands.) This command will move to the expected datum position and confirm that it really is at the datum. If the verification fails, then an error will be returned. This command will not be used as part of the normal configuration-change procedure. It will only be used if a possible hardware problem is suspected.

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## **4 Environment Control**

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### **4.1 Cooling Motor**

The closed-cycle coolers are driven by the same stepper-motor controller as the niri mechanisms.

### **4.2 Temperature regulation**

The NIRI and NIRI OIWFS temperature control is handled with three off-the-shelf Omega temperature controllers. These controllers accept ascii commands via their serial ports. A custom EPICS record is used to pass commands to these temperature controllers.

### **4.3 Heating Resistors**

Warm-up is handled by a combination of heating resistors, which can be disabled using the digital input/output board (controlled using an EPICS binary output record) and the Omega temperature controller.

It will be necessary to coordinate the warming of the components with the warming of the detector, to insure that the detector is always slightly warmer than the surrounding instrument. Overall control of the cooling motors and the temperature regulation is handled by a custom record.

### **4.4 Temperature Sensors**

An undetermined number of temperature sensors will monitor the behavior of the system if the temperature is out of the valid range an ERROR will be generated.

The temperature sensors will be read using an off-the-shelf Omega digital thermometer, which accepts commands via a serial port. A custom record has been provided to read the temperature.

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## **5 Special Calibration Programs**

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There will probably be a need for special purpose software for engineering purposes, e.g., to generate the look-up tables of filter positions. These programs will never be used in normal operation. After delivery of the instrument to gemini, the programs will only need to be run after daytime engineering in which significant changes are made to the NIRI hardware.

These programs will be documented in [5].

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## **6 The Gemini Interlock System**

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The appropriate hardware and software support will be provided for the Gemini Interlock System, but NIRI will never generate an interlock event, and will need to carry out no actions in response to an interlock demand.

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## **7 Special Epics Records**

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### **7.1 Modified records**

#### **7.1.1 Motor record (OMS VME8, VME44 device/driver)**

The EPICS motor and OMS VME8, VME44 device provides support for the Oregon Micro-systems VME8-8 board. We have produced a modified motor record and device support. (If necessary to avoid conflicts, this record will be called the “nirimotor” record.

- The record is modified to allow direct control over the AUX output. (The AUX output is used to control the power to the stepper motors from the stepper motor driver modules.) The PWR field has three possible values: “ON”, “OFF”, “AUTO”. “AUTO” is the default and is backwards compatible with the existing record.
- Constants and global variables have been renamed, to have a consistent “MOTOR\_” or “motor” prefix.

### **7.2 Custom Records, Device Support, and Subroutine Records**

The following will probably be necessary. They will be some combination of custom records, new device support for existing records, subroutine records, gensub records, and SNL code:

- Records/code to control the cooling and warming of the instrument;
- Records/code to optimize the motions of the filter wheels;
- Records/code to control the cooling motor;
- Records/code to control the temperature controller.



# Gemini Near Infra-Red Imager and Wavefront Sensor



# Near IR Imager (NIRI) and On-Instrument Wavefront Sensor (OIWFS)—Implementation Notes and Record Reference

**Hubert Yamada**

niri\_h ty\_011 / 03 / 1999-10-23

**This report contains a reference manual for several custom records and capfast symbols which are central to the implementation of The Gemini Near Infra-Red Imager and On Instrument Wavefront Sensor.**

---

## 1 Introduction

This report presents a brief overview of the implementation of the control software for the Near Infra-Red Imager and On Instrument Wavefront Sensor. It also contains reference manuals for several custom EPICS records and capfast symbols.

### 1.1 Purpose of the document

This report provides a description of the interfaces and behavior of various NIRI software components.

The intended audience of this document is:

- the maintainers of the NIRI software;
- the reviewers of the NIRI system;
- the reviewers of the NIRI software;
- anyone who may wish to understand the details of NIRI.

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## 2 Overview

The NIRI top-level directory (niri\_VXYY, where XYY is a release number in the form major release, minor release) is made up of the following subdirectories

**LIB.** Shared utilities and schematics

**CICSCC.** NIRI CC code, adapted from the CICS

**CICSWFS.** NIRI WFS code, adapted from the CICS

**ENG.** NIRI mechanisms, WFS mechanisms, and NIRI environment control

**DC.** The GNAAC DC (Disabled)

Standalone subsystems of this system can be built by modifying the appropriate Makefiles and startup scripts in the root directory to compile only the subdirectories containing the desired subsystem, the LIB subdirectory, and the ENG subdirectory. For example, a stand-alone wavefront sensor can be built by including just the “LIB”, “CICSWFS”, and “ENG” directory. The ENG subdirectory is designed to build a product which is capable of functioning by itself.

The NIRI top-level directory, the LIB directory, and the CICSCC directories are adapted from the corresponding CICS directories, with the following changes:

- The niri\_VXYY/ascii directory includes files from the ENG/ascii directory.
- The CICSCC directory has additional mechanisms, to simulate the entire NIRI instrument
- The channel-access links to the CICS motor simulator has been replaced with channel-access links to the ENG databases.

In addition, the CICSWFS directory is essentially a copy of the CICSCC directory, customized with the NIRI components.

## **2.1 The ENG directory**

Note: The ENG directory is designed to be run as a standalone package for low-level engineering of the niri mechanisms, or as a part of the complete niri system.

The ENG directory contains three independent databases for controlling different mechanisms: The CC database, the WFS database, and the environment-control database. It contains a single database, the GBL database, which provides low-level interlocks. It contains databases which may be loaded to simulate the corresponding subsystems. There are also some special records, device support, and other C code which are required to support the databases.

It is possible to simulate the NIRI mechanisms, the WFS mechanisms, and the NIRI environment controls by loading the appropriate simulation database(s), which are also included in the ENG directory. The top-level user interface should be have identically with the simulation or with the actual hardware. Some slight differences may be noted in the lower-level engineering interfaces.

### **2.1.1 The CC Database**

The components-controller database is made up of several instances of the same capfast hierarchical symbol, the mech symbol.

### **2.1.2 The WFS Database**



The wavefront-sensor database is structured identically to the CC database, with a different set of mechanisms.

### 2.1.3 The GBL Database

The global parameters database provides access to interlocks, to keep the system from moving when it is unsafe (e.g., the mechanisms are not at an equilibrium temperature) or unacceptable (e.g., during an observation) for mechanisms to move. It also restricts the number of mechanisms that are in motion. (The power supply is not able to provide power for all motors simultaneously.)

### 2.1.4 The ENV Database

The environment-control database is very simple, containing only a few records to monitor and to control the temperature sensors and temperature controllers.

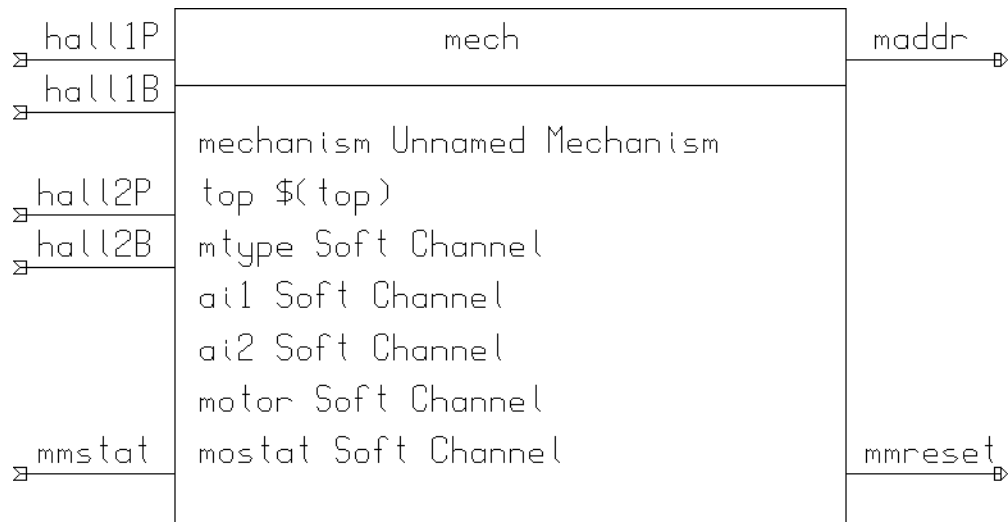
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## 3 The Mech Symbol and Diagram

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All mechanical components are controlled by instances of a single capfast hierarchal symbol and the associated capfast diagram. Custom parameters are passed in via input and output ports (mostly hardware addresses) and via macro substitution.

### 3.0.1 The Mech Symbol



The following table describes the macro variables, inputs, and outputs of the mech hierarchal symbol.

---

**TABLE 1.** Mech.sym macro variables and ports

| Macro/Port Name | Type                          | Value                                           |
|-----------------|-------------------------------|-------------------------------------------------|
| hall1P          | Input record hardware address | Primary Hall Effect sensor (wheel or main gear) |
| hall1B          | Input record hardware address | Backup Hall Effect sensor (wheel or main gear)  |

**TABLE 1.** Mech.sym macro variables and ports

| Macro/Port Name | Type                                  | Value                                                                              |
|-----------------|---------------------------------------|------------------------------------------------------------------------------------|
| hall2P          | Input record hardware address         | Primary Hall Effect sensor (linear sensor or unused)                               |
| hall2B          | Input record hardware address         | Secondary Hall Effect sensor (linear sensor or unused)                             |
| mmstat          | Input record hardware address         | Motor control module status (working or fault)                                     |
| maddr           | Motor control record hardware address | Motor controller board                                                             |
| mmreset         | Output record hardware address        | Motor control module reset (resets fault state)                                    |
| mechanism       | Macro variable                        | Text string (name of mechanism)                                                    |
| top             | Macro variable                        | Prefix for generating record names                                                 |
| mttype          | Macro variable                        | Type of motor controller (EPICS Device or Soft HS [simulator])                     |
| ai1             | Macro variable                        | Type of input sensor (EPICS Device or Soft HS [simulator])                         |
| ai2             | Macro variable                        | Type of input sensor (EPICS Device, Soft Channel [unused], or Soft HS [simulator]) |
| motor           | Macro variable                        | Type of motor (EPICS Device or Soft HS [simulator])                                |
| mostat          | Macro variable                        | Type of motor status input (EPICS Device or Soft Channel [unused])                 |

### 3.1 External interfaces (the NIRI/CICS interface)

This section describes the fields that the NIRI/CICS interface uses to control the mechanisms in the mech hierarchal symbol. These are the only fields that should be considered “public”<sup>1</sup>. All other fields are to be considered a private to the implementation, and may be changed at any time.

**TABLE 2.** External Interfaces

| Field             | Action          | Read or Write | Action                 | Comments                                                                                                                  |
|-------------------|-----------------|---------------|------------------------|---------------------------------------------------------------------------------------------------------------------------|
| \$(top)MoveTo.VAL | Demand Position | Write         | Move to new position   | If an operation is already in progress, a new operation will fail.                                                        |
| \$(top)Op.VAL     | Operation Code  | Write         | Perform operation.     | See Table 3 on page 5 for a list of operation codes<br>If an operation is already in progress, a new operation will fail. |
| \$(top)Stop.VAL   | Stop            | Write         | Interrupt operation    | Write a 1 to this field to interrupt an operation                                                                         |
| \$(top)Busy.VAL   | Busy            | Read          | Operation in progress? | Non-zero means operation in progress                                                                                      |

---

1. The low-level engineering interface interacts with other fields; do not use it as an example of the correct way to interface with the mech hierarchal symbol.

## The Mech Symbol and Diagram

**TABLE 2.** External Interfaces

| Field                | Action                       | Read or Write | Action                                  | Comments                                                                                                                                                                                                            |
|----------------------|------------------------------|---------------|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| \$(top)Done.VAL      | Done                         | Read          | Status after operation                  | Non-zero means error                                                                                                                                                                                                |
| \$(top)Hallstep.MESS | Completion message           | Read          | Error/Status message after an operation | Cleared by the next operation                                                                                                                                                                                       |
| \$(top)Hallstep.DATM | Datumed?                     | Read or Write | Mechanism datumed?                      | Can be set or cleared externally or internally at any time. Set after successful datum, cleared if datum failed, cleared upon reboot.<br>For informational purposes only; does not affect workings of record.       |
| \$(top)Hallstep.HSWN | Hall effect sensor disabled? | Read          | Hall-effect sensor disabled?            | Can be set or cleared externally or internally at any time.                                                                                                                                                         |
| \$(top)Moving.VAL    | Moving?                      | Read          | Mechanism moving?                       | Non-zero means that it is moving.<br>This is primarily for informational displays. Monitor \$(top)Busy.VAL instead, because the mechanism may briefly stop moving while an operation is still in progress.          |
| \$(top)Mpos.VAL      | Current Position?            | Read          | Mechanism current position              | This field takes approximately 0.1 seconds to update. It is intended to be used for informational displays. For precise positioning, wait at least 0.1 second after Moving.VAL becomes 0 before reading this field. |
| \$(top)Speed.VAL     | Desired speed?               | Read          | Mechanism speed                         | This is the desired speed after the mechanism has accelerated, <i>not</i> the current speed.                                                                                                                        |

The following table describes all operation codes. *These codes might be out of date. Check values in hsConsts.h.*

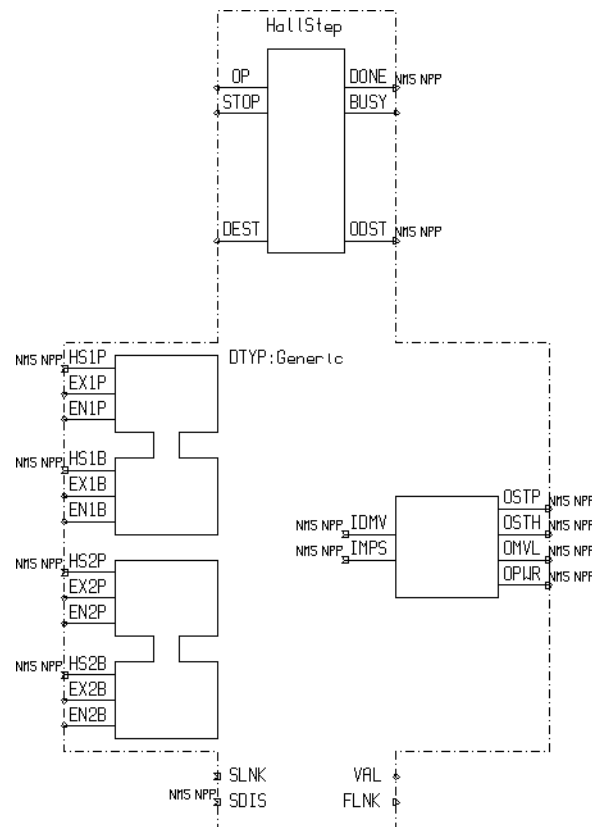
**TABLE 3.** Mech Hierarchical Symbol Operation codes

| Code | Action             | Public? | Comments                                                     |
|------|--------------------|---------|--------------------------------------------------------------|
| 0    | No Op              | No      | Reserved for internal use                                    |
| 1    | Move               | No      | Use \$(top)MoveTo.VAL instead                                |
| 2    | Datum mechanism    | No      | For engineering. Normal programs should use Datum + Diagnose |
| 3    | Redatum mechanism  | Yes     |                                                              |
| 4    | Cycle mechanism    | No      | For engineering                                              |
| 5    | Diagnose mechanism | Yes     |                                                              |
| 6    | Datum + Diagnose   | Yes     |                                                              |

## 4 The HallStep Record and Symbol

The “mech” diagram controls the hardware via the (custom) “hallStep” record. All commands to the hardware will pass through this record. (Hall effect sensor values may be read directly from the ai record.) Nothing should interface directly to the motor record, except for the records in the mech symbol.

*In general, it should never be necessary to interface to the HallStep Record. Use the public fields in the mech symbol instead.*



The following table describes the fields of the HallStep record.

**TABLE 4.** HallStep fields

| Name | Summary            | Description                                                  |
|------|--------------------|--------------------------------------------------------------|
| VAL  | Subr. return value | Error code returned if operation is rejected                 |
| OVAL | Old return value   |                                                              |
| PREC | Display Precision  |                                                              |
| STOP | Stop operation     | Setting this field to non-zero cancels the current operation |
| OP   | Operation          | See Table 5 on page 9                                        |
| DEST | Destination        | Destination for Move operation                               |

## The HallStep Record and Symbol

**TABLE 4.** HallStep fields

| Name  | Summary                 | Description                                                                       |
|-------|-------------------------|-----------------------------------------------------------------------------------|
| OPST  | Operation status        | Text description of current action                                                |
| MESS  | Operation error msg     | Text description of error or other important information                          |
| HOPR  | High Operating Range    | Upper limit of motion (Ignored during datum)                                      |
| LOPR  | Low Operating Range     | Lower limit of motion (Ignored during datum)                                      |
| RAW   | Raw mode                | Device dependant (e.g. ignore limits)                                             |
| DONE  | Operation error code    | Output link, writes zero [no error] or non-zero [error] after operation completes |
| BUSY  | Currently processing?   | Output link, writes 1 when operation begins, 0 when operation ends                |
| DSL P | “Datum slope P”         | Exact meaning depends on device support (Wheel or Stage), used for datuming       |
| DSL B | “Datum slope B”         | Exact meaning depends on device support (Wheel or Stage), used for datuming       |
| DOFP  | “Datum offset, rough P” | Exact meaning depends on device support (Wheel or Stage), used for datuming       |
| DOFB  | “Datum offset, rough B” | Exact meaning depends on device support (Wheel or Stage), used for datuming       |
| DBSH  | “Datum backlash”        | Exact meaning depends on device support (Wheel or Stage), used for datuming       |
| DOF1  | “Sensor offset, 1P-1B”  | Exact meaning depends on device support (Wheel or Stage), used for datuming       |
| WTHP  | “Peak width 1P”         | Estimated width of sensor peak (gear or wheel)                                    |
| PKP   | “Peak height 1P”        | Estimated height of sensor peak (gear or wheel)                                   |
| WTHB  | “Peak width 1B”         | Exact meaning depends on device support (Wheel or Stage)                          |
| PKB   | “Peak height 1B”        | Exact meaning depends on device support (Wheel or Stage)                          |
| CSTP  | “Cycle stop”            | Stop cycle test                                                                   |
| CTOL  | “Cycle tolerance”       | Sensor error for automatic stop                                                   |
| CSET  | “Cycle set values”      | (Flag) On next cycle, set expected values for cycle test from observed value      |
| CDSA  | “Cycle A dest.”         | First position for cycle test                                                     |
| CDSB  | “Cycle B dest.”         | Second position for cycle test                                                    |
| CDSC  | “Cycle C dest.”         | Third position for cycle test (normally 0)                                        |
| E1PA  | “Cycle A expect”        | Expected primary (gear or wheel) sensor value at first test position              |
| E1PB  | “Cycle B expect”        | Expected primary (gear or wheel) sensor value at second test position             |
| E1PC  | “Cycle C expect”        | Expected primary (gear or wheel) sensor value at third test position              |
| V1PA  | “Cycle A read”          | Actual primary (gear or wheel) sensor value at first test position                |
| V1PB  | “Cycle B read”          | Actual primary (gear or wheel) sensor value at first test position                |
| V1PC  | “Cycle C read”          | Actual primary (gear or wheel) sensor value at first test position                |
| E1BA  | “Cycle A expect”        | Expected secondary (gear or wheel) sensor value at first test position            |
| E1BB  | “Cycle B expect”        | Expected secondary (gear or wheel) sensor value at second test position           |
| E1BC  | “Cycle C expect”        | Expected secondary (gear or wheel) sensor value at third test position            |
| V1BA  | “Cycle A read”          | Actual secondary (gear or wheel) sensor value at first test position              |
| V1BB  | “Cycle B read”          | Actual secondary (gear or wheel) sensor value at second test position             |
| V1BC  | “Cycle C read”          | Actual secondary (gear or wheel) sensor value at third test position              |
| E2PA  | “Cycle A expect”        | Expected primary (linear) sensor value at first test position                     |
| E2PB  | “Cycle B expect”        | Expected primary (linear) sensor value at second test position                    |

## The HallStep Record and Symbol

**TABLE 4.** HallStep fields

| Name              | Summary                 | Description                                                               |
|-------------------|-------------------------|---------------------------------------------------------------------------|
| E2PC              | “Cycle C expect”        | Expected primary (linear) sensor value at third test position             |
| V2PA              | “Cycle A read”          | Actual primary (linear) sensor value at first test position               |
| V2PB              | “Cycle B read”          | Actual primary (linear) sensor value at second test position              |
| V2PC              | “Cycle C read”          | Actual primary (linear) sensor value at third test position               |
| E2BA              | “Cycle A expect”        | Expected secondary (linear) sensor value at first test position           |
| E2BB              | “Cycle B expect”        | Expected secondary (linear) sensor value at second test position          |
| E2BC              | “Cycle C expect”        | Expected secondary (linear) sensor value at third test position           |
| V2BA              | “Cycle A read”          | Actual secondary (linear) sensor value at first test position             |
| V2BB              | “Cycle B read”          | Actual secondary (linear) sensor value at second test position            |
| V2BC              | “Cycle C read”          | Actual secondary (linear) sensor value at third test position             |
| HSWN              | “Warn HS”               | Flag set when any hall effect sensor is disabled                          |
| HS1P              | “Input Link HS1P”       | Link to Hall-effect sensor input record                                   |
| EX1P              | “Expect HS1P”           | Expected value of sensor after next move command                          |
| EN1P              | “Enable HS1P”           | Enable/Disable Hall-effect sensor                                         |
| HS1B              | “Input Link HS1B”       | Link to Hall-effect sensor input record                                   |
| EX1B              | “Expect HS1B”           | Expected value of sensor after next move command                          |
| EN1B              | “Enable HS1B”           | Enable/Disable Hall-effect sensor                                         |
| HS2P              | “Input Link HS2P”       | Link to Hall-effect sensor input record                                   |
| EX2P              | “Expect HS2P”           | Expected value of sensor after next move command                          |
| EN2P              | “Enable HS2P”           | Enable/Disable Hall-effect sensor                                         |
| HS2B              | “Input Link HS2B”       | Link to Hall-effect sensor input record                                   |
| EX2B              | “Expect HS2B”           | Expected value of sensor after next move command                          |
| EN2B              | “Enable HS2B”           | Enable/Disable Hall-effect sensor                                         |
| IDMV <sup>a</sup> | “Motor finished moving” | Input link, used to monitor motion flag on motor-control record           |
| IMPS <sup>a</sup> | “Motor Cur. position”   | Input link, used to monitor motion flag on motor-control record           |
| OMVL <sup>a</sup> | “Motor New position”    | Output link, used to write new position on motor-control record           |
| OSTH <sup>a</sup> | “Motor Set home”        | Output link, used to trigger “set home” operation on motor-control record |
| OSTP <sup>a</sup> | “Motor Stop motion”     | Output link, used to write “stop” on motor-control record                 |
| OPWR <sup>a</sup> | “Motor Power”           | Output link, used to control power to on motor-control record             |
| ODST              | “Destination Link”      | Output link, after a move, writes the current position                    |

a. IDMV, IMPS, OMVL, OSTH, OSTP, and OPWR are very closely tied with the behavior of the motor record. If the motor record is modified, the behavior of these fields may need to change.

The following table describes all operation codes. *These codes might be out of date. Check values in `hsConsts.h`.*

---

**TABLE 5.****HallStep Record Operation Codes**

| Code | Action             |
|------|--------------------|
| 0    | No Op              |
| 1    | Move               |
| 2    | Datum mechanism    |
| 3    | Redatum mechanism  |
| 4    | Cycle mechanism    |
| 5    | Diagnose mechanism |
| 6    | Datum + Diagnose   |

#### 4.1 Record processing

This record is processed only when an operation code is written to the OP field.

##### 4.1.1 When an operation code is processed:

If the record is processing another operation, the new operation fails, and the VAL field is set to a non-zero value.

If the operation code is invalid, the new operation fails and the VAL field is set to a non-zero value.

If the operation code is valid, the operation code is passed to the completion task, and the completion task is awakened.

Record processing completes, and 0 is returned in the VAL field.

##### 4.1.2 When the completion task is given a new operation

The task writes a non-zero value to the BUSY link.

The task performs the desired operation (described below), reading sensor values from the sensor input records, writing control codes to the motor-control record. If there is an error, the task sets the MESS field to a descriptive message.

The task writes an error code (or 0 if there is no error) to the OPST field.

The task writes a 0 to the BUSY link.

For the move command:

The DEST field is read.

The hardware is moved to the new location.

The sensor values are compared to the expected sensor values (not yet implemented).

For the datum command:

The DATM field is set to 0

The hardware performs whatever operations are necessary to datum the hardware (see Section 3.1 on page 5 of [3]).

If the operation succeeded, the new datum is set.

If the operation succeeded, the DATM field is set to 1

For the redatum operation

The hardware performs whatever operations are necessary to datum the hardware (see Section 3.1 on page 5 of [3]).

The new datum is *not* set; the MESS field indicates how different the new datum is from the last datum

For the cycle operation

The hardware moves to the first test position (set by the CDSA, field. At the first test position, it reads the Hall effect sensors and compares them to the expected values.

If the values are not within the stated tolerance, the operation fails and processing terminates.

The hardware moves to the second test position (set by the CDSB, field.

At the second test position, it reads the Hall effect sensors and compares them to the expected values.

If the values are not within the stated tolerance, the operation fails and processing terminates.

The hardware moves to the third test position (set by the CDSC, field.

At the third test position, it reads the Hall effect sensors and compares them to the expected values.

If the values are not within the stated tolerance, the operation fails and processing terminates.

The process begins again, starting with the first test position, until the operation fails, or until the CSTP field is set to a non-zero value.

For the diagnose operation

The hardware moves to the first test position (set by the CDSA, field.

At the first test position, it reads the Hall effect sensors and compares them to the expected values.

If the values are not within the stated tolerance, the operation fails and processing terminates.

The hardware moves to the second test position (set by the CDSB, field.

At the second test position, it reads the Hall effect sensors and compares them to the expected values.

If the values are not within the stated tolerance, the operation fails and processing terminates.

The hardware moves to the third test position (set by the CDSC, field.

At the third test position, it reads the Hall effect sensors and compares them to the expected values.

If the values are not within the stated tolerance, the operation fails and processing terminates.



For the datum + diagnose operation

The DATM field is set to 0

The hardware performs whatever operations are necessary to datum the hardware (see Section 3.1 on page 5 of [3]).

If the operation succeeded, the new datum is set.

The hardware moves to the first test position (set by the CDSA, field).

At the first test position, it reads the Hall effect sensors and compares them to the expected values.

If the values are not within the stated tolerance, the operation fails and processing terminates.

The hardware moves to the second test position (set by the CDSB, field).

At the second test position, it reads the Hall effect sensors and compares them to the expected values.

If the values are not within the stated tolerance, the operation fails and processing terminates.

The hardware moves to the third test position (set by the CDSC, field).

At the third test position, it reads the Hall effect sensors and compares them to the expected values.

If the values are not within the stated tolerance, the operation fails and processing terminates.

If the operation succeeded, the DATM field is set to 1



# Gemini Near Infra-Red Imager and Wavefront Sensor



# Near IR Imager (NIRI) and On-Instrument Wavefront Sensor (OIWFS)—Calibration and Maintenance Tools

Hubert Yamada

niri\_h ty\_004 / 08 / 2000-01-25

This document describes the software calibration programs and procedure for NIRI and the Near-Infrared OIWFS.

## 1 Introduction

### 1.1 Purpose

This document describes the software calibration programs and procedure for NIRI.

The intended audience of this document is:

- The NIRI reviewers;
- The developers of the Gemini Observatory Control System (OCS);
- Groups which will be using the design of the NIRI OIWFS (e.g. GNIRS);
- The maintainers of NIRI after it is delivered to Gemini.

### 1.2 Definitions, Acronyms and Abbreviations

See “Near IR Imager (NIRI)—Definitions, Abbreviations, and References”.

### 1.3 Stylistic Conventions

References to documents in Section 1.5, “References,” on page 2 of “Near IR Imager (NIRI)—Definitions, Abbreviations, and References” are given like this; [1].

### 1.4 References

See “Near IR Imager (NIRI)—Definitions, Abbreviations, and References”.

---

## 2 When is it necessary to recalibrate a mechanism?

---

### 2.1 What is the difference between normal calibration and low-level calibration.

Normal calibration is a procedure which must be carried out for many different reasons (listed below). It is often necessary: if the zero point of the analog input board drifts, if wires are replaced, if the mechanism is partially disassembled.

### 2.2 When is normal calibration necessary?

Normal calibration is necessary if a mechanism appears to be working and datums itself at the correct location, but the final verification phase (diagnose) fails. It is also necessary if the mechanism is moving to the correct positions, but returns an error when it moves.

The procedure for normal calibration is described in Section 5.3 and Section 6.4.

### 2.3 When is low-level calibration necessary?

Low-level calibration is necessary only if a mechanism has been disassembled and reassembled; appears to be operating correctly for the most part; and cannot datum itself or datums itself at an insane position. Low-level calibration should be considered only after all alternatives have been excluded. Note that only the focus stages and the gimbal mirror require recalibration. For all other mechanisms, all constants are well-defined by mechanical means.

### 2.4 When should the mechanism not be calibrated?

If the mechanism is not functioning properly, recalibrating it will mask problems or cause it to return false error messages.

---

## 3 Running the tools

---

### 3.1 The Engineering User Interface

You must know how to operate the NIRI / OIWFS Engineering user interface before you begin the calibration. See the software release notes [7] and the User's Guide for the Engineering User Interface [8] for more details.

---

## 4 Scan Tool

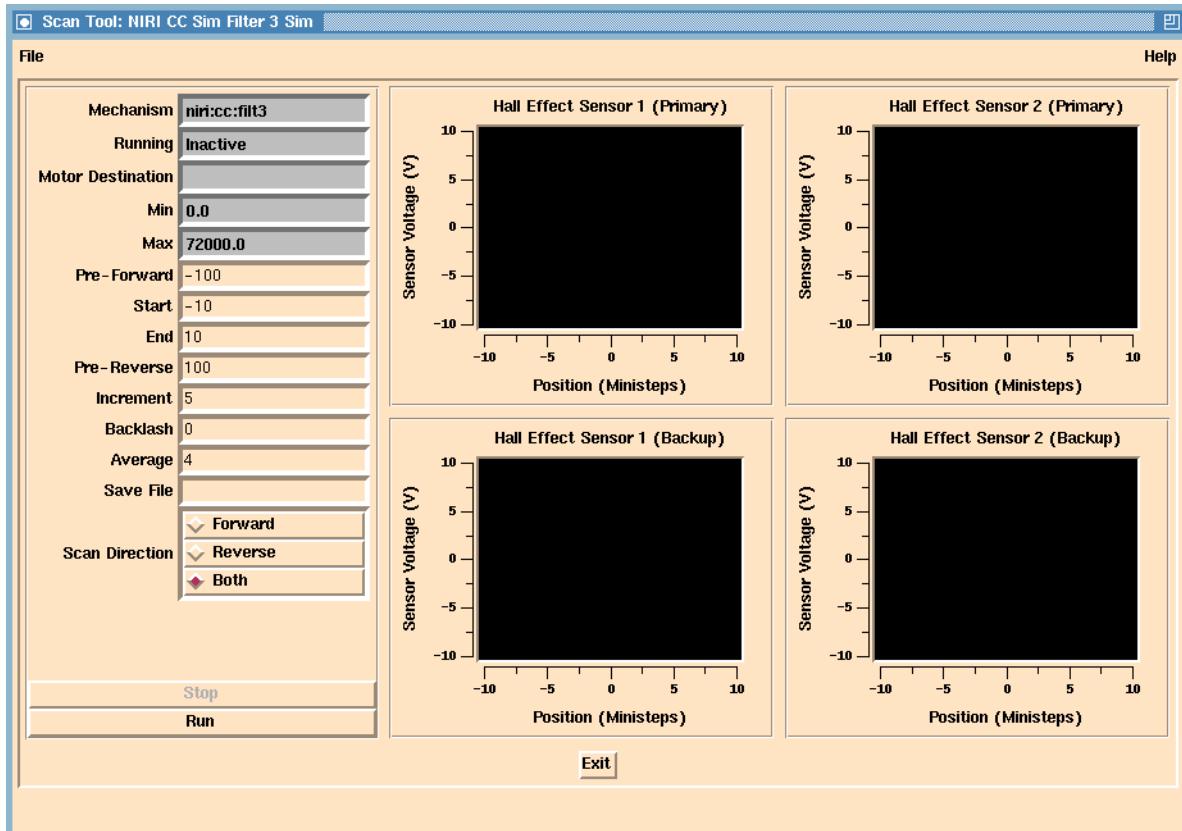
---

The "scan tool" is used for characterizing hardware, for measuring backlash, and for diagnosing problems. It is a useful tool for diagnosing problems with mechanisms, particularly with mechanisms that are not moving freely. It can be helpful in low-level calibration of mechanisms. It is not used in normal operation.

#### 4.1 Starting the “Scan Tool”

From the low-level engineering screen for the mechanism, press the “External Tools” button using the right mouse button, then select “Scan Tool.”

#### 4.2 Using The “Scan Tool”



The normal mode of operation is as follows:

Forward Pass:

Move to “Pre-forward”.  
 Move to “Start”, and read the Hall effect sensors  
 Move to “Start” + “Increment”, and read the Hall effect sensors  
 (Repeat until passes “End”)

Reverse Pass:

Move to “Pre-reverse”  
 Move to “End”, and read the Hall effect sensors  
 Move to “End” - “Increment”, and read the Hall effect sensors  
 (Repeat until passes “Start”)

The difference between the values on the forward and reverse pass provides an indication of the amount of backlash in the system. Significant differences between the forward and reverse passes indicate a serious hardware problem, i.e., something is slipping or sticking.

---

## **5 The “Cycle Tool”**

---

NIRI and the OIWFS are designed to provide quick verification that a mechanism is operating properly. This is done by placing analog Hall-effect sensors and magnets at various locations. When the mechanism is in a known, good state, the mechanism is moved to three test positions. The Hall-effect sensor values are read at each location.

The diagnose command verifies that the mechanism is operating correctly by moving it in sequence to each of these test positions and verifying that the sensor values are close to the which were recorded when the system was known to be operating correctly. The Cycle command is identical to the diagnose command, except that it repeats the test, until told to stop.

The “Cycle Tool” is a program that aids in this part of the calibration procedure. It moves the system to each of the test positions, and then records the values, which may then be written to a file. The file is read whenever the system is booted.

Warning: A side effect of running the “Cycle Tool” is that the in-memory table of expected values is changed. This is generally desirable, but can be misleading.

Warning: If the “Cycle Tool” is run while the mechanism is not working correctly then the results returned by the Cycle Test and Diagnose Test are suspect.

Note: The “Diagnose Test” is automatically performed when the “Datum” command is called from the high-level, interface. This is equivalent to the Datum/Diag command in the low-level engineering interface.

### **5.1 Starting the “Cycle Tool”**

From the low-level engineering screen for the mechanism, press the “External Tools” button using the right mouse button, then select “Cycle Tool.”

## 5.2 Using the “Cycle Tool”

The screenshot shows a window titled "et\_wish" with a menu bar (File, Mechanism, Tools, Help). The main area contains two columns of input fields for sensor data: V1 PA, V1 BA, V1 PB, V1 BB, V1 PC, V1 BC on the left and V2 PA, V2 BA, V2 PB, V2 BB, V2 PC, V2 BC on the right. Below these are fields for "Mech", ".pv base directory" (containing "/home/polaris/yamada/prj/xiri/ENG/pv"), ".pv subdirectory", "Ident", and "Out File". At the bottom are three buttons: "ReScan", "Write", and "Exit".

The fields / buttons of the cycle tool are:

**V1PA.** Voltage at test position A, sensor pair 1, primary sensor.

**V1BA.** Voltage at test position A, sensor pair 1, backup sensor

**V2PA.** Voltage at test position A, sensor pair 2, primary sensor.

**V1BA.** Voltage at test position A, sensor pair 2, backup sensor

**V1PB.** Voltage at test position B, sensor pair 1, primary sensor.

**V1BB.** Voltage at test position B, sensor pair 1, backup sensor

**V2PB.** Voltage at test position B, sensor pair 2, primary sensor.

**V1BB.** Voltage at test position B, sensor pair 2, backup sensor

**V1PC.** Voltage at test position C, sensor pair 1, primary sensor.

**V1BC.** Voltage at test position C, sensor pair 1, backup sensor

**V2PC.** Voltage at test position C, sensor pair 2, primary sensor.

**V1BC.** Voltage at test position C, sensor pair 2, backup sensor

**Mech.** The EPICS prefix for all fields related to this mechanism

**.lut base directory.** Where output files are created.

**.lut subdirectory.** Subdirectory of .lut base directory where files are created

**Ident.** The name of the default output file.

**Out File.** Name of the output file (user editable).

**ReScan.** Collect data.

**Write.** Write data file.

**Exit.** Exit from the “Cycle Tool.”

### 5.3 Normal Calibration (Phase 1)

1. Press the “ReScan” button to move the mechanism to the three test positions.
2. Examine the values. All values should be reasonable. (See Section 5.4.)
3. Examine the “Out File” field. Verify that the correct file will be created. Change the “Out File” field if you do not want to overwrite the normal file.
4. Press the “Write” button to write the data.
5. Press the “Exit” button to exit the “Cycle Tool.”
6. From a shell on the software server, change into the “.pv base directory” (from the “Cycle Tool”).
7. Run “gmake” in the “.pv base directory.”

Note: If you forget to press “Write” before pressing “Exit,” you will not be warned, but the file on disk will not be updated.

### 5.4 Expected Values (Phase 1)

The following tables describe the expected sensor values for each mechanism at the test positions.

The window cover has no sensors, and never needs to be calibrated. All sensor values read as 1 or 2 at the limits and 0 between limits.

**TABLE 1.**

Filter Wheels, The Pupil Mask Wheel, The Focal Plane Mask Wheel, The Entrance Beam Splitter, the Steering mirrors

| Abbreviation | Sensor      | Position | Significance   | Value (V) |
|--------------|-------------|----------|----------------|-----------|
| V1PA         | 1 (Primary) | A        | First Position | < -0.5    |
| V1BA         | 1 (Backup)  |          |                | < -0.5    |
| V2PA         | 2 (Primary) |          |                | 0         |
| V2BA         | 2 (Backup)  |          |                | 0         |
| V1PB         | 1 (Primary) | B        | Last Position  | < -0.5    |
| V1BB         | 1 (Backup)  |          |                | < -0.5    |
| V2PB         | 2 (Primary) |          |                | 0         |
| V2BB         | 2 (Backup)  |          |                | 0         |



---

## The “Cycle Tool”

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| Abbreviation | Sensor      | Position | Significance | Value (V) |
|--------------|-------------|----------|--------------|-----------|
| V1PC         | 1 (Primary) | C        | Home         | > 1       |
| V1BC         | 1 (Backup)  |          |              | > 1       |
| V2PC         | 2 (Primary) |          |              | 0         |
| V2BC         | 2 (Backup)  |          |              | 0         |

**TABLE 2.**

The Pupil Viewer

| Abbreviation | Sensor      | Position | Significance      | Value (V) |
|--------------|-------------|----------|-------------------|-----------|
| V1PA         | 1 (Primary) | A        | Alternate         | < -1      |
| V1BA         | 1 (Backup)  |          |                   | < -1      |
| V2PA         | 2 (Primary) |          |                   | 0         |
| V2BA         | 2 (Backup)  |          |                   | 0         |
| V1PB         | 1 (Primary) | B        | Home              | > 1       |
| V1BB         | 1 (Backup)  |          |                   | > 1       |
| V2PB         | 2 (Primary) |          |                   | 0         |
| V2BB         | 2 (Backup)  |          |                   | 0         |
| V1PC         | 1 (Primary) | C        | Home <sup>a</sup> | > 1       |
| V1BC         | 1 (Backup)  |          |                   | > 1       |
| V2PC         | 2 (Primary) |          |                   | 0         |
| V2BC         | 2 (Backup)  |          |                   | 0         |

a. If the corresponding values for positions B and C are not nearly identical, then the mechanism may have moved unexpectedly.

**TABLE 3.**

The Focus Stage

| Abbreviation | Sensor      | Position | Significance   | Value (V)        |
|--------------|-------------|----------|----------------|------------------|
| V1PA         | 1 (Primary) | A        | Large Positive | Near 0           |
| V1BA         | 1 (Backup)  |          |                | Near 0           |
| V2PA         | 2 (Primary) |          |                | Any <sup>a</sup> |
| V2BA         | 2 (Backup)  |          |                | Any              |
| V1PB         | 1 (Primary) | B        | Large Negative | Near 0           |
| V1BB         | 1 (Backup)  |          |                | Near 0           |
| V2PB         | 2 (Primary) |          |                | Any              |
| V2BB         | 2 (Backup)  |          |                | Any              |
| V1PC         | 1 (Primary) | C        | Home           | > 1              |
| V1BC         | 1 (Backup)  |          |                | > 1              |
| V2PC         | 2 (Primary) |          |                | Any              |
| V2BC         | 2 (Backup)  |          |                | Any              |

a. The #2 sensor values should change significantly between the A, B, and C positions. Either  $A > C > B$  or  $B > C > A$ .

TABLE 4.

The Gimbal Mirror

| Abbreviation | Sensor      | Position | Significance   | Value (V)        |
|--------------|-------------|----------|----------------|------------------|
| V1PA         | 1 (Primary) | A        | Large positive | Any              |
| V1BA         | 1 (Backup)  |          |                | Any              |
| V2PA         | 2 (Primary) |          |                | Any <sup>a</sup> |
| V2BA         | 2 (Backup)  |          |                | Any              |
| V1PB         | 1 (Primary) | B        | Large Negative | Any              |
| V1BB         | 1 (Backup)  |          |                | Any              |
| V2PB         | 2 (Primary) |          |                | Any              |
| V2BB         | 2 (Backup)  |          |                | Any              |
| V1PC         | 1 (Primary) | C        | Home           | Any              |
| V1BC         | 1 (Backup)  |          |                | Any              |
| V2PC         | 2 (Primary) |          |                | Any              |
| V2BC         | 2 (Backup)  |          |                | Any              |

a. The #2 sensors should change significantly between the A, B, and C positions.

## 5.5 Configuration file

The test positions are defined in ENG/pv/cc.lut and ENG/pv/wfs.lut.

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# 6 The “Position Tool”

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## 6.1 Starting the “Position Tool”

From the low-level engineering screen for the mechanism, press the “External Tools” button using the right mouse button, then select “Position Tool.”

## 6.2 Using The “Position Tool”

NIRI and the OIWFS are designed to provide verification that a mechanism is in the proper configuration. This is done by placing analog Hall-effect sensors and magnets at various locations. When the mechanism is in a known, good state, the mechanism is moved to all standard positions. (For mechanisms with a continuous range of travel, the standard positions are at positions that are convenient for checking mechanism operation.)The Hall-effect sensor values are read at each position.

Each time that the mechanism is moved to a standard position, the expected sensor values are compared to the actual sensor values. If the values do not agree, then an error is returned.

Furthermore, any time that the mechanism is moved, whether to a standard position or not, the software examines the step count and sensor values, and attempts to determine if the mechanism is in a known position.

---

## The “Position Tool”

---

The “Position Tool” is a program that aids in this part of the calibration procedure. It moves the system to each of the standard positions, and then records the values, which may then be written to a file. The file is read whenever the system is booted.

Warning: The “Cycle Tool” does not change the in-memory version of the table that it produces. To update the in-memory version of the table

1. Bring up the “Parameters” window (right-click the “Related Screens” button);
2. Install the updated table with gmake;
3. Press the “Load LUT” and “Load ILUT” buttons.

- 6.3 Warning: If the “Cycle Tool” is run while the mechanism is not working correctly, and if the values are written to disk, then the automatic verification of a move command will be suspect.

The screenshot shows a window titled "et\_wish #2" with a menu bar (File, Mechanism, Tools, Help). Below the menu is a table with five columns: "count", "count tol", "val tol", "sens (p)", and "sens (b)". The table has 15 empty rows. Below the table are five input fields labeled "Mech", ".lut base directory", ".lut Subdirectory", "Ident", and "Out File". The ".lut base directory" field contains the path "/home/polaris/yamada/prj/xiri/ENG/pv". At the bottom of the window are three buttons: "ReScan", "Write", and "Exit".

Each line of this display represents a single defined mechanism position. The fields in each line are as follows:

**count.** Nominal step count as this standard position.

**count tol.** How close must the step count be to the nominal step count for the mechanism to be considered to be at this standard position.

**val tol.** How close must the Hall-effect sensor voltages be to the expected voltages for the mechanism to be considered to be at this standard position.

**sens (p).** Voltage at this standard position on the primary sensor.

**sens (b).** Voltage at this standard position on the secondary sensor.

Furthermore, the following values are defined.

**Mech.** The EPICS prefix for all fields related to this mechanism

**.lut base directory.** Where output files are created.

**.lut subdirectory.** Subdirectory of .lut base directory where files are created

**Ident.** The name of the default output file.

**Out File.** Name of the output file (user editable).

**ReScan.** Collect data.

**Write.** Write data file.

**Exit.** Exit from the “Cycle Tool.”

### 6.4 Calibration Procedure (Phase 2)

1. Press the “ReScan” button to move the mechanism to all defined positions.
2. Examine the values. All values should be reasonable. (See Section 6.5.)
3. Examine the “Out File” field. Verify that the correct file will be created. Change the “Out File” field if you do not want to overwrite the normal file.
4. Press the “Write” button to write the data.
5. Press the “Exit” button to exit the “Cycle Tool.”
6. From a shell on the software server, change into the “.pv base directory” (from the “Cycle Tool”).
7. Run “gmake” in the “.pv base directory.”

Note: If you forget to press “Write” before pressing “Exit,” you will not be warned, but the file on disk will not be updated.

### 6.5 Expected Values (Phase 2)

The following tables describe the expected sensor values for each mechanism at the standard positions.

The window cover has no sensors, and never needs to be calibrated. All sensor values read as 1 or 2 at the limits and 0 between limits.

TABLE 5.

Expected values

| Position | Significance    | Primary Value (V) | Backup Value (V) |
|----------|-----------------|-------------------|------------------|
| pos00    | First Position  | any <sup>a</sup>  | any              |
| ...      | ...             | ...               | ...              |
| posNN    | Last Position   | any               | any              |
| home     | Datum Position  | > 1               | > 1              |
| park     | Parked Position | any <sup>b</sup>  | any              |

- a. These values should change significantly from position to position. If the same pair of values (within a few hundred millivolts) appears repeatedly, then the mechanism is not moving. If too many value pairs are near zero, then the mechanism is not moving correctly. If more than one positive value pair appears (other than the home or park position), then the mechanism is stuck at the home position.
- b. The park position is arbitrary, and is placed where it is most convenient. Its value might be positive, negative, or zero.

## 6.6 Configuration file

All constants that are defined in the position tool are defined in ENG/pv/cfg.tcl. See the comments in that file for more details.

---

## 7 Low Level Calibration Procedure

---

Whenever a mechanism is modified, it will be necessary to recalibrate it. Note that this procedure should only be used under very unusual conditions. It is not a standard maintenance procedure.

This section will describe the calibration procedure which is required for each mechanism.

### 7.1 The Wheels and Wheel-like Mechanisms, the Pupil Viewer, and the Window Cover

Low-level calibration is never necessary for the filter wheels, for the pupil-mask wheel, for the entrance beam splitter, for the focal-plane mask wheel, for the pupil viewer, for the window cover, and for the steering mirrors. There are no constants which require adjustment.

### 7.2 The Focus Stages

1. Start the Engineering Screen for the Science Focus Stage or OIWFS Focus Stage.
  2. Adjust the “New Position” field until the value of sensor 2P is less than 0.005 V. Call this *position A*.
  3. Repeat for sensor 2B. Call this *position B*.
  4. While observing the hardware, adjust the “New Position” field until the sensor is adjacent to the magnet. Call this *position C*.
  5. Compute the value of *position C* - *position A*. This is the value of *DOFP*.
-

6. Compute the value of *position C - position B*. This is the value of *DOFB*.
7. Bring up the parameter window. In the *DOFP* field, enter the value of *DOFP*. In the *DOFB* field, enter the value of *DOFB*. Close the parameter window.
8. Press the Datum Button, and let the system datum.
9. Set the “New Position” field to 0. You should find that sensor 1P has a signal between 1 and 5 V. If this is not true, then there is a hardware problem or the calibration was performed incorrectly.
10. While observing the stage, adjust the “New Position” field until you know the safe limits of motion of the mechanism. It is better to be slightly conservative here. Call the lower limit *LOPR*. Call the upper limit *HOPR*.
11. Bring up the parameter window. In the *HOPR* field, enter the value of *HOPR*. In the *LOPR* field, enter the value of *LOPR*.
12. Edit *niri/ENG/pv/cfg.tcl*. Enter the value of *LOPR* in the *pos01* field for *ccFoc*. Enter the value of *HOPR* in the *pos02* field for *ccFoc*.
13. Run Cycle Tool.
14. Run Position Tool.
15. Bring up the parameter window. Enter 0 in the *DOF1* field.
16. Disable sensor 1P.
17. Press the Redatum button in the engineering window. You should eventually see “Peak at (some value)”. Multiply this value by -1 and enter it into the *DOF1* field.
18. Enter sensor 1P.
19. Calibration is complete. The values which were placed in the *DOF1*, *DOFP*, and *DOFB* fields should be entered into *ENG/pv/cc.pv* as the values for *focHallstep.DOF1*, *focHallstep.DOFP*, and *focHallstep.DOFB*.
20. Enter the lower limit in *ENG/pv/cc.pv* as the value of *focHallstep.LOPR* and *focHallstep.CDSB*. Enter the upper limit as *focHallstep.HOPR* and *focHallstep.CDSA*.
21. Run *gmake*, then reboot the IOC.

### 7.3 The Gimbal Mirror

This section has not been written yet.

---

## 8 Warnings

---

If the system needs to be recalibrated frequently, there is a problem with the mechanism. Recalibration can mask minor problems, particularly with mechanisms that are not moving freely.

## Gemini Near Infra-Red Imager



# Near IR Imager (NIRI)—The Engineering User Interface

**Hubert Yamada**

niri\_h ty\_007 / 05 / 1999-11-11

**This document describes the operation of the engineering user interface for NIRI.**

---

## 1 Introduction

---

The Engineering User Interface will be delivered with NIRI to Gemini. Its purposes are:

- to provide a user interface for testing and debugging NIRI;
- to provide a user interface before the standard NIRI user interface is available;
- to provide sample code which the designers of the NIRI standard user interface may inspect and reuse;
- to provide a user interface for the NIRI acceptance test procedure;
- to provide an interface to NIRI functions which the designers of the NIRI standard user interface may choose not to make available to the general users.

*It is extremely important to read the warnings in Section 2.1, “Warnings (Very Important!),” on page 2!*

### 1.1 Purpose

This document describes the operation of the NIRI engineering user interface.

The intended audience of this document is:

- the NIRI reviewers;
- the developers of the Gemini Observatory Control System (OCS) and the NIRI standard user interface;
- the developers of the GNIRS OIWFS components controller and other groups which will be using the design of the NIRI OIWFS
- the scientists, engineers, and technicians who will be designing and testing NIRI;

- the committee which will be reviewing the NIRI acceptance test procedure and who will be verifying that NIRI passes its acceptance tests;
- the early users of NIRI.

### **1.2 Definitions, Acronyms and Abbreviations**

See “Near IR Imager (NIRI)—Definitions, Abbreviations, and References”.

### **1.3 Stylistic Conventions**

References to documents in Section 1.5, “References,” on page 2 of “Near IR Imager (NIRI)—Definitions, Abbreviations, and References” are given like this; [1].

### **1.4 References**

See “Near IR Imager (NIRI)—Definitions, Abbreviations, and References”.

---

## **2 Warnings and General Information**

---

### **2.1 Warnings (Very Important!)**

There are a number of screens which contain functions which may damage NIRI or which may leave it in a state from which no software recovery is possible. These screens are easily distinguished from the standard NIRI screens, because they have a bright yellow title bar on a black background. The functions on these screens are intended for use by the qualified personnel only, and are not intended for general use. They include several functions which are capable of bypassing various safety features.



## **2.2 A Few Notes About Edd/Dm Interfaces**

---

# **3 General User Commands**

---

## **3.1 Starting the User Interface**

## **3.2 The Introduction Screen**

## **3.3 The Components Controller Main Screen**

## **3.4 The On-Instrument Wavefront Sensor Main Screen**

---

# **4 Restarting After the IOC Has Been Rebooted**

---

## **4.1 Wait for initialization to complete**

## **4.2 Restart screens**

## **4.3 Look for RED**

### **4.3.1 Set the Temperature Interlock**

### **4.3.2 Start the Closed Cycle Coolers (if needed)**

## **4.4 Look for YELLOW**

### **4.4.1 Datum Mechanisms**

---

# **5 Advanced User commands**

---

The commands which are described in this section can damage the instrument or leave it in an unusable state. Use with extreme caution. *It is extremely important to read the warnings in Section 2.1, “Warnings (Very Important!),” on page 2!*

*Notice: The screens which are described in this section do not meet the Gemini interface standards. They do not interact with the EPICS database using the mandatory CAD/CAR/APPLY/SIR interface. This is explicitly permitted in [33].*

## 5.1 Starting the Low-Level Engineering Interface



## 5.2 The Main Engineering Screens

FIGURE 1. The Top Level CC Screen

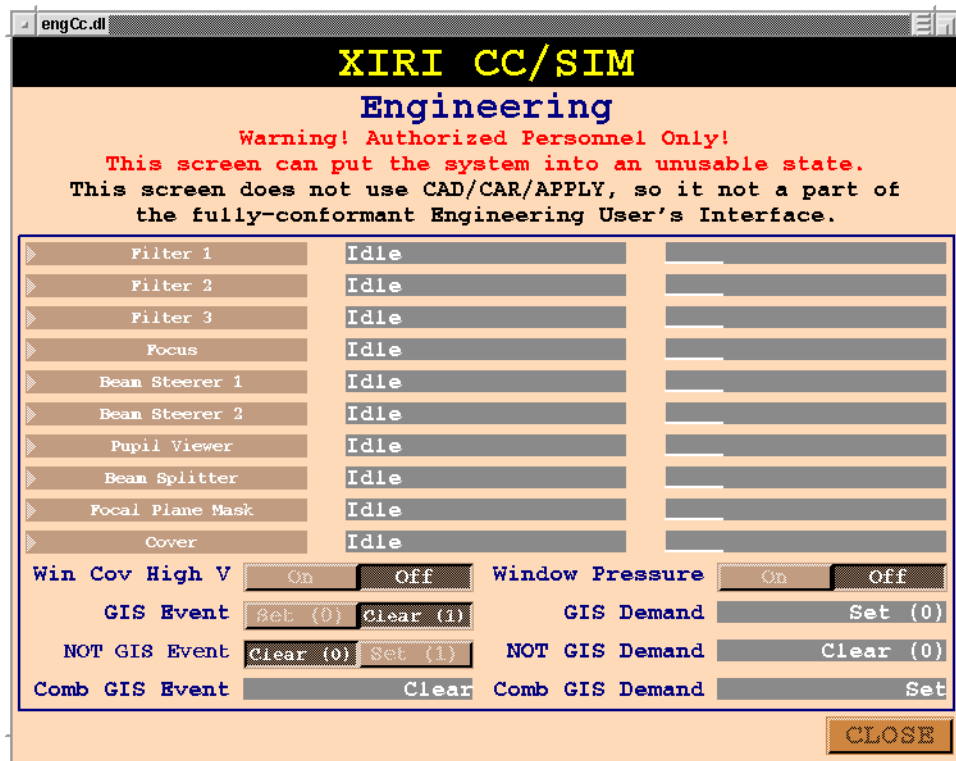
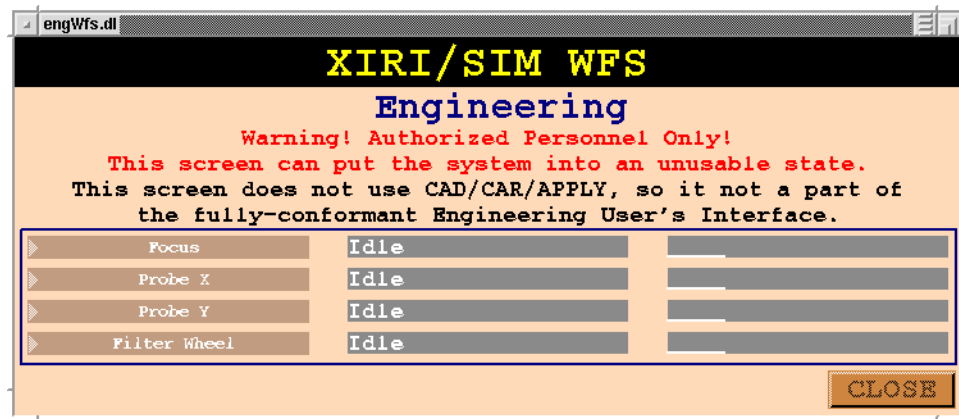


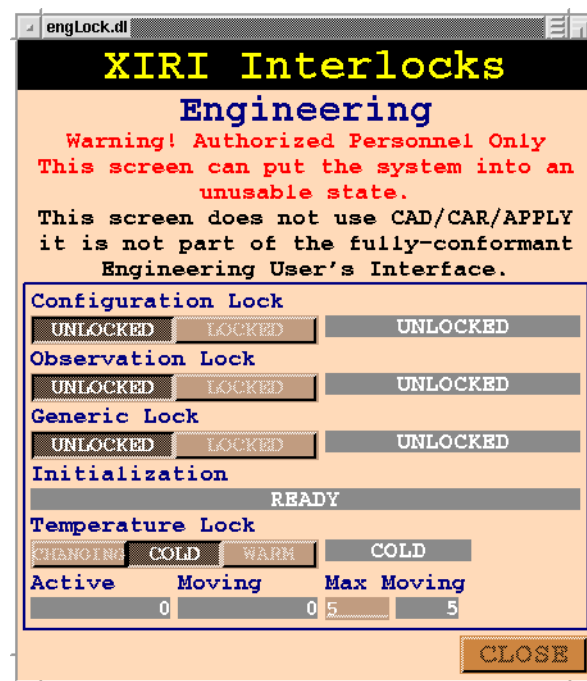
FIGURE 2. The Top Level OIWFS Screen



### 5.3 The Global Parameters Screen



### 5.4 The Engineering Interlocks Screen



## 5.5 The Low Level Mechanism Control Screen

engMech.dll

## XIRI CC/SIM Filter 1

### Engineering

Warning! Authorized Personnel Only!

This screen can put the system into an unusable state.  
This screen does not use CAD/CAR/APPLY so it is not a  
part of the fully-conformant Engineering User's  
Interface.

|                                                                                                                                                                                                                                                          |  |  |                                                                                                                                                                                                       |  |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|
| <b>Status</b><br>Idle<br>Error/Info 0                                                                                                                                                                                                                    |  |  | <b>Sensors</b><br>1P (V) 2P (V)<br>3.997 0.000<br><br>-5.000 5.000 -5.000 5.000<br>Disable Disable<br>Enable Enable |  |  |
| <b>Motor</b> User Raw Stop<br>New Pos. Cur Pos Pos Err<br>0 0 1<br>Low High Sens Tol<br>0 288000 0.300<br>Speed Acc T(s) Datumed<br>3000.0 1.0 0<br>Datum Redatum Datum/Diag<br>Cycle End Cycle Diagnose<br>Set ctr SW Count HW Count<br>Use Set 0.0 0.0 |  |  | 1B (V) 2B (V)<br>3.998 0.000<br><br>-5.000 5.000 -5.000 5.000<br>Disable Disable<br>Enable Enable                   |  |  |
| <b>Select</b><br>home home 0.000<br>Ex Md Ps Ex Sen P Ex Sen B<br>0 0.000 0.000<br>Mod Pos Rd Sen P Rd Sen B<br>0 3.997 3.998<br>Update                                                                                                                  |  |  | Stripchart<br>Related Screens<br>External Tools<br>CLOSE                                                                                                                                              |  |  |

## 5.6 The Parameter Screen

engParams.dll

## XIRI CC/SIM Filter 1

### Engineering Parameters

Warning! Authorized Personnel Only!  
This screen can put the system into an unusable state.

This screen does not use CAD/CAR/APPLY, so it is not a part of the fully-conformant Engineering User's Interface.

#### Cycle Parameters

|      | Toler  | Set Exp Value |       |       |       |   |
|------|--------|---------------|-------|-------|-------|---|
|      | 0.300  | NO            | YES   |       |       |   |
| Pos. | Exp1   | Val1          | Exp2  | Val2  |       |   |
| A    | 13400  | -0.779        | 0.000 | 0.000 | 0.000 | P |
|      |        | -0.779        | 0.000 | 0.000 | 0.000 | B |
| B    | -10600 | -3.119        | 0.000 | 0.000 | 0.000 | P |
|      |        | -2.339        | 0.000 | 0.000 | 0.000 | B |
| C    | 0      | 3.997         | 0.000 | 0.000 | 0.000 | P |
|      |        | 3.998         | 0.000 | 0.000 | 0.000 | B |

#### Motor Control

| Auto Off | Mot Pwr | Module  | Reset Module |
|----------|---------|---------|--------------|
| No       | Yes     | Auto On | Off          |
|          |         | Good    | Reset        |

#### Motion / Homing Parameters

| Backlash | DSLPL | DSLB   | THRP  | WTHP |
|----------|-------|--------|-------|------|
| 1000     | 0     | 0      | 0.000 | 400  |
| Timeout  | DOFP  | DOFB   | THRB  | WTHB |
| 120      | 0     | 0      | 2.000 | 400  |
| DOF1     | LOPR  | HOPR   | ZERO  |      |
| -30      | 0     | 288000 | 0     |      |

Reload LUTs:

## 5.7 The Temperature Control Screen

engTmp.dll

## XIRI/SIM Temperature control Engineering

Warning! Authorized Personnel Only!  
 This screen can put the system into an unusable state.  
 Center the science focusing stage before cooling or warming.  
 This screen does not use CAD/CAR/APPLY, so it not a part of  
 the fully-conformant Engineering User's Interface.

| Temperature Sensors |        | Temperature Controllers |                 |                   |        |
|---------------------|--------|-------------------------|-----------------|-------------------|--------|
|                     |        | OIWFS                   | Img Buf<br>Mass | Cold Pl<br>Center |        |
| Serial I/O          |        | Serial I/O              |                 |                   |        |
| Top Strap           | 300.00 | Temperature             | 300.00          | 300.00            | 300.00 |
| Bottom Strap        | 300.00 | Heater %                | 0.00            | 0.00              | 0.00   |
| Cryopump            | 300.00 | Set Point               | 0.00            | 0.00              | 0.00   |
| Cold Pl Edge        | 300.00 | Gain                    | 0.00            | 0.00              | 0.00   |
| CC Cabinet          | 300.00 | Rate                    | 0.00            | 0.00              | 0.00   |
| WFS Cabinet         | 300.00 | Reset                   | 0.00            | 0.00              | 0.00   |
| 7                   | 300.00 | Range                   | Off             | Off               | Off    |
| 8                   | 300.00 | Tune                    | Manual          | Manual            | Manual |
| Units               | K      | Units                   | K               | K                 | K      |

| Closed Cycle Coolers |        |     |      |             |        |
|----------------------|--------|-----|------|-------------|--------|
| Cooling              | Off    | Low | High | State       | Off    |
| Low                  | 30000  |     |      | Pos         | 0      |
| High                 | 55000  |     |      | Vel         | 1000   |
| Enable 1             | Auto   | On  | Off  | Jog         | 0      |
| Enable 2             | Auto   | On  | Off  | Dmov        | 1      |
|                      |        |     |      | Abort       |        |
| Timer                | Cancel |     |      | Heater Mode | Normal |

CLOSE





## ICD 1.9a/3.1

### “Near IR Imager (NIRI) to Observatory Control System”

niri\_h ty\_003

Hubert Yamada  
1999-11-11

Version: 0.99.02

Issued By: Hubert Yamada

Approved By: \_\_\_\_\_ Date: \_\_\_\_\_  
Group Manager  
\_\_\_\_\_ Group Manager (if req) \_\_\_\_\_  
\_\_\_\_\_ Systems Engineering \_\_\_\_\_

## Revision Control

|                             |                               |
|-----------------------------|-------------------------------|
| 1. Revision Version #       | 1.00alpha0                    |
| Date:                       | 1999-10-05                    |
| Revised by:                 | Hubert Yamada                 |
| Reason for / items changed: | Draft for 1.00 alpha release. |

 Last Modified: Tuesday, May 23, 2000 12:15 pm

# Chapter 1 Introduction

---

## 1–1.0 Description

---

### 1–1.1 Purpose

This Interface Control Document (ICD) serves the following purposes:

- To describe the commands and parameters that will pass between the OCS and the Near Infra-Red Imager (NIRI) sequencer and subsystems;
- To provide the necessary declarations of commands, parameters and Status and Information records which is required in an “Instrument to OCS ICD.” This document provides the required tables, using the required “Parameter Definition Format”, described in [11].

This document only describes the ways in which NIRI differs from the prototypical Gemini instrument, described in ICD 1.9/3.1 [16]. It is assumed that the reader is familiar with that document. *Much of this document is copied verbatim from ICD 1.9/3.1, without further acknowledgment.*

In addition to the documents mentioned in the previous paragraphs, the specific interfaces described in this document are governed by the following parent Interface Control Documents, which describe the general properties of particular kinds of interface:

- ICD 1a [24], which describes how the OCS deals with commands and attributes and the behavior the OCS expects from the systems it commands.
- ICD 1b [25], which gives a general description of the way the Observatory Control System communicates with the other principal systems. It is also related to ICD 7a, [27], which describes the way an Instrument Control System communicates with its subsystems.
- ICD 2 [26], which describes the way in which an Instrument Control System makes information available to the outside world (and the OCS) by means of “Status Information Records”.

The intended audience for this document is:

- the NIRI reviewers;
- the developers of the Gemini Observatory Control System (OCS).

### 1–1.2 Scope

This document describes only aspects of the NIRI software system which differ from, or are more specific than, the Gemini standard instrument, described in [16]. It is assumed that the reader is familiar with that documents.

This document does *not* describe the following topics:

- The science detector, except for sequence commands which will be forwarded to the science detector controller (see [18]);
- The NIRI hardware and electronics (see [11] and [12], summarized in [3]);

- The NIRI On Instrument Wavefront Sensor (see [9]);
- The names of the NIRI mechanism positions (see [36]);
- The NIRI Interlock system (see [6])<sup>1</sup>;
- The software design (see [2], and [3]).

---

## 1–2.0 Related Documents and Drawings

### 1–2.1 References

See “Near IR Imager (NIRI)—Definitions, Abbreviations, and References”.

### 1–2.2 Definitions, Acronyms and Abbreviations

See “Near IR Imager (NIRI)—Definitions, Abbreviations, and References”.

### 1–2.3 Stylistic Conventions

References to documents in Section 1.5, “References,” on page 2 of “Near IR Imager (NIRI)—Definitions, Abbreviations, and References” are given like this; [1].

---

1. All interlock commands are available from within the components controller, by using a niri:cc: prefix, instead of the niri:wfs: prefix.

## Chapter 2 Details of the NIRI to OCS Interface

---

### 2–1.0 Introduction

---

### 2–2.0 Physical System Interfaces

---

#### 2–2.1 Mechanical Interface

*Not Applicable for a software ICD.*

#### 2–2.2 Optical Interface

*Not Applicable for a software ICD.*

#### 2–2.3 Electronic Interface

The system hardware architecture for the NIRI control software is the same as described in [16], with the following exception:

- The Instrument Sequencer, Components Controller, and OIWFS Components Controller will share the same IOC.

There will also be the following electronic VME modules:

- The XVME-240 Digital I/O Board, (shared between the NIRI CC, the NIRI OIWFS, and the interlock bus):
  - To support the interlock bus (Gemini standard, see [29]);
  - To monitor the NIRI and NIRI OIWFS stepper-motor control modules;
  - To reset the NIRI and NIRI OIWFS stepper-motor control modules;
  - To control the heating system.
- A XVME 490 four-port serial I/O board:
  - To control the NIRI/NIRI OIWFS temperature controllers (three ports);
  - To control the NIRI/NIRI OIWFS temperature sensor (one port);
- Four custom-built VME sensor-support boards:
  - To read out the NIRI Hall-effect sensors;
  - To read out the NIRI OIWFS Hall-effect sensors;
- Two XVME-566 Analog Input Modules
  - To read out the VME sensor-support boards for the NIRI Hall-effect sensors;
  - To read out the VME sensor-support boards for the NIRI OIWFS Hall-effect sensors;
- Two Oregon Microsystems VME8-8 Stepper-motor controller boards
  - To control the NIRI stepper-motors
  - To control the NIRI OIWFS stepper-motors;

**2–2.4 Mass/Balance**

*Not Applicable for a software ICD.*

**2–2.5 Thermal Interface**

*Not Applicable for a software ICD.*

---

**2–3.0 Software / Control Function Interface**


---

**2–3.1 Overview****2–3.1.1 Communications infrastructure**

The communications infrastructure is the same as described in [16].

**2–3.1.2 Software architecture**

The software architecture is described in [2].

**2–3.2 Behavior**

The behavior of the NIRI/OCS interface is the same as described in [16].

**2–3.3 Implementation**

This section contains tables summarizing the EPICS records used for the interface. The next sections contain the details of what those records do.

It is assumed in all cases that the record names given in this document are prefixed by the string “niri:”<sup>2</sup>. Records in the SAD database are prefixed by “niri:”<sup>3</sup>.

**2–3.3.1 Command (CAD/CAR) records**

The columns of the command record tables are as follows:

**Description.** A brief description of the command.

**Attribute.** The name of the attribute which the Observatory Control System associates with a particular parameter of a particular command. Component position names are listed separately in Table 13.

**CAD record / CAD record.FIELD.** The name of the EPICS CAD record (minus the “niri:” prefix) together with the field in that CAD record associated with each parameter of the command.

**CAD ordering number.** The recommended order in which this command should be executed. CAD records should be connected to their associated APPLY record in the order 1, 2, 3, 4, 5, etc. CAD records within the same ordering number set can be connected to the APPLY record in any order in that set.

---

2. Or some other Gemini-assigned name, which must be at most four alpha-numeric characters, followed by a colon.

3. Or some other Gemini-assigned name, which must be at most four alpha-numeric characters, followed by one or two colons.

**CAR record.** The CAR record associated with the command. Note that several CAD records may share the same CAR record.

#### 2-3.3.1.1 Sequence command CAD/CAR records

The NIRI sequencer recognizes all the OCS sequence commands and uses the CAD/CAR records specified in ICD 1b [25]. See also ICD 1a [24], for a description of what these commands do.

**TABLE 1.** NIRI sequence command CAD/CAR records

| Description       | Attribute <sup>a</sup>                                                  | CAD record /<br>CAD record.FIELD<br>("niri:" +) | CAR record <sup>b</sup><br>("niri:" +) |
|-------------------|-------------------------------------------------------------------------|-------------------------------------------------|----------------------------------------|
| Test.             | (none)                                                                  | test                                            | testC                                  |
| Initialize.       | (none)                                                                  | init                                            | initC                                  |
| Datum mechanisms. | (none)                                                                  | datum                                           | datumC                                 |
| Verify            | (none)                                                                  | verify                                          | verifyC                                |
| End verify        | (none)                                                                  | endVerify                                       | endVerifyC                             |
| Guide             | (none)                                                                  | guide                                           | guideC                                 |
| End guide         | (none)                                                                  | endGuide                                        | endGuideC                              |
| Observe           | Data Label<br>(reserved) <sup>c</sup><br>Observation Count <sup>d</sup> | observe.A<br>observe.B<br>observe.C             | observeC                               |
| Pause             | (none)                                                                  | pause                                           |                                        |
| Continue          | (none)                                                                  | continue                                        |                                        |
| Stop              | (none)                                                                  | stop                                            |                                        |
| Abort             | (none)                                                                  | abort                                           |                                        |
| End Observe       | (none)                                                                  | endObserve                                      | endObserveC                            |
| Park              | (none)                                                                  | park                                            | parkC                                  |
| Setup             | (none)                                                                  | setup                                           | setupC                                 |

- a. "(none)" means that there are no arguments
- b. Currently, it is required to have separate TestC, InitC, etc. CAR records. There some disagreement over this requirement, and all of these CAR records may disappear, to be replaced by the global applyC.
- c. This field is reserved for implementation purposes, and should not be modified.
- d. I.e., the number of successive integrations / read-out cycles that should be made.

#### 2-3.3.1.2 General observing command CAD/CAR records

The commands in Table 2 are the only ones that are used during routine observation.

**TABLE 2.** NIRI general observing command CAD/CAR records

| Description                              | Attribute <sup>a</sup> | CAD record /<br>CAD record.FIELD<br>("niri:" +) | CAD<br>ordering<br>number | CAR<br>record<br>("niri:" +) |
|------------------------------------------|------------------------|-------------------------------------------------|---------------------------|------------------------------|
| Select the beam-splitter wheel position. | position name          | cc:spltSel.A                                    |                           | cc:spltC                     |
| Park the beam-splitter wheel.            | (none)                 | cc:spltPark                                     |                           |                              |

**TABLE 2.** NIRI general observing command CAD/CAR records

| Description                                               | Attribute <sup>a</sup> | CAD record /<br>CAD record.FIELD<br>("niri:" +) | CAD<br>ordering<br>number | CAR<br>record<br>("niri:" +) |
|-----------------------------------------------------------|------------------------|-------------------------------------------------|---------------------------|------------------------------|
| Select a filter in filter wheel 1.                        | filter name            | cc:flt1Sel.A                                    |                           | cc:flt1C                     |
| Park filter wheel 1.                                      | (none)                 | cc:flt1Park                                     |                           |                              |
| Select a filter in filter wheel 2.                        | filter name            | cc:flt2Sel.A                                    |                           | cc:flt2C                     |
| Park filter wheel 2.                                      | (none)                 | cc:flt2Park                                     |                           |                              |
| Select a pupil mask or a filter in the pupil-plane wheel. | position name          | cc:flt3Sel.A                                    |                           | cc:flt3C                     |
| Park the pupil-plane wheel.                               | (none)                 | cc:flt3Park                                     |                           |                              |
| Select the focal-plane mask.                              | position name          | cc:foplSel.A                                    |                           | cc:foplC                     |
| Park the focal-plane mask wheel.                          | (none)                 | cc:foplPark                                     |                           |                              |
| Select the beam-steerer position                          | camera name            | cc:sterSel.A                                    |                           | cc:sterC                     |
| Park the beam-steerers.                                   | (none)                 | cc:sterPark                                     |                           |                              |
| Select the pupil-viewer position                          | position name          | cc:puvwSel.A                                    |                           | cc:puvwSelC                  |
| Park the pupil-viewer.                                    | (none)                 | cc:puvwPark                                     |                           |                              |
| Select the focus-stage position.                          | position name          | cc:focSel.A                                     |                           | cc:focC                      |
| Set the focus-stage position.                             | position               | cc:focMove.A                                    |                           |                              |
| Park the focus stage.                                     | (none)                 | cc:focPark                                      |                           |                              |
| Select the window-cover position.                         | position name          | cc:covSel.A                                     |                           | cc:covC                      |
| Park the window cover. <sup>b</sup>                       | (none)                 | cc:covPark                                      |                           |                              |

a. "(none)" means that there are no arguments

b. Equivalent to closing the window cover.

### 2-3.3.1.3 Engineering command CAD/CAR records—System Commands

The following commands are used for engineering purposes, and are not expected to be used during normal observing. These are normally activated by the engineering user interface.

These commands affect the overall behavior of NIRI and the NIRI OIWFS. Note, in particular, that it is not possible to reboot NIRI and the NIRI OIWFS separately

**TABLE 3.** NIRI system engineering command CAD/CAR records

| Description                       | Attribute <sup>a</sup> | CAD record /<br>CAD record.FIELD<br>("niri:" +) | CAD<br>ordering<br>number | CAR record |
|-----------------------------------|------------------------|-------------------------------------------------|---------------------------|------------|
| Reboot.                           | (none)                 | reboot                                          | N/A                       | N/A        |
| Select debug mode.                | Debug mode             | debug.A                                         |                           | debugC     |
| Flush data from internal buffers. | (none)                 | flush                                           |                           | flushC     |

a. "(none)" means that there are no arguments.



#### 2-3.3.1.4 CC Sequence command CAD/CAR records

The NIRI components controller control software recognizes all the OCS sequence commands and uses the CAD/CAR records specified in ICD 1b, [25]. See also ICD 1a, [24], for a description of what these commands do.

**TABLE 4.** NIRI CC sequence command CAD/CAR records

| Description            | Attribute <sup>a</sup> | CAD record /<br>CAD record.FIELD<br>("niri:" +) | CAR record<br>("niri:" +) |
|------------------------|------------------------|-------------------------------------------------|---------------------------|
| Test.                  | (none)                 | cc:test                                         | cc:testC                  |
| Initialize.            | (none)                 | cc:init                                         | cc:initC                  |
| Datum mechanisms.      | (none)                 | cc:datum                                        | cc:datumC                 |
| Verify                 | (none)                 | cc:verify                                       | cc:verifyC                |
| End verify (no action) | (none)                 | cc:endVerify                                    | cc:endVerifyC             |
| Guide                  | (none)                 | cc:guide                                        | cc:guideC                 |
| End guide (no effect)  | (none)                 | cc:endGuide                                     | cc:endGuideC              |
| Park                   | (none)                 | cc:park                                         | cc:parkC                  |

a. "(none)" means that there are no arguments.

### 2-3.3.1.5 Engineering command NIRI CC CAD/CAR records

The following commands are used for engineering purposes, and are not expected to be used during normal observing. Note that, although these commands are connected to the APPLY record, they are normally activated directly by the engineering user interface.

**TABLE 5.** NIRI CC engineering command CAD/CAR records

| Description                                                                                                          | Attribute <sup>a</sup> | CAD record /<br>CAD record.FIELD<br>("niri:" +) | CAD<br>ordering<br>number | CAR<br>record<br>("niri:" +) |
|----------------------------------------------------------------------------------------------------------------------|------------------------|-------------------------------------------------|---------------------------|------------------------------|
| Datum filter wheel 1.                                                                                                | (none)                 | cc:flt1Datm                                     |                           | cc:flt1C                     |
| Set the filter-wheel 1 position using position numbers <sup>b</sup> .                                                | position               | cc:flt1Move.A                                   |                           |                              |
| Set the filter wheel 1 position in engineering units <sup>c</sup> .                                                  | motor position         | cc:flt1EngMove.A                                |                           |                              |
| Confirm that filter wheel 1 has not lost its datum position.                                                         | (none)                 | cc:flt1Rdtm                                     |                           |                              |
| Confirm that filter wheel 1 and its Hall-effect sensors are operational.                                             | (none)                 | cc:flt1Diag                                     |                           |                              |
| Datum filter wheel 2.                                                                                                | (none)                 | cc:flt2Datm                                     |                           | cc:flt2C                     |
| Set the filter-wheel 2 position using position number <sup>d</sup> .                                                 | position               | cc:flt2Move.A                                   |                           |                              |
| Set the filter wheel 2 position in engineering units <sup>e</sup> .                                                  | motor position         | cc:flt2EngMove.A                                |                           |                              |
| Confirm that filter wheel 2 has not lost its datum position.                                                         | (none)                 | cc:flt2Rdtm                                     |                           |                              |
| Confirm that filter wheel 2 and its Hall-effect sensors are operational                                              | (none)                 | cc:flt2Diag                                     |                           |                              |
| Datum the pupil-plane wheel (pupil mask wheel / filter wheel 3).                                                     | (none)                 | cc:flt3Datm                                     |                           | cc:flt3C                     |
| Set the pupil-plane wheel position using position number <sup>f</sup> .                                              | position               | cc:flt3Move.A                                   |                           |                              |
| Set the pupil-plane wheel (pupil mask wheel / filter wheel 3) position in engineering units <sup>g</sup> .           | motor position         | cc:flt3EngMove.A                                |                           |                              |
| Confirm that the pupil-plane wheel (pupil-mask wheel / filter wheel 3) has not lost its datum position.              | (none)                 | cc:flt3Rdtm                                     |                           |                              |
| Confirm that pupil-plane wheel (pupil-mask wheel / filter wheel 3) wheel and its Hall-effect sensors are operational | (none)                 | cc:flt3Diag                                     |                           |                              |
| Datum the focal-plane mask wheel.                                                                                    | (none)                 | cc:foplDatm                                     |                           | cc:foplC                     |
| Set the focal-plane mask wheel position using position number <sup>h</sup> .                                         | position               | cc:foplMove.A                                   |                           |                              |
| Set the focal-plane mask wheel position in engineering units <sup>i</sup> .                                          | motor position         | cc:foplEngMove.A                                |                           |                              |
| Confirm that the focal-plane mask wheel has not lost its datum position.                                             | (none)                 | cc:foplRdtm                                     |                           |                              |
| Confirm that the focal-plane mask wheel and its Hall-effect sensors are operational                                  | (none)                 | cc:foplDiag                                     |                           |                              |
| Datum the steering mirrors.                                                                                          | (none)                 | cc:sterDatm                                     |                           | cc:sterC                     |

**TABLE 5.** NIRI CC engineering command CAD/CAR records

| Description                                                                | Attribute <sup>a</sup> | CAD record /<br>CAD record.FIELD<br>("niri:" +) | CAD<br>ordering<br>number | CAR<br>record<br>("niri:" +) |
|----------------------------------------------------------------------------|------------------------|-------------------------------------------------|---------------------------|------------------------------|
| Datum steering-mirror 1.                                                   | (none)                 | cc:ster1Datm                                    |                           | cc:ster1C                    |
| Select the steering-mirror 1 position                                      | position name          | cc:ster1Sel.A                                   |                           |                              |
| Set the steering-mirror 1 position using position number <sup>j</sup> .    | position               | cc:ster1Move.A                                  |                           |                              |
| Set the steering-mirror 1 position in engineering units <sup>k</sup> .     | motor position         | cc:ster1EngMove.A                               |                           |                              |
| Confirm that steering mirror 1 has not lost its datum position.            | (none)                 | cc:ster1Rdtm                                    |                           |                              |
| Confirm that steering mirror 1 and its Hall-effect sensors are operational | (none)                 | cc:ster1Diag                                    |                           | cc:ster2C                    |
| Datum steering-mirror 2.                                                   | (none)                 | cc:ster2Datm                                    |                           |                              |
| Select the steering-mirror 2 position                                      | position name          | cc:ster2Sel.A                                   |                           |                              |
| Set the steering-mirror 2 position using position number <sup>l</sup> .    | position               | cc:ster2Move.A                                  |                           |                              |
| Set the steering-mirror 2 position in engineering units <sup>m</sup> .     | motor position         | cc:ster2EngMove.A                               |                           |                              |
| Confirm that steering mirror 2 has not lost its datum position.            | (none)                 | cc:ster2Rdtm                                    |                           | cc:puvwC                     |
| Confirm that steering mirror 2 and its Hall-effect sensors are operational | (none)                 | cc:ster2Diag                                    |                           |                              |
| Datum the pupil-viewer                                                     | (none)                 | cc:puvwDatm                                     |                           |                              |
| Set the pupil-viewer position <sup>n</sup> .                               | position               | cc:puvwMove.A                                   |                           |                              |
| Set the pupil-viewer position in engineering units <sup>o</sup> .          | motor position         | cc:puvwEngMove.A                                |                           |                              |
| Confirm that the pupil viewer and its Hall-effect sensors are operational  | (none)                 | cc:puvwDiag                                     |                           | cc:focC                      |
| Datum the focus stage                                                      | (none)                 | cc:focDatm                                      |                           |                              |
| Set the focus-stage position in engineering units.                         | motor position         | cc:focEngMove.A                                 |                           |                              |
| Confirm that the focus stage has not lost its datum position.              | (none)                 | cc:focRdtm                                      |                           |                              |
| Confirm that the focus stage and its Hall-effect sensors are operational   | (none)                 | cc:focDiag                                      |                           | cc:covC                      |
| Datum the window cover.                                                    | (none)                 | cc:covDatm                                      |                           |                              |
| Set the window-cover position.                                             | position               | cc:covMove.A                                    |                           |                              |
| Set the window cover position in engineering units.                        | motor position         | cc:covEngMove.A                                 |                           |                              |
| Confirm that the window cover has not lost its datum position.             | (none)                 | cc:covRdtm                                      |                           |                              |
| Confirm that the window cover and its Hall-effect sensors are operational  | (none)                 | cc:covDiag                                      |                           |                              |

- a. "(none)" means that a command accepts no arguments
- b. Only the standard positions (excluding datum and park) are repeatable.
- c. Only the standard positions (excluding datum and park) are repeatable.
- d. Only the standard positions (excluding datum and park) are repeatable.
- e. Only the standard positions (excluding datum and park) are repeatable.
- f. Only the standard positions (excluding datum and park) are repeatable.
- g. Only the standard positions (excluding datum and park) are repeatable.

- h. Only the standard positions (excluding datum and park) are repeatable.
- i. Only the standard positions (excluding datum and park) are repeatable.
- j. Only the standard positions (excluding datum and park) are repeatable.
- k. Only the standard positions (excluding datum and park) are repeatable.
- l. Only the standard positions (excluding datum and park) are repeatable.
- m. Only the standard positions (excluding datum and park) are repeatable.
- n. *This is strongly discouraged.* Only the standard positions (excluding datum and park) are repeatable. The predefined locations are only stable when approached in the correct fashion.
- o. *This is strongly discouraged.* Only the standard positions (excluding datum and park) are repeatable. The predefined locations are only stable when approached in the correct fashion.

### 2-3.3.2 Status Information Records

The following tables (Table 6 and Table 7) show the status information provided by NIRI, together with the subset of that information which is written to the FITS header. The tables distinguish status information provided by the Instrument Sequencer and Components Controller, but all the information shown is available over the OCS-to-NIRI interface. The tables have the following columns:

**SIR record.** The part of the name of the EPICS SIR record which follows the “niri:” prefix<sup>4</sup>.

**Units / Values.** The units in which the quantity is stored and / or the range of values that are taken.

- Ustep refers to motor microsteps.
- “User units” or “real-world” units refer to the units that are stored in the Units SIR variable.

**Comments.** A description of the item. The first paragraph of this will go into the SIR record description and also the comment field in the FITS header.

*Note that only the type, units and comments are actually contained in the SIR records. The FITS keyword is provided by the DHS.*

#### 2-3.3.2.1 NIRI Instrument Sequencer Status Information

Table 6 contains status information for all of NIRI, particularly, the instrument sequencer.

**TABLE 6.** NIRI Instrument Sequencer (IS) Status Information

| SIR record<br>(“niri:” or<br>“nirs:” +) | FITS<br>Keyword | FITS<br>Inclu<br>ded | Type   | Units/Values | Comments                                   |
|-----------------------------------------|-----------------|----------------------|--------|--------------|--------------------------------------------|
| dcAbort                                 | DCABRT          | never                | long   | 0/1          | Shows that an abort is in progress         |
| debugMode                               | DMODE           | never                | string |              | Debugging mode [NOLOG   NONE   MIN   FULL] |
| health                                  | HLTH            | never                | string |              | Instrument health [GOOD   WARNING   BAD]   |

4. Note that the actual system has been designed so the Status Alarm Database can easily be given a different prefix from the rest of the NIRI database (for example “niri:”). It is assumed in this document that the Status Alarm Database prefix is “niri:”.

**TABLE 6.** NIRI Instrument Sequencer (IS) Status Information

| SIR record<br>("niri:" or<br>"nirs:" +) | FITS<br>Keyword | FITS<br>Inclu<br>ded | Type                          | Units/Values | Comments                                                               |
|-----------------------------------------|-----------------|----------------------|-------------------------------|--------------|------------------------------------------------------------------------|
| heartBeat                               | HRTB            | never                | long                          | count        | Heartbeat: Shows that IS is still alive by increasing at a steady rate |
| historyLog                              | HLOG            | never                | array of<br>char <sup>a</sup> |              | History-log record <sup>b</sup>                                        |
| issPort                                 | ISSPT           | start                | long                          | number       | ISS port on which instrument is installed                              |
| label                                   | LABEL           | never                | string                        |              | The FDSC field describes the Instrument                                |
| name                                    | NAME            | never                | string                        |              | Instrument Name                                                        |
| simMode                                 | SMODE           | start                | string                        |              | Instrument simulation mode [NONE   FULL   FAST   VSM]                  |
| state                                   | STATE           | never                | string                        |              | Instrument state [BOOTING   INITIALIZING   RUNNING   CONFIGURING]      |
| wfsBeam                                 | WFSBM           | start                | long                          |              | Is the WFS detector unobstructed [0=YES   1=NO]                        |

a. should be "bigstring"

b. This record is not particularly useful, because of various EPICS limitations.

### 2–3.3.2.2 The Components Controller Status Information

Table 7 contains the same information as Table 6, but for the Components Controller, not the sequencer.

**TABLE 7.** NIRI Components Controller (CC) Overall Status Information

| SIR record (“niri:” or “nirs:” +) | FITS Keyword | FITS Included | Type                       | Units  | Comments                                                                        |
|-----------------------------------|--------------|---------------|----------------------------|--------|---------------------------------------------------------------------------------|
| cc:debugMode                      | CCDMODE      | never         | string                     |        | Debugging mode [NOLOG   NONE   MIN   FULL]                                      |
| cc:health                         |              | never         | string                     |        | CC Health [GOOD   WARNING   BAD]                                                |
| cc:heartBeat                      | CCHRTB       | never         | long                       | count  | Heart beat record: Shows that CC is still alive by increasing at a steady rate. |
| cc:historyLog                     | CCHLOG       | never         | array of char <sup>a</sup> |        | History log record <sup>b</sup>                                                 |
| cc:issPort                        | CCISSPT      | start         | long                       | number | ISS port on which instrument is installed                                       |
| cc:label                          | CCLABEL      | never         | string                     |        | The FDSC field describes the Components Controller                              |
| cc:name                           | CCNAME       | never         | string                     |        | Components Controller Name                                                      |
| cc:simMode                        | CCSMODE      | start         | string                     |        | Simulation mode [NONE   FULL   FAST   VSM]                                      |
| cc:state                          | CCSTATE      | never         | string                     |        | Instrument state [BOOTING   INITIALIZING   RUNNING   CONFIGURING]               |
| cc:wfsBeam                        | CCWFSBM      | start         | long                       | 0/1    | Is the WFS detector unobstructed? [YES=0   NO=1] <sup>c</sup>                   |

- Should be “bigstring”
- This record is not useful, because of various EPICS limitations.
- The cc:wfsBeam record only indicates whether any components controller mechanisms are obstructing the wavefront sensor. For most purposes, wfsBeam (which combines cc:wfsBeam and wfs:wfsBeam) should be used.

### 2–3.3.2.3 The Components Controller Component Status Information

The following table summarizes the information for the various components.

**TABLE 8.** NIRI Components Controller (CC) Component Status Information

| SIR record (“niri:” or “nirs:” +) | FITS Keyword | FITS Included | Type | Units | Comments                                          |
|-----------------------------------|--------------|---------------|------|-------|---------------------------------------------------|
| cc:covDatumed                     | CCCVDTM      | never         | long | 0/1   | Is the window cover datumed? [YES=0   NO=1]       |
| cc:covEngCyclic                   | CCCVECY      | never         | long | 0/1   | Ignore limits for the window cover [YES=0   NO=1] |

**TABLE 8.** NIRI Components Controller (CC) Component Status Information

| SIR record<br>("niri:" or "nirs:"<br>+) | FITS<br>Keyword | FITS<br>Included | Type   | Units            | Comments                                                                        |
|-----------------------------------------|-----------------|------------------|--------|------------------|---------------------------------------------------------------------------------|
| cc:covEngMax                            | CCCVEMX         | never            | long   | motor microsteps | The maximum position of the window cover in engineering units                   |
| cc:covEngMin                            | CCCVEMN         | never            | long   | motor microsteps | The minimum position of the window cover in engineering units                   |
| cc:covEngName                           | CCCVENM         | start            | string |                  | Engineering name of the window cover position                                   |
| cc:covEngPos                            | CCCVEPS         | start            | long   | motor microsteps | The position of the window cover in engineering units                           |
| cc:covLabel                             | CCCVLBL         | never            | string |                  | The FDSC field describes the window cover                                       |
| cc:covMax                               | CCCVMAX         | never            | double | position number  | The maximum position of the window cover in real-world units (position number). |
| cc:covMin                               | CCCVMIN         | never            | double | position number  | The minimum position of the window cover in real-world units (position number). |
| cc:covName                              | CCCVNAM         | start            | string |                  | User name of the window cover position                                          |
| cc:covParked                            | CCCVPRK         | never            | long   | 0/1              | Is the window cover parked? [YES=0   NO=1]                                      |
| cc:covPos                               | CCCVPOS         | start            | double | position number  | The window cover position in real-world units (position number).                |
| cc:covRdtmVal                           | CCCVRDT         | never            | long   | motor microsteps | Returned value from the window cover redatum command                            |
| cc:covReject                            | CCCVREJ         | never            | long   | 0/-1             | Combined window cover CAR state [GOOD=0   REJECT=-1]                            |
| cc:covState                             | CCCVSTE         | never            | string |                  | the window cover SNL state                                                      |
| cc:covUnits                             | CCCVUNT         | start            | string |                  | Definition of real-world units for the window cover ("position number").        |
| cc:filt1Datumed                         | CCFL1DTM        | never            | long   | 0/1              | Is Filter Wheel 1 datumed? [YES=0   NO=1]                                       |
| cc:filt1EngCyclic                       | CCFL1ECY        | never            | long   | 0/1              | Ignore limits for Filter Wheel 1? [YES=0   NO=1]                                |
| cc:filt1EngMax                          | CCFL1EMX        | never            | long   | motor microsteps | The maximum position of filter wheel 1 in engineering units                     |
| cc:filt1EngMin                          | CCFL1EMN        | never            | long   | motor microsteps | The minimum position of filter wheel 1 in engineering units                     |
| cc:filt1EngName                         | CCFL1ENM        | start            | string |                  | Engineering name of filter wheel 1 position                                     |
| cc:filt1EngPos                          | CCFL1EPS        | start            | long   | motor microsteps | The position of filter wheel 1 in engineering units                             |
| cc:filt1Label                           | CCFL1LBL        | never            | string |                  | The FDSC field describes the Filter Wheel 1                                     |
| cc:filt1Max                             | CCFL1MAX        | never            | double | position number  | The maximum position of filter wheel 1 in real-world units (position number).   |

**TABLE 8.** NIRI Components Controller (CC) Component Status Information

| SIR record<br>("niri:" or "nirs:"<br>+) | FITS<br>Keyword | FITS<br>Included | Type   | Units            | Comments                                                                      |
|-----------------------------------------|-----------------|------------------|--------|------------------|-------------------------------------------------------------------------------|
| cc:flt1Min                              | CCFL1MIN        | never            | double | position number  | The minimum position of filter wheel 1 in real-world units (position number). |
| cc:flt1Name                             | CCFL1NAM        | start            | string |                  | User name of filter wheel 1 position                                          |
| cc:flt1Parked                           | CCFL1PRK        | never            | long   | 0/1              | Is Filter Wheel 1 parked? [YES=0   NO=1]                                      |
| cc:flt1Pos                              | CCFL1POS        | start            | double | position number  | The position of filter wheel 1 in real-world units (position number).         |
| cc:flt1RdtmVal                          | CCFL1RDT        | never            | long   | motor microsteps | Returned value from Filter Wheel 1 redatum command                            |
| cc:flt1Reject                           | CCFL1REJ        | never            | long   | 0/-1             | Combined Filter Wheel 1 CAR state [GOOD=0   REJECT=-1]                        |
| cc:flt1State                            | CCFL1STE        | never            | string |                  | Filter Wheel 1 SNL state                                                      |
| cc:flt1Units                            | CCFL1UNT        | start            | string |                  | Definition of real-world units for Filter Wheel 1 ("pos num").                |
| cc:flt2Datumed                          | CCFL2DTM        | never            | long   | 0/1              | Is Filter Wheel 2 datumed? [YES=0   NO=1]                                     |
| cc:flt2EngCyclic                        | CCFL2ECY        | never            | long   | 0/1              | Ignore limits for Filter Wheel 2? [YES=0   NO=1]                              |
| cc:flt2EngMax                           | CCFL2EMX        | never            | long   | motor microsteps | The maximum position of filter wheel 2 in engineering units                   |
| cc:flt2EngMin                           | CCFL2EMN        | never            | long   | motor microsteps | The minimum position of filter wheel 2 in engineering units                   |
| cc:flt2EngName                          | CCFL2ENM        | start            | string |                  | The engineering name of filter wheel 2 position                               |
| cc:flt2EngPos                           | CCFL2EPS        | start            | long   | motor microsteps | The position of filter wheel 2 in engineering units                           |
| cc:flt2Label                            | CCFL2LBL        | never            | string |                  | The FDSC field describes the Filter Wheel 2                                   |
| cc:flt2Max                              | CCFL2MAX        | never            | double | position number  | The maximum position of filter wheel 2 in real-world units (position number). |
| cc:flt2Min                              | CCFL2MIN        | never            | double | position number  | The minimum position of filter wheel 2 in real-world units (position number)  |
| cc:flt2Name                             | CCFL2NAM        | start            | string |                  | User name of filter wheel 2 position                                          |
| cc:flt2Parked                           | CCFL2PRK        | never            | long   | 0/1              | Is Filter Wheel 2 parked? [YES=0   NO=1]                                      |
| cc:flt2Pos                              | CCFL2POS        | start            | double | position number  | Filter wheel 2 position in real-world units (position number)                 |
| cc:flt2RdtmVal                          | CCFL2RDT        | never            | long   | motor microsteps | Returned value from Filter Wheel 2 redatum command                            |
| cc:flt2Reject                           | CCFL2REJ        | never            | long   | 0/-1             | Combined Filter Wheel 2 CAR state [GOOD=0   REJECT=-1]                        |
| cc:flt2State                            | CCFL2STE        | never            | string |                  | Filter Wheel 2 SNL state                                                      |
| cc:flt2Units                            | CCFL2UNT        | start            | string |                  | Definition of real-world units for Filter Wheel 2 ("pos num").                |



**TABLE 8.** NIRI Components Controller (CC) Component Status Information

| SIR record<br>("niri:" or "nirs:"<br>+) | FITS<br>Keyword | FITS<br>Included | Type   | Units            | Comments                                                                                                                |
|-----------------------------------------|-----------------|------------------|--------|------------------|-------------------------------------------------------------------------------------------------------------------------|
| cc:filt3Datumed                         | CCFL3DTM        | never            | long   | 0/1              | Is the pupil-plane wheel (pupil mask wheel / filter wheel 3) datumed? [YES=0   NO=1]                                    |
| cc:filt3EngCyclic                       | CCFL3ECY        | never            | long   | 0/1              | Ignore limits for the pupil-plane wheel (pupil mask wheel / filter wheel 3)? [YES=0   NO=1]                             |
| cc:filt3EngMax                          | CCFL3EMX        | never            | long   | motor microsteps | The maximum position of the pupil-plane wheel (pupil mask wheel / filter wheel 3) in engineering units                  |
| cc:filt3EngMin                          | CCFL3EMN        | never            | long   | motor microsteps | The minimum position of the pupil-plane wheel (pupil mask wheel / filter wheel 3) in engineering units                  |
| cc:filt3EngName                         | CCFL3ENM        | start            | string |                  | The engineering name of the pupil-plane wheel (pupil mask wheel / filter wheel 3) position                              |
| cc:filt3EngPos                          | CCFL3EPS        | start            | long   | motor microsteps | The position of the pupil-plane wheel (pupil mask wheel / filter wheel 3) in engineering units                          |
| cc:filt3Label                           | CCFL3LBL        | never            | string |                  | The FDSC field describes the pupil-plane wheel (pupil mask wheel / filter wheel 3)                                      |
| cc:filt3Max                             | CCFL3MAX        | never            | double | position number  | The maximum position of the pupil-plane wheel (pupil mask wheel / filter wheel 3) in real-world units (position number) |
| cc:filt3Min                             | CCFL3MIN        | never            | double | position number  | The minimum position of the pupil-plane wheel (pupil mask wheel / filter wheel 3) in real-world units (position number) |
| cc:filt3Name                            | CCFL3NAM        | start            | string |                  | User name of the pupil-plane wheel (pupil mask wheel / filter wheel 3) position                                         |
| cc:filt3Parked                          | CCFL3PRK        | never            | long   | 0/1              | Is the pupil-plane wheel (pupil mask wheel / filter wheel 3) parked? [YES=0   NO=1]                                     |
| cc:filt3Pos                             | CCFL3POS        | start            | double | position number  | The pupil plane wheel (pupil mask wheel / filter wheel 3) position in real-world units (position number)                |
| cc:filt3RdtmVal                         | CCFL3RDT        | never            | long   | motor microsteps | Returned value from the pupil-plane wheel (pupil mask wheel / filter wheel 3) redatum command                           |
| cc:filt3Reject                          | CCFL3REJ        | never            | long   | 0/-1             | Combined pupil-plane wheel (pupil mask wheel / filter wheel 3) CAR state [GOOD=0   REJECT=-1]                           |
| cc:filt3State                           | CCFL3STE        | never            | string |                  | The pupil-plane wheel (pupil mask wheel / filter wheel 3) SNL state                                                     |
| cc:filt3Units                           | CCFL3UNT        | start            | string |                  | Definition of real-world units for the pupil-plane wheel (pupil mask wheel / filter wheel 3) ("pos num").               |
| cc:focDatumed                           | CCFOCDTM        | never            | long   | 0/1              | Is the focus stage datumed? [YES=0   NO=1]                                                                              |

**TABLE 8.** NIRI Components Controller (CC) Component Status Information

| SIR record<br>("niri:" or "nirs:"<br>+) | FITS<br>Keyword | FITS<br>Included | Type   | Units            | Comments                                                                                  |
|-----------------------------------------|-----------------|------------------|--------|------------------|-------------------------------------------------------------------------------------------|
| cc:focEngCyclic                         | CCFOCECY        | never            | long   | 0/1              | Ignore limits for the focus stage? [YES=0   NO=1]                                         |
| cc:focEngMax                            | CCFOCEMX        | never            | long   | motor microsteps | The maximum position of the focus stage in engineering units                              |
| cc:focEngMin                            | CCFOCEMN        | never            | long   | motor microsteps | The minimum position of the focus stage in engineering units                              |
| cc:focEngName                           | CCFOCENM        | start            | string |                  | The engineering name of the focus stage position <sup>a</sup>                             |
| cc:focEngPos                            | CCFOCEPS        | start            | long   | motor microsteps | The position of the focus stage in engineering units                                      |
| cc:focLabel                             | CCFOCLBL        | never            | string |                  | The FDSC field describes the focus stage                                                  |
| cc:focMax                               | CCFOCMAX        | never            | double | mm               | The maximum position of the focus stage in real-world units (mm of translation).          |
| cc:focMin                               | CCFOCMIN        | never            | double | mm               | The minimum position of the focus stage in real-world units (mm of translation).          |
| cc:focName                              | CCFOCNAM        | start            | string |                  | User name of the focus stage position <sup>b</sup>                                        |
| cc:focParked                            | CCFOCPRK        | never            | long   | 0/1              | Is the focus stage parked? [YES=0   NO=1]                                                 |
| cc:focPos                               | CCFOCPOS        | start            | double | mm               | The focus stage position in real-world units (mm of translation).                         |
| cc:focRdtmVal                           | CCFOCRDT        | never            | long   | motor microsteps | Returned value from the focus stage redatum command                                       |
| cc:focReject                            | CCFOCREJ        | never            | long   | 0/-1             | Combined the focus stage CAR state [GOOD=0   REJECT=-1]                                   |
| cc:focState                             | CCFOCSTE        | never            | string |                  | the focus stage SNL state                                                                 |
| cc:focUnits                             | CCFOCUNT        | start            | string |                  | Definition of real-world units for the focus stage ("mm").                                |
| cc:foplDatumed                          | CCFPDTM         | never            | long   | 0/1              | Is the focal-plane mask wheel datumed? [YES=0   NO=1]                                     |
| cc:foplEngCyclic                        | CCFPECY         | never            | long   | 0/1              | Ignore limits for the focal-plane mask wheel? [YES=0   NO=1]                              |
| cc:foplEngMax                           | CCFPPEMX        | never            | long   | motor microsteps | The maximum position of the focal-plane mask wheel in engineering units                   |
| cc:foplEngMin                           | CCFPPEMN        | never            | long   | motor microsteps | The minimum position of the focal-plane mask wheel in engineering units                   |
| cc:foplEngName                          | CCFPENM         | start            | string |                  | The engineering name of the focal-plane mask wheel position                               |
| cc:foplEngPos                           | CCFPPEPS        | start            | long   | motor microsteps | The position of the focal-plane mask wheel in engineering units                           |
| cc:foplLabel                            | CCFPPLBL        | never            | string |                  | The FDSC field describes the focal-plane mask wheel                                       |
| cc:foplMax                              | CCFPMAX         | never            | double | position number  | The maximum position of the focal-plane mask wheel in real-world units (position number). |

**TABLE 8.** NIRI Components Controller (CC) Component Status Information

| SIR record<br>("niri:" or "nirs:"<br>+) | FITS<br>Keyword | FITS<br>Included | Type   | Units            | Comments                                                                                  |
|-----------------------------------------|-----------------|------------------|--------|------------------|-------------------------------------------------------------------------------------------|
| cc:foplMin                              | CCFPMIN         | never            | double | position number  | The minimum position of the focal-plane mask wheel in real-world units (position number). |
| cc:foplName                             | CCFPNAM         | start            | string |                  | User name of the focal-plane mask wheel position                                          |
| cc:foplParked                           | CCFPPRK         | never            | long   | 0/1              | Is the focal-plane mask wheel parked? [YES=0   NO=1]                                      |
| cc:foplPos                              | CCFPPOS         | start            | double | position number  | The focal-plane mask wheel position in real-world units (position number).                |
| cc:foplRdtmVal                          | CCFPRDT         | never            | long   | motor microsteps | Returned value from the focal-plane mask wheel redatum command                            |
| cc:foplReject                           | CCFPREJ         | never            | long   | 0/-1             | Combined the focal-plane mask wheel CAR state [GOOD=0   REJECT=-1]                        |
| cc:foplState                            | CCFPSTE         | never            | string |                  | the focal-plane mask wheel SNL state                                                      |
| cc:foplUnits                            | CCFPUNT         | start            | string |                  | Definition of real-world units for the focal-plane mask wheel ("pos num").                |
| cc:puvwDatumed                          | CCPVDTM         | never            | long   | 0/1              | Is the pupil viewer datumed? [YES=0   NO=1]                                               |
| cc:puvwEngCyclic                        | CCPVECY         | never            | long   | 0/1              | Ignore limits for the pupil viewer? [YES=0   NO=1]                                        |
| cc:puvwEngMax                           | CCPVEMX         | never            | long   | motor microsteps | The maximum position of the pupil viewer in engineering units                             |
| cc:puvwEngMin                           | CCPVEMN         | never            | long   | motor microsteps | The minimum position of the pupil viewer in engineering units                             |
| cc:puvwEngName                          | CCPVENM         | start            | string |                  | The engineering name of the pupil viewer position                                         |
| cc:puvwEngPos                           | CCPVEPS         | start            | long   | motor microsteps | The position of the pupil viewer in engineering units                                     |
| cc:puvwLabel                            | CCPVLBL         | never            | string |                  | The FDSC field describes the pupil viewer                                                 |
| cc:puvwMax                              | CCPVMAX         | never            | double | position number  | The maximum position of the pupil viewer in real-world units (position number).           |
| cc:puvwMin                              | CCPVMIN         | never            | double | position number  | The minimum position of the pupil viewer in real-world units (position number).           |
| cc:puvwName                             | CCPVNAM         | start            | string |                  | User name of the pupil viewer position                                                    |
| cc:puvwParked                           | CCPVPRK         | never            | long   | 0/1              | Is the pupil viewer parked? [YES=0   NO=1]                                                |
| cc:puvwPos                              | CCPVPOS         | start            | double | position number  | The pupil viewer position in real-world units (position number).                          |
| cc:puvwRdtmVal                          | CCPVRDT         | never            | long   | motor microsteps | Returned value from the pupil viewer redatum command                                      |
| cc:puvwReject                           | CCPVREJ         | never            | long   | 0/-1             | Combined the pupil viewer CAR state [GOOD=0   REJECT=-1]                                  |
| cc:puvwState                            | CCPVSTE         | never            | string |                  | the pupil viewer SNL state                                                                |

**TABLE 8.** NIRI Components Controller (CC) Component Status Information

| SIR record<br>("niri:" or "nirs:"<br>+) | FITS<br>Keyword | FITS<br>Included | Type   | Units            | Comments                                                                               |
|-----------------------------------------|-----------------|------------------|--------|------------------|----------------------------------------------------------------------------------------|
| cc:pupwUnits                            | CCPVUNT         | start            | string |                  | Definition of real-world units for the pupil viewer ("pos num").                       |
| cc:spltDatumed                          | CCSPDTM         | never            | long   | 0/1              | Is the beam-splitter wheel datumed? [YES=0   NO=1]                                     |
| cc:spltEngCyclic                        | CCSPECY         | never            | long   | 0/1              | Ignore limits for the beam-splitter wheel? [YES=0   NO=1]                              |
| cc:spltEngMax                           | CCSPEMX         | never            | long   | motor microsteps | The maximum position of the beam-splitter wheel in engineering units                   |
| cc:spltEngMin                           | CCSPEMN         | never            | long   | motor microsteps | The minimum position of the beam-splitter wheel in engineering units                   |
| cc:spltEngName                          | CCSPENM         | start            | string |                  | The engineering name of the beam-splitter wheel position                               |
| cc:spltEngPos                           | CCSPEPS         | start            | long   | motor microsteps | The position of the beam-splitter wheel in engineering units                           |
| cc:spltLabel                            | CCSPLBL         | never            | string |                  | The FDSC field describes the beam-splitter wheel                                       |
| cc:spltMax                              | CCSPMAX         | never            | double | position number  | The maximum position of the beam-splitter wheel in real-world units (position number). |
| cc:spltMin                              | CCSPMIN         | never            | double | position number  | The minimum position of the beam-splitter wheel in real-world units (position number). |
| cc:spltName                             | CCSPNAM         | start            | string |                  | User name of the beam-splitter wheel position                                          |
| cc:spltParked                           | CCSPPRK         | never            | long   | 0/1              | Is the beam-splitter wheel parked? [YES=0   NO=1]                                      |
| cc:spltPos                              | CCSPPOS         | start            | double | position number  | The beam-splitter wheel position in real-world units (position number).                |
| cc:spltRdtmVal                          | CCSPRDT         | never            | long   | motor microsteps | Returned value from the beam-splitter wheel redatum command                            |
| cc:spltReject                           | CCSPREJ         | never            | long   | 0/-1             | Combined the beam-splitter wheel CAR state [GOOD=0   REJECT=-1]                        |
| cc:spltState                            | CCSPSTE         | never            | string |                  | the beam-splitter wheel SNL state                                                      |
| cc:spltUnits                            | CCSPUNT         | start            | string |                  | Definition of real-world units (position number). for Beam Splitter                    |
| cc:ster1Datumed                         | CCST1DTM        | never            | long   | 0/1              | Is steering mirror 1 datumed? [YES=0   NO=1]                                           |
| cc:ster1EngCyclic                       | CCST1ECY        | never            | long   | 0/1              | Ignore limits for steering mirror 1? [YES=0   NO=1]                                    |
| cc:ster1EngMax                          | CCST1EMX        | never            | long   | motor microsteps | The maximum position of steering mirror 1 in engineering units                         |
| cc:ster1EngMin                          | CCST1EMN        | never            | long   | motor microsteps | The minimum position of steering mirror 1 in engineering units                         |

**TABLE 8.** NIRI Components Controller (CC) Component Status Information

| SIR record<br>("niri:" or "nirs:"<br>+) | FITS<br>Keyword | FITS<br>Included | Type   | Units            | Comments                                                                         |
|-----------------------------------------|-----------------|------------------|--------|------------------|----------------------------------------------------------------------------------|
| cc:ster1EngName                         | CCST1ENM        | start            | string |                  | The engineering name of steering mirror 1 position                               |
| cc:ster1EngPos                          | CCST1EPS        | start            | long   | motor microsteps | The position of steering mirror 1 in engineering units                           |
| cc:ster1Label                           | CCST1LBL        | never            | string |                  | The FDSC field describes the steering mirror 1                                   |
| cc:ster1Max                             | CCST1MAX        | never            | double | position number  | The maximum position of steering mirror 1 in real-world units (position number). |
| cc:ster1Min                             | CCST1MIN        | never            | double | position number  | The minimum position of steering mirror 1 in real-world units (position number). |
| cc:ster1Name                            | CCST1NAM        | start            | string |                  | User name of steering mirror 1 position                                          |
| cc:ster1Parked                          | CCST1PRK        | never            | long   | 0/1              | Is steering mirror 1 parked? [YES=0   NO=1]                                      |
| cc:ster1Pos                             | CCST1POS        | start            | double | position number  | Steering mirror 1 position in real-world units (position number).                |
| cc:ster1RdtmVal                         | CCST1RDT        | never            | long   | motor microsteps | Returned value from steering mirror 1 redatum command                            |
| cc:ster1Reject                          | CCST1REJ        | never            | long   | 0/-1             | Combined steering mirror 1 CAR state [GOOD=0   REJECT=-1]                        |
| cc:ster1State                           | CCST1STE        | never            | string |                  | steering mirror 1 SNL state                                                      |
| cc:ster1Units                           | CCST1UNT        | start            | string |                  | Definition of real-world units for steering mirror 1 ("pos num").                |
| cc:ster2Datumed                         | CCST2DTM        | never            | long   | 0/1              | Is steering mirror 2 datumed? [YES=0   NO=1]                                     |
| cc:ster2EngCyclic                       | CCST2ECY        | never            | long   | 0/1              | Ignore limits for steering mirror 2? [YES=0   NO=1]                              |
| cc:ster2EngMax                          | CCST2EMX        | never            | long   | motor microsteps | The maximum position of steering mirror 2 in engineering units                   |
| cc:ster2EngMin                          | CCST2EMN        | never            | long   | motor microsteps | The minimum position of steering mirror 2 in engineering units                   |
| cc:ster2EngName                         | CCST2ENM        | start            | string |                  | The engineering name of steering mirror 2 position                               |
| cc:ster2EngPos                          | CCST2EPS        | start            | long   | motor microsteps | The position of steering mirror 2 in engineering units                           |
| cc:ster2Label                           | CCST2LBL        | never            | string |                  | The FDSC field describes the steering mirror 2                                   |
| cc:ster2Max                             | CCST2MAX        | never            | double | position number  | The maximum position of steering mirror 2 in real-world units (position number). |
| cc:ster2Min                             | CCST2MIN        | never            | double | position number  | The minimum position of steering mirror 2 in real-world units (position number). |
| cc:ster2Name                            | CCST2NAM        | start            | string |                  | User name of steering mirror 2 position                                          |
| cc:ster2Parked                          | CCST2PRK        | never            | long   | 0/1              | Is steering mirror 2 parked? [YES=0   NO=1]                                      |

**TABLE 8.** NIRI Components Controller (CC) Component Status Information

| SIR record<br>("niri:" or "nirs:"<br>+) | FITS<br>Keyword | FITS<br>Included | Type   | Units            | Comments                                                          |
|-----------------------------------------|-----------------|------------------|--------|------------------|-------------------------------------------------------------------|
| cc:ster2Pos                             | CCST2POS        | start            | double | position number  | Steering mirror 2 position in real-world units (position number). |
| cc:ster2RdtmVal                         | CCST2RDT        | never            | long   | motor microsteps | Returned value from steering mirror 2 redatum command             |
| cc:ster2Reject                          | CCST2REJ        | never            | long   | 0/-1             | Combined steering mirror 2 CAR state [GOOD=0   REJECT=-1]         |
| cc:ster2State                           | CCST2STE        | never            | string |                  | steering mirror 2 SNL state                                       |
| cc:ster2Units                           | CCST2UNT        | start            | string |                  | Definition of real-world units for steering mirror 2 ("pos num"). |
| cc:sterDatumed                          | CCSTD TM        | never            | long   | 0/1              | Are steering mirrors datumed? [YES=0   NO=1]                      |
| cc:sterLabel                            | CCSTLBL         | never            | string |                  | The FDSC field describes the steering mirrors                     |
| cc:sterName                             | CCSTNAM         | start            | string |                  | steering mirrors name of current position                         |
| cc:sterParked                           | CCSTPRK         | never            | long   | 0/1              | Are steering mirrors parked? [YES=0   NO=1]                       |
| cc:sterReject                           | CCSTREJ         | never            | long   | 0/1              | Combined reject state for steering mirrors                        |

- For the focus stage, the position returned by cc:focEngName will often be "INVALID". This should not be treated as an error, merely as an indication that only a few positions (e.g., the PARK position) have names.
- For the focus stage, the position returned by cc:focName will often be "INVALID". This should not be treated as an error, merely as an indication that only a few positions (e.g., the PARK position) have names.

#### 2-3.3.2.4 Temperature Control and Monitoring Subsystem, Status Information

The following table summarizes the provided information for the temperature subsystem.

**TABLE 9.** Components Controller (CC) Temperature Control and Monitoring Subsystem, Status Information

| SIR record<br>("niri:" or "nirs:"<br>+) | FITS<br>Keyword | FITS<br>Included | Type   | Units                | Comments                                 |
|-----------------------------------------|-----------------|------------------|--------|----------------------|------------------------------------------|
| cc:tmpC1Setp                            | CCTPC1S         | start            | double | K                    | OIWFS Set-point temperature              |
| cc:tmpC1Tmp                             | CCTPC1T         | never            | double | K                    | OIWFS temperature                        |
| cc:tmpC2Setp                            | CCTPC2S         | start            | double | K                    | Imager buffer-mass set-point temperature |
| cc:tmpC2Tmp                             | CCTPC2T         | never            | double | K                    | Imager buffer-mass temperature           |
| cc:tmpC3Setp                            | CCTPC3S         | start            | double | K                    | Cold-plate center set-point temperature  |
| cc:tmpC3Tmp                             | CCTPC3T         | never            | double | K                    | Cold-plate center temperature            |
| cc:tmpCoolBusy                          | CCTPCBY         | never            | string |                      | Cooling motor activity                   |
| cc:tmpCoolName                          | CCTPCNM         | never            | string |                      | Cooling motor rate                       |
| cc:tmpCoolRate1                         | CCTPCR1         | never            | double | motor-microsteps / s | Cooling motor 1 rate                     |
| cc:tmpCoolRate2                         | CCTPCR2         | never            | double | motor-microsteps / s | Cooling motor 2 rate                     |

**TABLE 9.** Components Controller (CC) Temperature Control and Monitoring Subsystem, Status Information

| SIR record<br>("niri:" or<br>"nirs:" +) | FITS<br>Keyword | FITS<br>Included | Type   | Units | Comments                                           |
|-----------------------------------------|-----------------|------------------|--------|-------|----------------------------------------------------|
| cc:tmpLabel                             | CCTPLBL         | never            | string |       | The FDSC field describes the temperature Subsystem |
| cc:tmpS1Tmp1                            | CCTPST1         | never            | double | K     | The temperature at the top of strap                |
| cc:tmpS1Tmp2                            | CCTPST2         | never            | double | K     | The temperature at the bottom of strap             |
| cc:tmpS1Tmp3                            | CCTPST3         | never            | double | K     | The engineering test point temperature             |
| cc:tmpS1Tmp4                            | CCTPST4         | never            | double | K     | The temperature at the cold-plate edge             |
| cc:tmpS1Tmp5                            | CCTPST5         | never            | double | K     | An unused temperature-sensor channel (5)           |
| cc:tmpS1Tmp6                            | CCTPST6         | never            | double | K     | An unused temperature-sensor channel (6)           |
| cc:tmpS1Tmp7                            | CCTPST7         | never            | double | K     | An unused temperature-sensor channel (7)           |
| cc:tmpS1Tmp8                            | CCTPST8         | never            | double | K     | An unused temperature-sensor channel (8)           |

### 2-3.3.2.5 Component Health Information

The following table summarizes the health information records. All records return one of the strings: "GOOD", "WARNING", or "BAD". Each record describes one aspect of the health of a component.

**TABLE 10.** NIRI Components Controller (CC) Component Health Information

| SIR record ("niri:" or<br>"nirs:" +) | FITS<br>Keyword | FITS<br>Included | Type   | Units | Comments                                                   |
|--------------------------------------|-----------------|------------------|--------|-------|------------------------------------------------------------|
| cc:covDatumHealth                    |                 | never            | string |       | The health of the window cover datum state                 |
| cc:covHallHealth                     |                 | never            | string |       | The health of the window cover Hall-effect sensors         |
| cc:covHealth                         |                 | never            | string |       | The overall health of Window Cover                         |
| cc:covModuleHealth                   |                 | never            | string |       | The health of the window-cover stepper motor driver module |
| cc:covPosErrHealth                   |                 | never            | string |       | The health of the window cover backlash correction         |
| cc:covSnIHealth                      |                 | never            | string |       | The health of the window cover State code                  |
| cc:flt1DatumHealth                   |                 | never            | string |       | The health of the filter wheel 1 datum state               |
| cc:flt1HallHealth                    |                 | never            | string |       | The health of the filter wheel 1 Hall-effect sensors       |
| cc:flt1Health                        |                 | never            | string |       | The overall health of filter wheel 1                       |
| cc:flt1ModuleHealth                  |                 | never            | string |       | The health of filter wheel 1 driver module                 |
| cc:flt1PosErrHealth                  |                 | never            | string |       | The health of the filter wheel 1 backlash correction       |
| cc:flt1SnIHealth                     |                 | never            | string |       | The health of the filter wheel 1 state code                |
| cc:flt2DatumHealth                   |                 | never            | string |       | The health of the filter wheel 2 datum state               |
| cc:flt2HallHealth                    |                 | never            | string |       | The health of the filter wheel 2 Hall-effect sensors       |
| cc:flt2Health                        |                 | never            | string |       | The overall health of filter wheel 2                       |
| cc:flt2ModuleHealth                  |                 | never            | string |       | The health of filter wheel 2 driver module                 |
| cc:flt2PosErrHealth                  |                 | never            | string |       | The health of the filter wheel 2 backlash correction       |
| cc:flt2SnIHealth                     |                 | never            | string |       | The health of filter wheel 2 state code                    |

**TABLE 10.** NIRI Components Controller (CC) Component Health Information

| SIR record ("niri:" or "nirs:" +) | FITS Keyword | FITS Included | Type   | Units | Comments                                                                                    |
|-----------------------------------|--------------|---------------|--------|-------|---------------------------------------------------------------------------------------------|
| cc:filt3DatumHealth               |              | never         | string |       | The health of the pupil-plane wheel (pupil mask wheel / filter wheel 3) datum state         |
| cc:filt3HallHealth                |              | never         | string |       | The health of the pupil-plane wheel (pupil mask wheel / filter wheel 3) Hall-effect sensors |
| cc:filt3Health                    |              | never         | string |       | The overall health of the pupil-plane wheel (pupil mask wheel / filter wheel 3)             |
| cc:filt3ModuleHealth              |              | never         | string |       | The health of the pupil-plane wheel (pupil mask wheel / filter wheel 3) driver module       |
| cc:filt3PosErrHealth              |              | never         | string |       | The health of the pupil plane wheel (pupil mask wheel / filter wheel 3) backlash correction |
| cc:filt3SnIHealth                 |              | never         | string |       | The health of the pupil-plane wheel (pupil mask wheel / filter wheel 3) state code          |
| cc:focDatumHealth                 |              | never         | string |       | The health of the focus stage datum state                                                   |
| cc:focHallHealth                  |              | never         | string |       | The health of the focus stage Hall-effect sensors                                           |
| cc:focHealth                      |              | never         | string |       | The overall health of the focus stage                                                       |
| cc:focModuleHealth                |              | never         | string |       | The health of the focus stage driver module                                                 |
| cc:focPosErrHealth                |              | never         | string |       | The health of the focus stage backlash correction                                           |
| cc:focSnIHealth                   |              | never         | string |       | The health of the focus stage state code                                                    |
| cc:foplDatumHealth                |              | never         | string |       | The health of the focal-plane mask wheel datum state                                        |
| cc:foplHallHealth                 |              | never         | string |       | The health of the focal-plane mask wheel Hall-effect sensors                                |
| cc:foplHealth                     |              | never         | string |       | The overall health of the focal-plane mask wheel                                            |
| cc:foplModuleHealth               |              | never         | string |       | The health of the focal-plane mask wheel driver module                                      |
| cc:foplPosErrHealth               |              | never         | string |       | The health of the focal-plane mask wheel backlash correction                                |
| cc:foplSnIHealth                  |              | never         | string |       | The health of the focal-plane mask wheel state code                                         |
| cc:puvwDatumHealth                |              | never         | string |       | The health of the pupil viewer datum state                                                  |
| cc:puvwHallHealth                 |              | never         | string |       | The health of the pupil viewer Hall-effect sensors                                          |
| cc:puvwHealth                     |              | never         | string |       | The overall health of the pupil viewer                                                      |
| cc:puvwModuleHealth               |              | never         | string |       | The health of the pupil viewer driver module                                                |
| cc:puvwPosErrHealth               |              | never         | string |       | The health of the pupil viewer backlash correction                                          |
| cc:puvwSnIHealth                  |              | never         | string |       | The health of the pupil viewer state code                                                   |
| cc:spltDatumHealth                |              | never         | string |       | The health of the beam-splitter wheel datum state                                           |
| cc:spltHallHealth                 |              | never         | string |       | The health of the beam-splitter wheel Hall-effect sensors                                   |
| cc:spltHealth                     |              | never         | string |       | The overall health of the beam-splitter wheel                                               |
| cc:spltModuleHealth               |              | never         | string |       | The health of the beam-splitter wheel driver module                                         |
| cc:spltPosErrHealth               |              | never         | string |       | The health of the beam-splitter wheel backlash correction                                   |
| cc:spltSnIHealth                  |              | never         | string |       | The health of the beam-splitter wheel state code                                            |
| cc:ster1DatumHealth               |              | never         | string |       | The health of the steering mirror 1 datum state                                             |
| cc:ster1HallHealth                |              | never         | string |       | The health of the steering mirror 1 Hall-effect sensors                                     |



**TABLE 10.** NIRI Components Controller (CC) Component Health Information

| SIR record ("niri:" or "nirs:" +) | FITS Keyword | FITS Included | Type   | Units | Comments                                                |
|-----------------------------------|--------------|---------------|--------|-------|---------------------------------------------------------|
| cc:ster1Health                    |              | never         | string |       | The overall health of steering mirror 1                 |
| cc:ster1ModuleHealth              |              | never         | string |       | The health of steering mirror 1 driver module           |
| cc:ster1PosErrHealth              |              | never         | string |       | The health of the steering mirror 1 backlash correction |
| cc:ster1SnIHealth                 |              | never         | string |       | The health of steering mirror 1 state code              |
| cc:ster2DatumHealth               |              | never         | string |       | The health of the steering mirror 2 datum state         |
| cc:ster2HallHealth                |              | never         | string |       | The health of the steering mirror 2 Hall-effect sensors |
| cc:ster2Health                    |              | never         | string |       | The overall health of steering mirror 2                 |
| cc:ster2ModuleHealth              |              | never         | string |       | The health of steering mirror 2 driver module           |
| cc:ster2PosErrHealth              |              | never         | string |       | The health of the steering mirror 2 backlash correction |
| cc:ster2SnIHealth                 |              | never         | string |       | The health of steering mirror 2 state code              |
| cc:sterHealth                     |              | never         | string |       | The overall health of the steering mirrors              |

#### 2-3.3.2.6 The Temperature Control and Monitoring Subsystem Health Information

Table 11 summarizes the health messages involving the temperature monitoring and control subsystem. All records return one of the strings: "GOOD", "WARNING", or "BAD".

**TABLE 11.** NIRI Components Controller (CC) Temperature Subsystem Health Information

| SIR record ("niri:" or "nirs:" +) | FITS Keyword | FITS Included | Type   | Units | Comments                                                                                |
|-----------------------------------|--------------|---------------|--------|-------|-----------------------------------------------------------------------------------------|
| cc:tmpC1Health                    |              | never         | string |       | The health of temperature controller 1, controlling the On-Instrument Wavefront Sensor. |
| cc:tmpC2Health                    |              | never         | string |       | The health of temperature controller 2, controlling the imager buffer mass.             |
| cc:tmpC3Health                    |              | never         | string |       | The health of temperature controller 3, controlling the center of the cold plate.       |
| cc:tmpCoolModeHealth              |              | never         | string |       | The health of cooling motor on/off state                                                |
| cc:tmpHealth                      |              | never         | string |       | The overall health of the temperature subsystem                                         |
| cc:tmpModeHealth                  |              | never         | string |       | The health of Heater / motor mode                                                       |
| cc:tmpS1Health                    |              | never         | string |       | The health of the temperature-sensor                                                    |
| cc:tmpTmpCheckHealth              |              | never         | string |       | The health of the temperature-in-range indicator                                        |

### 2-3.3.2.7 The World Coordinate System Interface

Table 12 summarizes the available World Coordinate System information.

**TABLE 12.** NIRI Components Controller (CC) World Coordinate System Information<sup>a</sup>

| SIR record ("niri:" or "nirs:" +) | FITS Keyword | FITS Included | Type   | Units | Comments                                                   |
|-----------------------------------|--------------|---------------|--------|-------|------------------------------------------------------------|
| cc:wcs:ctype1                     | CTYPE1       | Start         | string |       | Projection type for axis 1                                 |
| cc:wcs:crpix1                     | CRPIX1       | Start         | double |       | Pixel coordinate reference for axis 1                      |
| cc:wcs:crval1                     | CRVAL1       | Start         | double |       | World coordinate reference for axis 1                      |
| cc:wcs:ctype2                     | CTYPE2       | Start         | string |       | Projection type for axis 2                                 |
| cc:wcs:crpix2                     | CRPIX2       | Start         | double |       | Pixel coordinate reference for axis 2                      |
| cc:wcs:crval2                     | CRVAL2       | Start         | double |       | World coordinate reference for axis 2                      |
| cc:wcs:cd11                       | CD1_1        | Start         | double |       | $x_i$ rotation/skew matrix element                         |
| cc:wcs:cd12                       | CD1_2        | Start         | double |       | $x_j$ rotation/skew matrix element                         |
| cc:wcs:cd21                       | CD2_1        | Start         | double |       | $y_i$ rotation/skew matrix element                         |
| cc:wcs:cd22                       | CD2_2        | Start         | double |       | $y_j$ rotation/skew matrix element                         |
| cc:wcs:radecsys                   | RADECSYS     | Start         | string |       | Type of RA/Dec (for celestial coordinates)                 |
| cc:wcs:equinox                    | EQUINOX      | Start         | double |       | Epoch of mean equator and equinox (celestial coordinates). |
| cc:wcs:mjdobs                     | MJD-OBS      | Start         | double |       | Modified Julian Date                                       |

a. Derived from [38].

### 2-3.3.3 Position names

Table 13 contains a list of valid position names. Note that these names are case sensitive and must be entered exactly as presented here, including underscores. In addition to the listed positions, all mechanisms respond to the special names "datum" and "park" which may or may not be distinct from the listed positions.

**TABLE 13.** Position Names (observing commands)

| Component                          | Position Names | Engineering Position <sup>a</sup> | Comments |
|------------------------------------|----------------|-----------------------------------|----------|
| Window Cover                       | open           | (tbd)                             |          |
|                                    | closed         | (tbd)                             |          |
| Filter Wheel 1                     | See [36]       |                                   |          |
| Filter Wheel 2                     | See [36]       |                                   |          |
| Pupil Plane Wheel (Filter Wheel 3) | See [36]       |                                   |          |
| Focal Plane Mask Wheel             | See [36]       |                                   |          |
| Pupil Viewer                       | in             | pos01                             |          |
|                                    | out            | pos00                             |          |

**TABLE 13.** Position Names (observing commands)

| Component              | Position Names | Engineering Position <sup>a</sup> | Comments                |
|------------------------|----------------|-----------------------------------|-------------------------|
| Steering Mirrors       | f6             | (comp)                            | f/6                     |
|                        | f13.9          | (comp)                            | f/13.9                  |
|                        | f32            | (comp)                            | f/32                    |
| Steering Mirror 1      | f6             | pos03                             | f/6                     |
|                        | f13.9          | pos02                             | f/13.9                  |
|                        | f32            | pos00                             | f/32                    |
| Steering Mirror 2      | f6             | pos03                             | f/6                     |
|                        | f13.9          | pos02                             | f/13.9                  |
|                        | f32            | pos00                             | f/32                    |
| Entrance Beam Splitter | f6             | pos00                             | f/6                     |
|                        | f13.9          | pos01                             | f/13.9                  |
|                        | f32            | pos02                             | f/32                    |
|                        | pinhole_mask   | pos03                             | Pin-Hole Mask           |
| Focus Stage            | Neutral        | pos00                             | Center of travel        |
|                        | Minimum        | pos01                             | Minimum travel location |
|                        | Maximum        | pos02                             | Maximum travel location |

a. “(tbd)” means that the mapping from user positions to engineering positions has yet to be determined. “(comp)” means that the positions are mapped into component positions, and not sent directly to a mechanism.

## 2–3.4 Detailed Command Description

A general table of the commands accepted by the NIRI control software, together with their arguments, is given in Table 2 on page 7. This section contains a detailed description of each of the commands that can be sent between the OCS and the NIRI control software. It is based on the required behavior of the commands specified in [21] and [32].

Any directive asserted on the top-level APPLY record will be forwarded to all the CAD records in the sequencer, CC, and DC (perhaps via lower-level APPLY records). Only the marked CAD records will respond. If the directive is accepted it will cause the NIRI control software to change its configuration. The “applyC” CAR record will go “BUSY” while the NIRI control software is changing its configuration and go to “IDLE” if completed successfully or to “ERR” if the configuration change fails.

### 2–3.4.1 Sequence Commands

#### 2–3.4.1.1 TEST

On receipt of this command, the NIRI IS will insure that an observation is not taking place, and then will forward this command to the DC and CC. The “testC” CAR record will go “BUSY” while the tests are being carried out and go to “IDLE” if completed successfully or to “ERR” if the tests fail.

The CC accepts, but ignores this command.

The behavior of the DC is documented in [18].

*Because the TEST command is not permitted to move any mechanisms, it is unable to detect any problems with the components controller. Use the DIAGNOSE and REDATUM commands instead.*

#### 2-3.4.1.2    INIT

On receipt of this command, the NIRI IS will insure that an observation is not taking place. It will then execute its initialization sequence, in which it will: Read its hardware setup files and lookup tables, and reset itself to the start-up state. This command will be forwarded to the CC and the DC. The “initC” CAR record will go “BUSY” while the initialization is being carried out and go to “IDLE” if completed successfully or to “ERR” if the initialization fails.

The CC will read its hardware setup files and lookup tables, and reset itself to the start-up state. No mechanisms will be moved.

The behavior of the DC is documented in [18].

#### 2-3.4.1.3    DATUM

On receipt of this command, the NIRI IS will insure that an observation is not taking place and then forward the command to the CC and DC. The “datumC” CAR record will go “BUSY” while the datum search is being carried out and go to “IDLE” if completed successfully or to “ERR” if the datum searched fails.

The CC will locate the datum (i.e. index, home or reference point) for all its mechanisms.

The behavior of the DC is documented in [18].

#### 2-3.4.1.4    VERIFY

On receipt of this command, the NIRI IS will insure that an observation is not taking place and then forward the command to the CC and DC. The “verifyC” CAR record will be changed to “BUSY”. The command will complete, and the “verifyC” CAR changed back to “IDLE” when the state of the NIRI control software can be changed safely.

The CC accepts, but ignores this command.

The behavior of the DC is documented in [18].

#### 2-3.4.1.5    ENDVERIFY

On receipt of this command, the NIRI IS will insure that an observation is not taking place and then forward the command to the CC and DC. The command will change the “endVerifyC” CAR status to “BUSY” and then back to “IDLE”.

The CC accepts but ignores this command.

The behavior of the DC is documented in [18].

#### 2-3.4.1.6    GUIDE

On receipt of this command, the NIRI IS will insure that an observation is not taking place. It will then set a flag as a reminder that guiding is taking place (as this information may affect its interaction with any on-instrument wavefront sensor). The sequencer will also forward this command on to the CC and DC. The “guideC” CAR record will be changed to “BUSY” and will change to “IDLE” when all the mechanisms are tracking within specification.

The CC accepts but ignores this command.

The behavior of the DC is documented in [18].

#### 2-3.4.1.7    ENDGUIDE

On receipt of this command, the NIRI IS will insure that an observation is not taking place. It will then reset the guiding flag to indicate guiding has stopped. The sequencer will then forward this command on to the CC and DC. The “endGuideC” CAR record will be changed to “BUSY” and then back to “IDLE”.

The CC accepts but ignores this command.

The behavior of the DC is documented in [18].

#### 2-3.4.1.8    SETUP

On receipt of this command, the NIRI IS will insure that an observation is not taking place and then issue the necessary commands to the DC to accomplish the setup operation. This assumes that the appropriate parameters have already been written to the GNAAC detector controller. The IS will reject the command if its control software or the CC control software is still configuring its mechanisms. If accepted, the “setupC” CAR record will be changed to “BUSY”. It will go back to “IDLE” when the setup observation completes (or “ERR” if the observation was not successful).

#### 2-3.4.1.9    OBSERVE

On receipt of this command, the NIRI IS will insure that an observation is not taking place and then forward the command to the DC. The IS will reject the command if its control software or the CC control software is still configuring its mechanisms. If accepted, the “observeC” CAR record will be changed to “BUSY”. It will go back to “IDLE” when the data set observation completes (or “ERR” if the observation was not successful).

The CC will not respond to this command.

The behavior of the DC is documented in [18].

#### 2-3.4.1.10    ENDOBSERVE

On receipt of this command, the NIRI IS will forward the command to the DC. No changes will be made to the “observeC” CAR record.

The CC will not respond to this command.

The behavior of the DC is documented in [18].

2-3.4.1.11 PAUSE

On receipt of this command, the NIRI IS will forward the command to the DC.

The CC will not respond to this command.

In the DC, this will cause the observeC CAR record to change to “PAUSED”. Further details of the behavior of the DC is documented in [18].

2-3.4.1.12 CONTINUE

On receipt of this command, the NIRI IS will forward the command to the DC. The command will be rejected if an observation has not been paused. If successful, the “observeC” CAR record will be changed to “BUSY”.

The CC will not respond to this command.

The behavior of the DC is documented in [18].

2-3.4.1.13 STOP

On receipt of this command, the NIRI IS will forward the command to the DC. The command will be rejected if an observation is not being made. If successful, the “observeC” CAR record will be changed to “IDLE”.

The CC will not respond to this command.

The behavior of the DC is documented in [18].

2-3.4.1.14 ABORT

On receipt of this command, the NIRI IS will forward the command to the DC. If successful, the “observeC” CAR record will be changed to “IDLE”.

The CC will not respond to this command.

The behavior of the DC is documented in [18].

2-3.4.1.15 PARK

On receipt of this command, the NIRI IS will insure that an observation is not taking place and then forward the command to the DC and the CC.

The CC will move its mechanisms to a configuration in which it can safely be switched off. It will utilize all the low-level CAD records dedicated to parking the individual mechanisms. The behavior of the DC is documented in [18]. The “parkC” CAR record will go “BUSY” while the mechanisms are being parked and go to “IDLE” if completed successfully or to “ERR” if any mechanism fails to park successfully.

The behavior of the DC is documented in [18].

**2-3.4.2 General Engineering Commands**2-3.4.2.1 REBOOT

On receipt of this command, the NIRI control software will reboot its IOC and go through the same start-up procedure it would normally do when first switched on. It will reboot into simulation mode NONE and debug mode NONE. The prototype and simulator will reboot

into simulation mode FULL and debug mode NONE. There is no CAR record associated with this command because connection with the IOC will be lost while it reboots.

*This command will also reboot the OIWFS components controller.*

#### 2-3.4.2.2    DEBUG

On receipt of this command, the NIRI control software will change to the given debug mode (NONE, MIN or FULL). This mode determines the amount and frequency of information logged by the NIRI control software, as described in the “Debugging Modes” section of ICD 1a, [24]. The “debugC” CAR record will go briefly “BUSY” and then back to “IDLE”.

#### 2-3.4.2.3    FLUSH

This command will be forwarded to the DC.

The CC does not recognize this command.

On receipt of this command, the DC control software will transmit whatever is contained within its data buffers to the DHS data storage server for storage to disk. The action can be used for recovering data after a software problem. The “flushC” CAR record will go “BUSY” while the data are being transmitted and switch to “IDLE” if the data transfer succeeded or “ERR” if it failed. The behavior of the DC is documented in [18].

#### 2-3.4.2.4    Temperature-controller commands

All temperature-controller commands are part of the engineering interface, and are not within the scope of this document.

### **2-3.4.3    NIRI Standard Mechanisms**

NIRI will have two filter wheels, a pupil-plane wheel (pupil-mask wheel / additional filter wheel), a focal-plane mask wheel, two steering mirrors (treated as a single, multi-axis mechanism), a focus stage, a window cover, and a pupil viewer. The filter wheels, pupil-plane wheel, focal-plane mask wheel, steering mirrors, window cover, and pupil viewer are considered “discrete” mechanisms, because they normally are moved only to one of a small number of locations. The focus stage is considered a continuous mechanism, because it can move to a wide range of locations.

The following commands will act on all of the above mentioned mechanisms; however, as noted below, the use of certain commands with certain mechanisms is discouraged.

#### 2-3.4.3.1    Sel—Select Position by Name

This is the principal interface for all continuous mechanisms. It is provided as a convenience for discrete mechanisms. Note that for multi-axis components (e.g., the steering mirrors), there are commands that act on the combined components and on the individual mechanisms.

On receipt of this command, the NIRI CC control software will:

Ensure an observation is not taking place.  
 Translate desired position name into an engineering position with the aid of a look-up table.  
 Look up the expected value for the Hall-effect sensors at the desired position location.  
 Check to insure that protective interlocks are not set.  
 Wait until the number of active mechanisms drops below the number of mechanisms that the power supply can support.  
 Move the mechanism to the engineering position corresponding to the desired position.  
 Verify that the expected sensor value matches the measured value.

#### 2-3.4.3.2 Move—Move to specified Position

This is the principal interface for all discrete components. It is available for continuous components, but is of limited use for normal use of the instrument<sup>5</sup>. Note that for multi-axis components (e.g., the steering mirrors), there are commands that act on the combined components and on the individual mechanisms.

On receipt of this command, the NIRI CC control software will:

Ensure an observation is not taking place.  
 Check to insure that protective interlocks are not set.  
 Wait until the number of active mechanisms drops below the number of mechanisms that the power supply can support.  
 Move the focus mechanism to the engineering position corresponding to the desired focus.  
 For some mechanisms, compute the expected values for the Hall-effect sensors.  
 For some mechanisms, verify that the final hall effect sensors match the expected values.

#### 2-3.4.3.3 Park—Park Mechanism

This command is used to prepare a mechanism for instrument shutdown. Note that for multi-axis components (e.g., the steering mirrors), there are commands that act on the combined components and on the individual mechanisms.

On receipt of this command, the NIRI CC control software will:

Ensure an observation is not taking place.  
 Check to insure that protective interlocks are not set.  
 Wait until the number of active mechanisms drops below the number of mechanisms that the power supply can support.  
 Move the mechanism to its parking position.  
 Verify that the expected sensor value matches the measured value.

The park command is not needed for most NIRI components, and is provided for uniformity. The single exception is the focus stage. The focus stage must be parked before disassembly.

#### 2-3.4.3.4 Datm—Datum mechanism

This command is used to find the datum (home) position which defines the mechanism locations. Note that for multi-axis components (e.g., the steering mirrors), there are commands that act on the combined components and on the individual mechanisms.

---

5. For discrete mechanisms, the units are position numbers; i.e., the first position is at 1.0, the second position is at 2.0.



On receipt of this command, the NIRI CC control software will:

Ensure an observation is not taking place.  
 Check to insure that protective interlocks are not set.  
 Wait until the number of active mechanisms drops below the number of mechanisms that the power supply can support.  
 Move the mechanism and search for its datum.  
 Set the hardware datum position.

#### 2-3.4.3.5 MoveEng—Set position in engineering units

This command is used to move the components to locations using units that are convenient for working directly with the hardware. Note that for multi-axis components (e.g., the steering mirrors), the command acts only on the individual mechanisms.

On receipt of this command, the NIRI CC control software will:

Ensure an observation is not taking place.  
 Check to insure that protective interlocks are not set.  
 Wait until the number of active mechanisms drops below the number of mechanisms that the power supply can support.  
 Move the mechanism to the specified engineering position.

The Hall-effect sensors will not be used for confirmation.

#### 2-3.4.3.6 Rdtm—Redatum mechanism.

This command provides a quick test that will detect a loss of positioning. It will detect small positioning errors, but can underestimate large positioning errors. Note that for multi-axis components (e.g., the steering mirrors), the command acts only on the individual mechanisms. It returns an estimate of the difference between the actual datum position and the current datum position (which can occur, e.g., if a mechanism is sticking, being driven too quickly, or being driven with too high an acceleration).

On receipt of this command, the NIRI CC control software will:

Ensure an observation is not taking place.  
 Check to insure that protective interlocks are not set.  
 Wait until the number of active mechanisms drops below the number of mechanisms that the power supply can support.  
 Move the mechanism and search for its datum.

The hardware datum position will be left unchanged. This command returns a value in the RdtmVal record for this mechanism.

*This command is not supported by the pupil viewer or the window cover, because those mechanisms use a home-finding algorithm which is incompatible with this command.*

#### 2-3.4.3.7 Diag—Diagnose mechanism

This is a quick test that will detect a loss of positioning. It will detect arbitrarily large positioning errors, but will fail to detect small positioning errors. Note that for multi-axis components (e.g., the steering mirrors), command acts only on the individual mechanisms.

On receipt of this command, the NIRI CC control software will:

Ensure that an observation is not taking place.  
 Check to insure that protective interlocks are not set.  
 Wait until the number of active mechanisms drops below the number of mechanisms that the power supply can support.  
 Move the mechanism and attempt to verify that the motors and Hall-effect sensors are operating.

If an error is detected, the diagnose command will fail and will return an error code in its CAR record.

## 2-3.5 Detailed Status information

### 2-3.5.1 General records

#### 2-3.5.1.1 name

This record contains a string constant describing the name of the instrument. For NIRI this record will contain "NIRI".

#### 2-3.5.1.2 state

This record will take the values BOOTING, INITIALISING, RUNNING and CONFIGURING<sup>6</sup> and reflect the state of the NIRI control software while it is initializing. The activities corresponding to these states are

- BOOTING — the state during IOC Init and while any database default values are being set.
- INITIALISING — the NIRI control software is checking its own hardware and reading its hardware configuration files during this phase.
- RUNNING — the NIRI control software is ready to accept commands.
- CONFIGURING — the NIRI control software is ready to accept commands but it is busy configuring and unable to make observations.

#### 2-3.5.1.3 health

This record will reflect the overall health of the NIRI control software (including the components controller and the detector controller). Expected values are GOOD, WARNING or BAD. If the overall health is BAD then the NIRI control software is unusable. If the Health is set to WARNING then the NIRI control software is able to execute some functions but not perhaps to specification.

#### 2-3.5.1.4 historyLog<sup>7</sup>

The record used to contain messages to be recorded in the history log.

### 2-3.5.2 Component Status Records

#### 2-3.5.2.1 Datumed

This record indicates that a component has been datumed. For multi-axis components, this flag will only be set if all mechanisms have been datumed. This record takes on the following values:

- 
- 6. Currently unimplemented
  - 7. Currently unusable

- 0 = Datumed
- 1 = Not Datumed

#### 2-3.5.2.2 EngCyclic

This record indicates that a mechanism may be set to a location beyond its physical limits (convenient for circular mechanisms like filter wheels). Note that this only affects the higher level commands; the low-level software may choose to ignore any invalid requests. This record takes on the following values:

- 0 = Limits can be exceeded
- 1 = Limits may not be exceeded

*This record is for internal use by the NIRI software. Its value should not be changed. It is documented here only for completeness.*

#### 2-3.5.2.3 EngMax

This record contains the maximum position that the mechanism can be moved to, in engineering units.

#### 2-3.5.2.4 EngMin

This record contains the minimum position that the mechanism can be moved to, in engineering units.

#### 2-3.5.2.5 EngName

This record contains the engineering name of the current mechanism position. This represents a physical configuration of a component, and will never change (unlike the “Name” record which may change, if filters or other optical elements are replaced or rearranged). The names of the positions are:

- park;
- home;
- pos00, pos01, pos02, ... , up to the maximum number of positions;
- INVALID (if not in a standard location).

#### 2-3.5.2.6 EngPos

This record contains the current position of a mechanism, in engineering units.

#### 2-3.5.2.7 Label

The FDSC field of this record contains a text string which describes a component. This is convenient in providing a title string for edd/dm screens.

#### 2-3.5.2.8 Max

This record contains the maximum position that the mechanism can be moved to, in real-world units. A text string describing the real-world units may be found in the Units record. For mechanisms such as a filter wheel in which there are no appropriate units, the position will typically be given as a position number, i.e., position 0 is 0.0, position 1 is 1.0.

#### 2-3.5.2.9 Min

This record contains the maximum position that the mechanism can be moved to, in real-world units. A text string describing the real-world units may be found in the Units record.

For mechanisms such as a filter wheel in which there are no appropriate units, the position will typically be given as a position number, i.e., position 0 is 0.0, position 1 is 1.0.

#### 2-3.5.2.10 Name

This record contains the name of the current component position. This name typically describes a filter or other optical element and will be redefined if filters or other optical elements are replaced or rearranged (unlike the “EngName” record which describes a component’s physical configuration). The current names are listed in [36].

Note that for multi-axis components, there will be a Name record for each individual mechanism, as well as for the component as a whole.

#### 2-3.5.2.11 Parked

This record indicates that a component is in a parked state. For multi-axis components, this record will be set only if all mechanisms are parked. This field takes the values:

- 0 = All mechanisms parked;
- 1 = Not all mechanisms are parked.

#### 2-3.5.2.12 Pos

This record contains the current position of a mechanism, in real-world units. A text string describing the real-world units may be found in the Units record. For mechanisms such as a filter wheel in which there are no appropriate units, the position will typically be given as a position number, i.e., position 0 is 0.0, position 1 is 1.0.

#### 2-3.5.2.13 RdtmVal

This record contains the result of the last redatum command, in engineering units. (The redatum command returns an estimate of the how far the current mechanism datum position is from the actual datum position; it is useful in diagnosing hardware problems.)

#### 2-3.5.2.14 Reject

This record contains a combination of the values of all CAD records for all mechanisms that make up a component. It contains the following values:

- -1 = At least one CAD record is in a reject state;
- 0 = No CAD records are in a reject state.

*This record exists as a convenience for the user interface. There is no reason for the OCS to use it. It is documented here only for completeness.*

#### 2-3.5.2.15 State

This record contains a string which describes the current state of the mechanism-support code.

*This record is for debugging the NIRI software. The possible values that this record may take may change, without notice. It is documented here only for completeness.*

#### 2-3.5.2.16 Units

This record contains a text string which describes the real world units (e.g., “mm” or “arc sec”)

### **2-3.5.3 Component Health Records**

All of these records return the standard health values: “GOOD”, “WARNING”, and “BAD”.

2-3.5.3.1 DatumHealth

This record reports “WARNING” if a mechanism (or if any mechanism of a multi-axis component) is not datumed. It is cleared by successful execution of a datum command.

2-3.5.3.2 HallHealth

This record reports “WARNING” if a Hall-effect sensor has been disabled. (The sensors can only be disabled from the engineering user interface, and are normally disabled only to deal with a failed sensor.)

This record reports “BAD” if both sensors of a Hall-effect sensor pair have been disabled. Note that a mechanism is still usable if both sensors are disabled, but cannot be datumed. (The sensors can only be disabled from the engineering user interface, and are normally disabled only to deal with a failed sensor.)

2-3.5.3.3 Health

This record reports the most severe error value of all other health records.

2-3.5.3.4 ModuleHealth

This record reports “BAD” if the stepper-motor driver module has entered a fault state. This normally indicates a serious hardware problem. It may be possible to clear the error state by issuing a driver-reset from the engineering user interface. If it is not possible to clear the error, then the stepper-motor driver module or other related hardware has failed, and must be repaired.

2-3.5.3.5 PosErrHealth

This record reports “WARNING” if a mechanism (or if any mechanism of a multi-axis component) has been moved in such a way that proper backlash correction has not been made. In this case, the reported mechanism position may not be the actual mechanism position. This situation can arise if a movement command has been canceled while a motion is in progress; if a movement command failed to complete correctly, if certain low-level engineering commands are executed, and immediately after the control software has been rebooted. It is cleared by successful execution of any motion command.

2-3.5.3.6 SnlHealth

This record reports “BAD” or “WARNING” if the underlying mechanism support code detects an error while a mechanism is in motion. The only situation which is likely to arise is “WARNING”, if an operation takes longer than expected.

**2-3.5.4 Temperature Monitoring and Control Status Records**2-3.5.4.1 C1Setp, C2Setp, C3Setp

These records report the current set-point temperature which for each temperature controller.

2-3.5.4.2 C1Tmp, C2Tmp, C3Tmp

These records report the current temperature which is reported by each temperature controller.

2-3.5.4.3 CoolBusy

This record returns one of the three strings: “Off”, “High”, or “Low”, to indicate whether the cooling motors are off or operating at high or low speed.

2-3.5.4.4 CoolRate1

This record returns the rate of cold-cycle cooler 1, in microsteps per second.

2-3.5.4.5 CoolRate2

This record returns the rate of cold-cycle cooler 2, in microsteps per second.

2-3.5.4.6 Label

The FDSC field of this record returns a string describing the temperature control and monitoring system, for use in the title of edd/dm screens.

2-3.5.4.7 S1Tmp1, S1Tmp2, S1Tmp3, ..., S1Tmp8

These records report the current temperature which is reported by each temperature sensor which is connected to the temperature-sensor controller.

**2-3.5.5 Temperature Monitoring and Control—Health Records**

All of these records return the standard health values: “GOOD”, “WARNING”, and “BAD”.

2-3.5.5.1 C1Health, C2Health, C3Health

This record will report “BAD” if a temperature controller is not functioning properly. This can indicate either a communications problem, a hardware failure with the temperature controller, or a faulty temperature sensor.

2-3.5.5.2 CoolModeHealth

This record will report “BAD” if the cooling motors are off, when the instrument should be cold, or if the cooling motors are on, when the instrument should be warm.

2-3.5.5.3 Health

This record combines all of the other health records for the temperature control and monitoring subsystem, and reports the most severe condition.

2-3.5.5.4 ModeHealth

This record will report “BAD” if the external mode-selector connector is not in the appropriate position for normal operation. (If the connector is in the “warm-up mode”, the motors cannot be operated.)

2-3.5.5.5 S1Health

This record will report “BAD” if the temperature sensor unit is not functioning properly. This can indicate either a communications problem or a hardware failure with the temperature sensor unit.

2-3.5.5.6 TmpCheckHealth

This record will report “BAD” if any temperature is unreasonably high while the system is expected to be cold or if a temperature is too low, while the system is expected to be warm.

**2-3.6 Alarm conditions**

The following situations will generate an alarm.

- Hall effect sensor disabled
- Motor-control module fault
- Mechanism not datumed
- Mechanism not backlash corrected

- Problem detected by SNL code (time-out)
- Temperature changing
- Temperature out of range

### 2–3.7 Debugging and Maintenance

This interface can be tested and debugged by means of an engineering user interface supplied as part of the NIRI.

#### 2–3.7.1 Debugging Modes

The NIRI control software will have the standard debugging modes described in ICD 1a, [24].

#### 2–3.7.2 Calibration

If a Hall effect sensor fails or if a mechanism must be disassembled for maintenance, the sensors will need to be recalibrated. A separate program is provided for this purpose. Calibration procedures are described in a separate document [5].

### 2–3.8 Simulation

The Gemini simulation modes are described in ICD 1a, [24]. The NIRI control software will use the FAST and FULL modes, and by default it begins in “NONE” mode. VSM mode may be supported, if time permits.

---

## 2–4.0 Safety Issues

---

It is necessary to avoid moving NIRI components while the mechanical components are not at thermal equilibrium. Records are provided to provide a temperature interlock. Details are described in [6]<sup>8</sup>.

---

8. All interlock commands are available from within the components controller, by using a niri:cc: prefix, instead of the niri:wfs: prefix.





**Gemini  
Near  
Infra-Red  
Imager  
and  
Wavefront  
Sensor**



# Near IR Imager (NIRI) — Component Position Names

**Hubert Yamada**

niri\_h ty\_013 / 01 / 1999-11-22

This document describes the position names for the NIRI mechanisms (i.e., filter names).

## 1 Introduction

### 1.1 Purpose

This document describes the position names for the NIRI mechanisms. In general, information which logically belongs within an ICD but which is subject to change with relatively trivial hardware changes are presented here. In particular, the names of the various filters contained within the different NIRI filter wheels are presented here.

The intended audience of this document is:

- The NIRI reviewers;
- The developers of the Gemini Observatory Control System (OCS);
- The maintainers of NIRI after it is delivered to Gemini.

### 1.2 Definitions, Acronyms and Abbreviations

See “Near IR Imager (NIRI)—Definitions, Abbreviations, and References”.

### 1.3 Stylistic Conventions

References to documents in Section 1.5, “References,” on page 2 of “Near IR Imager (NIRI)—Definitions, Abbreviations, and References” are given like this; [1].

### 1.4 References

See “Near IR Imager (NIRI)—Definitions, Abbreviations, and References”.

---

## 2 Tables

---

### 2.1 Filter Wheel 1

Table 1 describes the named positions for filter wheel 1.

---

**TABLE 1.** Named Positions for Filter Wheel 1

| Engineering Position Name | Position Name |
|---------------------------|---------------|
| pos00                     |               |
| pos01                     |               |
| pos02                     |               |
| pos03                     |               |
| pos04                     |               |
| pos05                     |               |
| pos06                     |               |
| pos07                     |               |
| pos08                     | Open          |
| pos09                     | PK-50         |
| pos10                     |               |
| pos11                     |               |

### 2.2 Filter Wheel 2

Table 2 describes the named positions for filter wheel 2

---

**TABLE 2.** Named Positions for Filter Wheel 2

| Engineering Position Name | Position Name |
|---------------------------|---------------|
| pos00                     | J             |
| pos01                     | K'            |
| pos02                     | K_S           |
| pos03                     | L'            |
| pos04                     | M'            |
| pos05                     | PN090299/3    |
| pos06                     |               |
| pos07                     |               |
| pos08                     |               |
| pos09                     |               |
| pos10                     | Open          |
| pos11                     | H             |

### 2.3 Filter Wheel 3

2.4 Table 3 describes the named positions for filter wheel 3

---

**TABLE 3.** Named Positions for Filter Wheel 3

| Engineering Position Name | Position Name   |
|---------------------------|-----------------|
| pos00                     |                 |
| pos01                     |                 |
| pos02                     |                 |
| pos03                     |                 |
| pos04                     |                 |
| pos05                     |                 |
| pos06                     | wollaston_prism |
| pos07                     | grid            |
| pos08                     | pupil_mask      |
| pos09                     | --              |
| pos10                     |                 |
| pos11                     |                 |

Table 3 describes the named positions for the focal-plane mask wheel.

---

**TABLE 4.** Named Positions for the Focal Plane Mask Wheel

| Engineering Position Name | Position Name |
|---------------------------|---------------|
| pos00                     | f6            |
| pos01                     |               |
| pos02                     |               |
| pos03                     |               |
| pos04                     |               |
| pos05                     |               |
| pos06                     |               |
| pos07                     |               |
| pos08                     |               |
| pos09                     | pinhole_mask  |
| pos11                     | f13.9         |
| pos10                     | f32           |

---

## Tables

---

## ICD 1.6/1.10[ab]

### “A&G to Niri/GNIRS On Instrument Wavefront Sensors”

niri\_h ty\_008

Hubert Yamada  
2001-06-27

Version: 1.01

Issued By: Hubert Yamada

Approved By: \_\_\_\_\_ Date: \_\_\_\_\_  
Group Manager  
\_\_\_\_\_ Group Manager (if req) \_\_\_\_\_  
\_\_\_\_\_ Systems Engineering \_\_\_\_\_

## Revision Control

1. Revision Version # 1.00  
Date: 2000-06-07  
Revised by: Hubert Yamada  
Reason for / items changed: Draft for 1.00 release.
2. Revision Version # 1.01  
Date: 2000-06-08  
Revised by: Hubert Yamada  
Reason for / items changed: Added information about missing fields, in response to questions from Chris Mayer.
3. Revision Version # 1.02  
Date: 2001-06-27  
Revised by: Hubert Yamada  
Reason for / items changed: Fixed some minor errors in response to questions from Richard Wolff. This are only corrections and clarifications, not changes in interfaces. Numerous references to NIRI have been corrected, as appropriate.  
Last Modified: Wednesday, June 27, 2001 4:24 pm

# Chapter 1 Introduction

---

## 1–1.0 Description

---

### 1–1.1 Purpose

This Interface Control Document (ICD) serves the following purposes:

- To describe the commands and parameters that will pass between the A&G system and the On-Instrument Wavefront Sensor (OIWFS) Components Controller (CC).
- To provide a prototype which may, with minimal modification, be used to describe the GNIRS OIWFS, and the OIWFS for any system which copies the NIRI OIWFS hardware and software.

This document only describes the ways in which the NIRI/GNIRS OIWFS differs from the prototypical OIWFS which is described in ICD 1.6/1.10 [19] and in [20].

*At some time in the future, the NIRI/GNIRS OIWFS information may be merged with ICD 1.6/1.10.*

In addition to the documents mentioned in the previous paragraphs, the specific interfaces described in this document are governed by the following parent Interface Control Documents, which describe the general properties of particular kinds of interface:

- ICD 1a [24], which describes how the OCS deals with commands and attributes and the behavior the OCS expects from the systems it commands.
- ICD 1b [25], which gives a general description of the way the Observatory Control System communicates with the other principal systems. It is also related to ICD 7a, [27], which describes the way an Instrument Control System communicates with its subsystems.
- ICD 2 [26], which describes the way in which an Instrument Control System makes information available to the outside world (and the OCS) by means of “Status Information Records”.
- ICD 7b [28], which describes the way that the Telescope Control System (TCS) interacts with its subsystems. (From a control standpoint, the OIWFS is a subsystem of the A&G, which is a subsystem of the TCS.)

The intended audience for this document is:

- The NIRI reviewers;
- The GNIRS reviewers;
- Any other groups which will be using the design of the NIRI/GNIRS OIWFS;
- The developers of the Gemini Acquisition and Guidance units;
- The developers of the GNIRS OIWFS components controller.

### 1–1.2 Scope

This document describes only aspects of the NIRI/GNIRS OIWFS software system which differ from, or are more specific than, the Gemini standard on-instrument wavefront sensor

(OIWFS), described in [19]. It is assumed that the reader is familiar with both of those documents.

This document does *not* describe the following topics:

- The OIWFS hardware and electronics (see [11] and [12], summarized in [3]);
- The OIWFS detector software, hardware, and electronics (see [12] and [13]);
- The GNIRS software, hardware, and electronics;
- Other instruments which will be using the NIRI/GNIRS OIWFS;
- The OIWFS Interlock system (see [6]);
- The names of the NIRI OIWFS mechanism positions (see [37]);
- The names of the GNIRS OIWFS mechanism positions;
- Interfacing an instrument to the NIRI/GNIRS OIWFS (see [6]);
- The software design (see [2], and [3]).

Unless otherwise stated, any references to the OIWFS refers only to the OIWFS mechanisms. The OIWFS detector is a completely separate system, and does not have an EPICS interface.

---

## 1–2.0 Related Documents and Drawings

### 1–2.1 References

See “Near IR Imager (NIRI)—Definitions, Abbreviations, and References”.

### 1–2.2 Definitions, Acronyms and Abbreviations

See “Near IR Imager (NIRI)—Definitions, Abbreviations, and References”.

### 1–2.3 Stylistic Conventions

References to documents in Section 1.5, “References,” on page 2 of “Near IR Imager (NIRI)—Definitions, Abbreviations, and References” are given like this; [1].



## Chapter 2 Details of the A&G to NIRI/GNIRS OIWFS Interface

---

### 2-1.0 Introduction

---

### 2-2.0 Physical System Interfaces

---

#### 2-2.1 Mechanical Interface

*Not Applicable for a software ICD.*

#### 2-2.2 Optical Interface

*Not Applicable for a software ICD.*

#### 2-2.3 Electronic Interface

##### 2-2.3.1 NIRI

The system hardware architecture for the NIRI OIWFS control software is the same as described in [16], with the following exception:

- The Instrument Sequencer, Components Controller, and OIWFS Components Controller will share the same IOC.

There will also be the following electronic VME modules:

- The XVME-240 Digital I/O Board:
  - To monitor the status of the stepper-motor control modules;
  - To reset the stepper-motor control modules;
- Four custom-built VME sensor-support boards:
  - To read out the NIRI OIWFS Hall-effect sensors;
- Two XVME-566 Analog Input Modules:
  - To read out the VME sensor-support boards for the NIRI OIWFS Hall-effect sensors;
- Two Oregon Microsystems VME8-8 Stepper-motor controller boards
  - To control the NIRI OIWFS stepper-motors.

Note that these boards will be shared between the NIRI OIWFS hardware and the NIRI hardware, and will perform other functions that are not listed above. In particular, the XVME-240 board is shared between the NIRI CC, the NIRI OIWFS, and is also used to interface to the interlock bus. See [3] for more details.

##### 2-2.3.2 GNIRS / Standalone OIWFS

The system hardware architecture for the GNIRS OIWFS control software is the same as described in [16], with the following exception:

- The OIWFS Components Controller and its supporting hardware will be in the same VME crate as other GNIRS hardware, but will have a separate bus.

There will also be the following electronic VME modules:

- The XVME-240 Digital I/O Board;
  - To monitor the status of the stepper-motor control modules;
  - To reset the stepper-motor control modules;
- One custom-built VME sensor-support board:
  - To read out the GNIRS OIWFS Hall-effect sensors;
- One XVME-566 Analog Input Modules:
  - To read out the VME sensor-support boards for the GNIRS OIWFS Hall-effect sensors;
- One Oregon Microsystems VME8-8 Stepper-motor controller boards
  - To control the GNIRS OIWFS stepper-motors.

These boards will not be shared between the OIWFS hardware and any other system; however, the XVME-240 board will be used to interface to the interlock bus.

#### 2-2.4 Mass/Balance

*Not Applicable for a software ICD.*

#### 2-2.5 Thermal Interface

*Not Applicable for a software ICD.*

### 2-3.0 Software/Control Function Interface

---

#### 2-3.1 Overview

##### 2-3.1.1 Communications infrastructure

The communications infrastructure is the same as described in [16].

##### 2-3.1.2 Software architecture

The software architecture is described in [2].

#### 2-3.2 Behavior

The behavior of the NIRI/GNIRS OIWFS/A&G interface is the same as described in [20] and [19].

#### 2-3.3 Implementation

This section contains tables summarizing the EPICS records used for the interface. The next sections (Section 2-3.4 on page 17 and Section 2-3.5 on page 21) contain the details of what those records do.

It is assumed in all cases that the record names given in this document will be prefixed by the string “niri:” or “nirs:”<sup>1</sup>, as appropriate when the database is implemented, except for the SAD database which will be prefixed by “niri:”, “niri::”, “nirs:”, or “nirs::”<sup>2</sup>.

### 2–3.3.1 Command (CAD/CAR) records

The columns of the command record tables are as follows:

**Description.** A brief description of the command.

**Attribute.** The name of the attribute which the Observatory Control System associates with a particular parameter of a particular command. Component position names are listed separately in Table 7.

**CAD Record / Cad Record.FIELD.** The name of the EPICS (minus the “niri:” or “nirs:” prefix) together with the field in that CAD record associated with each parameter of the command.

**CAD ordering number.** The recommended order in which this command should be executed. CAD records should be connected to their associated APPLY record in the order 1, 2, 3, 4, 5, etc... CAD records within the same ordering number set can be connected to the APPLY record in any order in that set.

**CAR record.** The CAR record associated with the command. Note that several CAD records may share the same CAR record.

#### 2–3.3.1.1 CAD/CAR records—System Commands

These commands affect the overall behavior of NIRI and the NIRI OIWFS; they affect the GNIRS OIWFS, but not the other GNIRS components. Note, in particular, that it is not possible to reboot NIRI and the NIRI OIWFS separately. It may be possible to reboot the GNIRS OIWFS without rebooting GNIRS.

**TABLE 1.** NIRI/GNIRS system engineering command CAD/CAR records

| Description                       | Attribute  | CAD Record /<br>CAD Record.FIELD<br>(“niri:” or “nirs:” +) | CAR record |
|-----------------------------------|------------|------------------------------------------------------------|------------|
| Reboot.                           |            | wfs:reboot                                                 | N/A        |
| Select debug mode.                | Debug mode | wfs:debug.A                                                | wfs:debugC |
| Flush data from internal buffers. |            | wfs:flush                                                  | wfs:flushC |

1. Or some other Gemini-assigned name, which must be at most four alpha-numeric characters, followed by a colon.
2. Or some other Gemini-assigned name, which must be at most four alpha-numeric characters, followed by one or two colons.

### 2-3.3.1.2 OIWFS command CAD/CAR records—OIWFS components

The commands in Table 2 are the only ones that are used during routine observation.

**TABLE 2.** OIWFS command CAD/CAR records

| Description                                                                                                                                                                                                                                  | Attribute <sup>a</sup> | CAD Record /<br>CAD Record.FIELD<br>("niri:" or "nirs:" +) | CAR<br>record<br>("niri:"<br>or<br>"nirs:"<br>+) |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|------------------------------------------------------------|--------------------------------------------------|
| Select a filter in the filter wheel <sup>b</sup> .                                                                                                                                                                                           | filter name            | wfs:filtSel.A                                              | wfs:filtC                                        |
| Park the filter wheel.                                                                                                                                                                                                                       | (none)                 | wfs:filtPark                                               |                                                  |
| Move gimbal mirrors, to given (x, y) position and leave it stationary. (x, y) is the position in the focal plane measured in mm in the instrument's frame of reference.                                                                      | x<br>y                 | wfs:prbMove.A<br>wfs:prbMove.B                             | wfs:prbC                                         |
| Park the gimbal mirror.                                                                                                                                                                                                                      | (none)                 | wfs:prbPark                                                |                                                  |
| Select the focus-stage position by name.                                                                                                                                                                                                     | position name          | wfs:focSel.A                                               | wfs:focC                                         |
| Set the focus-stage position in mm of translation.                                                                                                                                                                                           | position               | wfs:focMove.A                                              |                                                  |
| Park the focus stage.                                                                                                                                                                                                                        | (none)                 | wfs:focPark                                                |                                                  |
| The gimbal and focus stage are caused to follow a continuous stream of updated (x=gimbal x, y=gimbal y, z=focus) positions. These positions are not defined as CAD record arguments but in separate records (see ICD 1.6/1.10). <sup>c</sup> | (none)                 | wfs:folFollow                                              | folC                                             |
| "Follow" mode, initiated by the wfs:folFollow command, is terminated.                                                                                                                                                                        | (none)                 | wfs:folStop                                                |                                                  |
| Set the tolerance values for all three axes (x=gimbal x, y=gimbal y, z=focus) in mm. The tolerance is used to calculate the wfs:folInToler SIR record.                                                                                       | x<br>y<br>z            | wfs:folSetTol.A<br>wfs:folSetTol.B<br>wfs:folSetTol.C      | folSetC                                          |

a. "(none)" means that there are no arguments.

b. ICD 1.6/1.10 [19] has three separate components: a field-stop, a neutral-density filter, and a color filter. For the NIRI/GNIRS OIWFS, these are combined into a single wheel which may have multiple optical elements at each position.

c. The focus is not designed to be operated continuously, and it should not be necessary to do so. By default, the stream of focus offset values is ignored.

### 2-3.3.1.3 OIWFS Engineering commands CAD/CAR records—OIWFS components

The following commands are used for engineering purposes, and are not expected to be used during normal observing. Note that, although these commands are connected to the OIWFS APPLY record, they are normally activated directly by the engineering user interface.

The following commands affect the overall behavior of the OIWFS but do not affect the science detector.

**TABLE 3.**

NIRI/GNIRS OIWFS engineering command CAD/CAR records

| Description                                                                                   | Attribute      | CAD Record /<br>CAD Record.FIELD<br>("niri:" or "nirs:" +) | CAR<br>record<br>("niri:" or<br>"nirs:" +) |
|-----------------------------------------------------------------------------------------------|----------------|------------------------------------------------------------|--------------------------------------------|
| Datum filter wheel.                                                                           |                | wfs:filtDatm                                               | wfs:filtC                                  |
| Set filter wheel position in position numbers <sup>a</sup> .                                  | position       | wfs:filtMove.A                                             |                                            |
| Set filter wheel position in engineering units (motor microsteps) <sup>b</sup> .              | motor position | wfs:filtEngMove.A                                          |                                            |
| Confirm that filter wheel has not lost its datum position.                                    |                | wfs:filtRdtm                                               |                                            |
| Confirm that filter wheel and its Hall-effect sensors are operational                         |                | wfs:filtDiag                                               |                                            |
| Datum the gimbal mirror.                                                                      |                | wfs:prbDatm                                                | wfs:prbC                                   |
| Datum the x-axis of the gimbal mirror.                                                        |                | wfs:prbDatm                                                | wfs:prbxC                                  |
| Tip gimbal mirror along its x-axis, given x position in engineering units (motor microsteps). | motor position | wfs:prbxEngMove.A                                          |                                            |
| Confirm that the OIWFS gimbal mirror x-axis motor has not lost its datum position.            |                | wfs:prbxRdtm                                               |                                            |
| Confirm that the OIWFS gimbal mirror x-axis and its Hall-effect sensors are operational       |                | wfs:prbxDiag                                               |                                            |
| Datum the y-axis of the gimbal mirror.                                                        |                | wfs:prbDatm                                                | wfs:prbyC                                  |
| Tip gimbal mirror along its y-axis, given y position in engineering units (motor microsteps). | motor position | wfs:prbyEngMove.A                                          |                                            |
| Confirm that OIWFS gimbal mirror y-axis motor has not lost its datum position.                |                | wfs:prbyRdtm                                               |                                            |
| Confirm that OIWFS gimbal mirror y-axis motor and its Hall-effect sensors are operational     |                | wfs:prbyDiag                                               |                                            |
| Datum focus mechanism.                                                                        |                | wfs:focDatm                                                | wfs:focC                                   |
| Set focus stage translation in engineering units (motor microsteps).                          | motor position | wfs:focEngMove.A                                           |                                            |
| Confirm that focus stage has not lost its datum position.                                     |                | wfs:focRdtm                                                |                                            |
| Confirm that focus stage and its Hall-effect sensors are operational                          |                | wfs:focDiag                                                |                                            |

a. Only the standard positions (excluding datum and park) are fully repeatable.

b. Only the standard positions (excluding datum and park) are fully repeatable.

### 2-3.3.2 Status Information Records

The following tables show the status information provided by the OIWFS. The tables have the following columns:

**SIR record.** The part of the name of the EPICS SIR record which follows the "niri:" or "nirs:" prefix<sup>3</sup>.

**Units.** The units in which the quantity is stored. (Ustep refers to units of motor microsteps; “user units” or “real-world units” refer to the arbitrary units described by the Units SIR variable.)

**Comments.** A description of the item. The first paragraph of this will go into the SIR record description and also the comment field in the FITS header.

*Note that only the type, units and comments are actually contained in the SIR records. The FITS keyword is provided by the DHS.*

#### 2-3.3.2.1 WFS Status Information

Table 4 contains status information for all of the WFS.

**TABLE 4.** The WFS Overall Status Information

| SIR record<br>("niri:" or "nirs:" +) | FITS<br>Keyword | FITS<br>included | Type                          | Units  | Comments                                                                                        |
|--------------------------------------|-----------------|------------------|-------------------------------|--------|-------------------------------------------------------------------------------------------------|
| wfs:debugMode                        | WSDMODE         | Never            | string                        |        | Debugging mode [NOLOG   NONE   MIN   FULL]                                                      |
| wfs:health                           |                 | Never            | string                        |        | WFS Health [GOOD   WARNING   BAD]                                                               |
| wfs:heartBeat                        | WSHRTB          | Never            | long                          | count  | Heart beat record: Shows that WFS software is alive by returning a constantly increasing value. |
| wfs:historyLog                       | WSHLOG          | Never            | array of<br>char <sup>a</sup> |        | History log record <sup>b</sup>                                                                 |
| wfs:issPort                          | WSISSPT         | Start            | long                          | number | ISS port on which instrument is installed                                                       |
| wfs:label                            | WSLABEL         | Never            | string                        |        | The FDSC field describes the WFS Components Controller                                          |
| wfs:name                             | WSNAME          | Never            | string                        |        | WFS Components Controller Name                                                                  |
| wfs:simMode                          | WSSMODE         | Start            | string                        |        | Simulation mode [NONE   FULL   FAST   VSM]                                                      |
| wfs:state                            | WSSTATE         | Never            | string                        |        | Inst. state [BOOTING   INITIALIZING   RUNNING   CONFIGURING]                                    |
| wfs:wfsBeam                          | WSWFSBM         | Start            | long                          | 0/1    | Is the WFS detector unobstructed? [YES=0   NO=1] <sup>c</sup>                                   |

- Should be “bigstring”
- This record is not useful, because of various EPICS limitations.
- The wfs:wfsBeam record only indicates whether the OIWFS components controller mechanisms are obstructing the beam; it does not indicate that light is actually getting to the detector. Furthermore, it does not account for the obstruction caused by the pick-off mirror. The instrument needs to combine the wfs:wfsBeam with information about any other components in order to provide an instrument-wide wfsBeam record.

- Note that the actual system has been designed so the Status Alarm Database can easily be given a different prefix from the rest of the NIRI/GNIRS database(s) (for example “niri:” or “nirs:”). It is assumed in this document that the Status Alarm Database prefix will be “niri:” or “nirs:”.

### 2-3.3.2.2 The Wavefront Sensor Components Status Information

The following table summarizes the information for the various components.

**TABLE 5.** NIRI/GNIRS On-Instrument Wave-Front Sensor Status Records

| SIR record<br>("niri:" or "nirs:" +) | FITS Keyword | FITS<br>Include<br>d | Type   | Units            | Comments                                                                        |
|--------------------------------------|--------------|----------------------|--------|------------------|---------------------------------------------------------------------------------|
| wfs:filtDatumed                      | WSFLDTM      | Never                | long   | 0/1              | Is the filter wheel datumed? [YES=0   NO=1]                                     |
| wfs:filtEngCyclic                    | WSFLECY      | Never                | long   | 0/1              | Ignore limits for the filter wheel? [YES=0   NO=1]                              |
| wfs:filtEngMax                       | WSFLEMX      | Never                | long   | motor microsteps | Maximum position of the filter wheel in engineering units (motor microsteps)    |
| wfs:filtEngMin                       | WSFLEMN      | Never                | long   | motor microsteps | Minimum position of the filter wheel in engineering units (motor microsteps)    |
| wfs:filtEngName                      | WSFLENM      | Start                | string |                  | Engineering name of the filter wheel position                                   |
| wfs:filtEngPos                       | WSFLEPS      | Start                | long   | motor microsteps | Position of the filter wheel in engineering units (motor microsteps)            |
| wfs:filtLabel                        | WSFLLBL      | Never                | string |                  | The FDSC field describes the filter wheel                                       |
| wfs:filtMax                          | WSFLMAX      | Never                | double | position number  | Maximum position of the filter wheel in real-world units (position number)      |
| wfs:filtMin                          | WSFLMIN      | Never                | double | position number  | Minimum position of the filter wheel in real-world units (position number)      |
| wfs:filtName                         | WSFLNAM      | Start                | string |                  | User name of the filter wheel position                                          |
| wfs:filtParked                       | WSFLPRK      | Never                | long   | 0/1              | Is the filter wheel parked? [YES=0   NO=1]                                      |
| wfs:filtPos                          | WSFLPOS      | Start                | double | position number  | Position of the filter wheel position in real-world units (position number)     |
| wfs:filtRdtmVal                      | WSFLRDT      | Never                | long   | motor microsteps | Returned value from the filter wheel redatum command                            |
| wfs:filtReject                       | WSFLREJ      | Never                | long   | 0/-1             | Combined the filter wheel CAR state [GOOD=0   REJECT=-1]                        |
| wfs:filtState                        | WSFLSTE      | Never                | string |                  | the filter wheel SNL state                                                      |
| wfs:filtUnits                        | WSFLUNT      | Start                | string |                  | Definition of real-world units ("user units") for the filter wheel ("pos num"). |
| wfs:focAgDiff                        | WSFOCADF     | Never                | double | mm               | The focus stage offset error in real-world units (mm of translation)            |
| wfs:focAgEngDiff                     | WSFOCAEF     | Never                | double | motor microsteps | The focus stage offset error in engineering units (motor microsteps)            |
| wfs:focDatumed                       | WSFOCDTM     | Never                | long   | 0/1              | Is the focus stage datumed? [YES=0   NO=1]                                      |
| wfs:focEngCyclic                     | WSFOCECY     | Never                | long   | 0/1              | Ignore limits for the focus stage? [YES=0   NO=1]                               |
| wfs:focEngMax                        | WSFOCEMX     | Never                | long   | motor microsteps | Maximum position of the focus stage in engineering units (motor microsteps)     |
| wfs:focEngMin                        | WSFOCEMN     | Never                | long   | motor microsteps | Minimum position of the focus stage in engineering units (motor microsteps)     |

**TABLE 5.** NIRI/GNIRS On-Instrument Wave-Front Sensor Status Records

| SIR record<br>("niri:" or "nirs:" +) | FITS Keyword | FITS<br>Include<br>d | Type   | Units            | Comments                                                                     |
|--------------------------------------|--------------|----------------------|--------|------------------|------------------------------------------------------------------------------|
| wfs:focEngName                       | WSFOCENM     | Start                | string |                  | Engineering name of the focus stage position <sup>a</sup>                    |
| wfs:focEngPos                        | WSFOCEPS     | Start                | long   | motor microsteps | Position of the focus stage in engineering units (motor microsteps)          |
| wfs:focInToler                       | WSFOCINT     | Never                | long   | 0/1              | Is the focus stage offset in tolerance <sup>b</sup> ? [YES=0   NO=1]         |
| wfs:focLabel                         | WSFOCLBL     | Never                | string |                  | The FDSC field describes the focus stage                                     |
| wfs:focMax                           | WSFOCMAX     | Never                | double | mm               | Maximum position of the focus stage in real-world units (mm of translation)  |
| wfs:focMin                           | WSFOCMIN     | Never                | double | mm               | Minimum position of the focus stage in real-world units (mm of translation)  |
| wfs:focName                          | WSFOCNAM     | Start                | string |                  | User name of the focus stage position <sup>c</sup>                           |
| wfs:focParked                        | WSFOCPRK     | Never                | long   | 0/1              | Is the focus stage parked? [YES=0   NO=1]                                    |
| wfs:focPos                           | WSFOCPOS     | Always               | double | mm               | Position of the focus stage position in real-world units (mm of translation) |
| wfs:focRdtmVal                       | WSFOCRDT     | Never                | long   | motor microsteps | Returned value from the focus stage redatum command                          |
| wfs:focReject                        | WSFOCREJ     | Never                | long   | 0/-1             | Combined focus stage CAR state [GOOD=0   REJECT=-1]                          |
| wfs:focState                         | WSFOCSTE     | Never                | string |                  | the focus stage SNL state                                                    |
| wfs:focTol                           | WSFOCTOL     | Never                | double | mm               | Tolerance of the focus stage offset in real-world units (mm of translation)  |
| wfs:focUnits                         | WSFOCUNT     | Always               | string |                  | Definition of real-world units ("user units") for the focus stage. ("mm").   |
| wfs:folDatumed                       | WSFLDTM      | Never                | long   | 0/1              | Are all dynamic mechanisms datumed? [YES=0   NO=1]                           |
| wfs:folInToler                       | WSFOLINT     | Never                | long   | 0/1              | Are all dynamic mechanisms in tolerance <sup>d</sup> ? [YES=0   NO=1]        |
| wfs:folIsFollow                      | WSFLFOL      | Never                | long   | 0/1              | Are all dynamic components in follow mode? [YES=0   NO=1]                    |
| wfs:folIsIdle                        | WSFLIDL      | Never                | long   | 0/1              | Are all dynamic components idle? [YES=0   NO=1]                              |
| wfs:folLabel                         | WSFLLBL      | Never                | string |                  | The FDSC field describes the dynamic mechanisms                              |
| wfs:folName                          | WSFLNAM      | Start                | string |                  | Dynamic components: name of current position                                 |
| wfs:folParked                        | WSFLPRK      | Never                | long   | 0/1              | Are all dynamic mechanisms parked? [YES=0   NO=1]                            |
| wfs:folPresent                       | WSFLPRS      | Never                | double | seconds          | Current Gemini time                                                          |
| wfs:folReject                        | WSFLREJ      | Never                | long   | 0/1              | Combined reject state for Follow Command                                     |
| wfs:prbDatumed                       | WSPBDTM      | Never                | long   | 0/1              | Is/Are Gimbal Mirror datumed? [YES=0   NO=1]                                 |



**TABLE 5.** NIRI/GNIRS On-Instrument Wave-Front Sensor Status Records

| SIR record<br>("niri:" or "nirs:" +) | FITS Keyword | FITS<br>Include<br>d | Type   | Units            | Comments                                                                                 |
|--------------------------------------|--------------|----------------------|--------|------------------|------------------------------------------------------------------------------------------|
| wfs:prbLabel                         | WSPBLBL      | Never                | string |                  | The FDSC field describes the gimbal mirror                                               |
| wfs:prbName                          | WSPBNAM      | Start                | string |                  | the gimbal mirror name of current position <sup>e</sup>                                  |
| wfs:prbParked                        | WSPBPRK      | Never                | long   | 0/1              | Is the gimbal mirror parked? [YES=0   NO=1]                                              |
| wfs:prbReject                        | WSPBREJ      | Never                | long   | 0/1              | Combined reject state for the gimbal mirror                                              |
| wfs:prbSky:xPos                      | WSPBSXPS     | Never                | double | sky units        | Position of the gimbal mirror x in sky units                                             |
| wfs:prbSky:yPos                      | WSPBSYPS     | Never                | double | sky units        | Position of the gimbal mirror y in sky units                                             |
| wfs:prbSkyLabel                      | WSPBSLBL     | Never                | string |                  | The FDSC field describes the gimbal mirror                                               |
| wfs:prbSkyUnits                      | WSPBSUNT     | Start                | string |                  | Definition of "sky units"                                                                |
| wfs:prbxAgDiff                       | WSPBXADF     | Never                | double | mm               | The gimbal-mirror x-axis offset error in real-world units (mm in the focal plane)        |
| wfs:prbxAgEngDiff                    | WSPBXAEF     | Never                | double | motor microsteps | The gimbal-mirror x-axis offset error in engineering units (motor microsteps)            |
| wfs:prbxDatumed                      | WSPBXDTM     | Never                | long   | 0/1              | Is the gimbal-mirror x-axis datumed? [YES=0   NO=1]                                      |
| wfs:prbxEngCyclic                    | WSPBXECY     | Never                | long   | 0/1              | Ignore limits for the gimbal-mirror x-axis? [YES=0   NO=1]                               |
| wfs:prbxEngMax                       | WSPBXEMX     | Never                | long   | motor microsteps | Maximum position of the gimbal-mirror x-axis in engineering units (motor microsteps)     |
| wfs:prbxEngMin                       | WSPBXEMN     | Never                | long   | motor microsteps | Minimum position of the gimbal-mirror x-axis in engineering units (motor microsteps)     |
| wfs:prbxEngName                      | WSPBXENM     | Start                | string |                  | Engineering name of the gimbal-mirror x-axis position <sup>f</sup>                       |
| wfs:prbxEngPos                       | WSPBXEPS     | Start                | long   | motor microsteps | Position of the gimbal-mirror x-axis in engineering units (motor microsteps)             |
| wfs:prbxInToler                      | WSPBXINT     | Never                | long   | 0/1              | Is the gimbal-mirror x-axis offset in tolerance <sup>g</sup> ? [YES=0   NO=1]            |
| wfs:prbxLabel                        | WSPBXLBL     | Never                | string |                  | The FDSC field describes the gimbal-mirror x-axis                                        |
| wfs:prbxMax                          | WSPBXMAX     | Never                | double | mm               | Maximum position of the gimbal-mirror x-axis in real-world units (mm in the focal plane) |
| wfs:prbxMin                          | WSPBXMIN     | Never                | double | mm               | Minimum position of the gimbal-mirror x-axis in real-world units (mm in the focal plane) |
| wfs:prbxName                         | WSPBXNAM     | Start                | string |                  | User name of the gimbal-mirror x-axis position <sup>h</sup>                              |
| wfs:prbxParked                       | WSPBXPBK     | Never                | long   | 0/1              | Is the gimbal-mirror x-axis parked? [YES=0   NO=1]                                       |

**TABLE 5.** NIRI/GNIRS On-Instrument Wave-Front Sensor Status Records

| <b>SIR record<br/>("niri:" or "nirs:" +)</b> | <b>FITS Keyword</b> | <b>FITS<br/>Include<br/>d</b> | <b>Type</b> | <b>Units</b>     | <b>Comments</b>                                                                           |
|----------------------------------------------|---------------------|-------------------------------|-------------|------------------|-------------------------------------------------------------------------------------------|
| wfs:prbxPos                                  | WSPBXPOS            | Always                        | double      | mm               | Position of the gimbal-mirror x-axis position in real-world units (mm in the focal plane) |
| wfs:prbxRdtmVal                              | WSPBXRDT            | Never                         | long        | motor microsteps | Returned value from the gimbal-mirror x-axis redatum command                              |
| wfs:prbxReject                               | WSPBXREJ            | Never                         | long        | 0/-1             | Combined the gimbal-mirror x-axis CAR state [GOOD=0   REJECT=-1]                          |
| wfs:prbxState                                | WSPBXSTE            | Never                         | string      |                  | the gimbal-mirror x-axis SNL state                                                        |
| wfs:prbxTol                                  | WSPBXTOL            | Never                         | double      | mm               | Tolerance of the gimbal-mirror x-axis offset in real-world units (mm in the focal plane)  |
| wfs:prbxUnits                                | WSPBXUNT            | Always                        | string      |                  | Definition of real-world units ("user units") for the gimbal-mirror x-axis. ("mm").       |
| wfs:prbyAgDiff                               | WSPBYADF            | Never                         | double      | mm               | The gimbal-mirror y-axis offset error in real-world units (mm in the focal plane)         |
| wfs:prbyAgEngDiff                            | WSPBYAEF            | Never                         | double      | motor microsteps | The gimbal-mirror y-axis offset error in engineering units (motor microsteps)             |
| wfs:prbyDatumed                              | WSPBYDTM            | Never                         | long        | 0/1              | Is the gimbal-mirror y-axis datumed? [YES=0   NO=1]                                       |
| wfs:prbyEngCyclic                            | WSPBYECY            | Never                         | long        | 0/1              | Ignore limits for the gimbal-mirror y-axis? [YES=0   NO=1]                                |
| wfs:prbyEngMax                               | WSPBYEMX            | Never                         | long        | motor microsteps | Maximum position of the gimbal-mirror y-axis in engineering units (motor microsteps)      |
| wfs:prbyEngMin                               | WSPBYEMN            | Never                         | long        | motor microsteps | Minimum position of the gimbal-mirror y-axis in engineering units (motor microsteps)      |
| wfs:prbyEngName                              | WSPBYENM            | Start                         | string      |                  | Engineering name of the gimbal-mirror y-axis position <sup>i</sup>                        |
| wfs:prbyEngPos                               | WSPBYEPS            | Start                         | long        | motor microsteps | Position of the gimbal-mirror y-axis in engineering units (motor microsteps)              |
| wfs:prbyInToler                              | WSPBYINT            | Never                         | long        | 0/1              | Is the gimbal-mirror y-axis offset in tolerance? [YES=0   NO=1]                           |
| wfs:prbyLabel                                | WSPBYLBL            | Never                         | string      |                  | The FDSC field describes the gimbal-mirror y-axis                                         |
| wfs:prbyMax                                  | WSPBYMAX            | Never                         | double      | mm               | Maximum position of the gimbal-mirror y-axis in real-world units (mm in the focal plane)  |
| wfs:prbyMin                                  | WSPBYMIN            | Never                         | double      | mm               | Minimum position of the gimbal-mirror y-axis in real-world units (mm in the focal plane)  |
| wfs:prbyName                                 | WSPBYNAM            | Start                         | string      |                  | User name of the gimbal-mirror y-axis position <sup>k</sup>                               |

**TABLE 5.** NIRI/GNIRS On-Instrument Wave-Front Sensor Status Records

| SIR record<br>("niri:" or "nirs:" +) | FITS Keyword | FITS<br>Include<br>d | Type   | Units            | Comments                                                                                  |
|--------------------------------------|--------------|----------------------|--------|------------------|-------------------------------------------------------------------------------------------|
| wfs:prbyParked                       | WSPBYPRK     | Never                | long   | 0/1              | Is the gimbal-mirror y-axis parked?<br>[YES=0   NO=1]                                     |
| wfs:prbyPos                          | WSPBYPOS     | Always               | double | mm               | Position of the gimbal-mirror y-axis position in real-world units (mm in the focal plane) |
| wfs:prbyRdtmVal                      | WSPBYRDT     | Never                | long   | motor microsteps | Returned value from the gimbal-mirror y-axis redatum command                              |
| wfs:prbyReject                       | WSPBYREJ     | Never                | long   | 0/-1             | Combined the gimbal-mirror y-axis CAR state [GOOD=0   REJECT=-1]                          |
| wfs:prbyState                        | WSPBYSTE     | Never                | string |                  | the gimbal-mirror y-axis SNL state                                                        |
| wfs:prbyTol                          | WSPBYTOL     | Never                | double | mm               | Tolerance of the gimbal-mirror y-axis offset in real-world units (mm in the focal plane)  |
| wfs:prbyUnits                        | WSPBYUNT     | Always               | string |                  | Definition of real-world units ("user units") for the gimbal-mirror y-axis ("mm").        |

- a. For the focus stage the position returned by wfs:focEngName will often be "INVALID". This should not be treated as an error, merely as an indication that only a few positions (e.g., the PARK position) have names.
- b. Valid only when the system is in follow mode.
- c. For the focus stage the position returned by wfs:focName will often be "INVALID". This should not be treated as an error, merely as an indication that only a few positions (e.g., the PARK position) have names.
- d. Valid only when the system is in follow mode.
- e. For the gimbal mirror, the position returned by wfs:prbName will often be "INVALID". This should not be treated as an error, merely as an indication that only a few positions (e.g., the PARK position) have names.
- f. For the gimbal mirror x-axis, the position returned by wfs:prbxEngName will often be "INVALID". This should not be treated as an error, merely as an indication that only a few positions (e.g., the PARK position) have names.
- g. Valid only when the system is in follow mode.
- h. For the gimbal mirror x-axis, the position returned by wfs:prbxName will often be "INVALID". This should not be treated as an error, merely as an indication that only a few positions (e.g., the PARK position) have names.
- i. For the gimbal mirror y-axis, the position returned by wfs:prbyEngName will often be "INVALID". This should not be treated as an error, merely as an indication that only a few positions (e.g., the PARK position) have names.
- j. Valid only when the system is in follow mode.
- k. For the gimbal mirror y-axis, the position returned by wfs:prbyName will often be "INVALID". This should not be treated as an error, merely as an indication that only a few positions (e.g., the PARK position) have names.

### 2-3.3.2.3 Component Health Information

The following table summarizes the health information records. All records return one of the strings: “GOOD”, “WARNING”, or “BAD”. Each record describes one aspect of the health of a component

**TABLE 6.** OIWFS Components Controller Component Health Information

| SIR record<br>("niri:" or "nirs:" +) | FITS<br>Keyword | FITS<br>Included | Type   | Units | Comments                                                                         |
|--------------------------------------|-----------------|------------------|--------|-------|----------------------------------------------------------------------------------|
| wfs:fltDatumHealth                   |                 | Never            | string |       | The health of the filter wheel datum state                                       |
| wfs:fltHallHealth                    |                 | Never            | string |       | The health of the filter wheel Hall-effect sensors                               |
| wfs:fltHealth                        |                 | Never            | string |       | The overall health of the filter wheel                                           |
| wfs:fltModuleHealth                  |                 | Never            | string |       | The health of the filter wheel driver                                            |
| wfs:fltPosErrHealth                  |                 | Never            | string |       | The health of the filter wheel backlash correction                               |
| wfs:fltSnIHealth                     |                 | Never            | string |       | The health of the filter wheel state code                                        |
| wfs:focDatumHealth                   |                 | Never            | string |       | The health of the focus stage datum state                                        |
| wfs:focHallHealth                    |                 | Never            | string |       | The health of the focus stage Hall-effect sensors                                |
| wfs:focHealth                        |                 | Never            | string |       | The overall health of the focus stage                                            |
| wfs:focModuleHealth                  |                 | Never            | string |       | The health of the focus stage driver                                             |
| wfs:focPosErrHealth                  |                 | Never            | string |       | The health of the focus stage backlash correction                                |
| wfs:focSnIHealth                     |                 | Never            | string |       | The health of the focus stage state code                                         |
| wfs:folHealth                        |                 | Never            | string |       | The overall health of the dynamic mechanisms (the gimbal mirror and focus stage) |
| wfs:prbHealth                        |                 | Never            | string |       | The overall health of the gimbal mirror                                          |
| wfs:prbxDatumHealth                  |                 | Never            | string |       | The health of the gimbal-mirror x-axis datum state                               |
| wfs:prbxHallHealth                   |                 | Never            | string |       | The health of the gimbal-mirror x-axis Hall-effect sensors                       |
| wfs:prbxHealth                       |                 | Never            | string |       | The overall health of the gimbal mirror x-axis                                   |
| wfs:prbxModuleHealth                 |                 | Never            | string |       | The health of the gimbal-mirror x-axis driver                                    |
| wfs:prbxPosErrHealth                 |                 | Never            | string |       | The health of the gimbal-mirror x-axis backlash correction                       |
| wfs:prbxSnIHealth                    |                 | Never            | string |       | The health of the gimbal-mirror x-axis state code                                |
| wfs:prbyDatumHealth                  |                 | Never            | string |       | The health of the gimbal-mirror y-axis datum state                               |
| wfs:prbyHallHealth                   |                 | Never            | string |       | The health of the gimbal-mirror y-axis Hall-effect sensors                       |
| wfs:prbyHealth                       |                 | Never            | string |       | The overall health of the gimbal mirror y-axis                                   |
| wfs:prbyModuleHealth                 |                 | Never            | string |       | The health of the gimbal-mirror y-axis driver                                    |
| wfs:prbyPosErrHealth                 |                 | Never            | string |       | The health of the gimbal-mirror y-axis backlash correction                       |
| wfs:prbySnIHealth                    |                 | Never            | string |       | The health of the gimbal-mirror y-axis state code                                |

### 2-3.3.3 Position names

Table 7 contains a list of valid position names. Note that these names are case sensitive and must be entered exactly as presented here, including underscores. In addition to the listed

positions, all mechanisms respond to the special names “datum” and “park” which may or may not be distinct from the listed positions.

**TABLE 7.** Position Names (observing commands)

| Component            | Position Names | Engineering Position <sup>a</sup> | Comments                |
|----------------------|----------------|-----------------------------------|-------------------------|
| Filter Wheel         | See [37]       |                                   |                         |
| Gimbal Mirror        | Neutral        | (comp)                            |                         |
|                      | Minimum        | (comp)                            |                         |
|                      | Maximum        | (comp)                            |                         |
| Gimbal Mirror X-Axis | Neutral        | pos03                             | Mirror centered         |
|                      | Minimum        | pos02                             | Minimum travel location |
|                      | Maximum        | pos00                             | Maximum travel location |
| Gimbal Mirror Y-Axis | Neutral        | pos03                             | Mirror centered         |
|                      | Minimum        | pos02                             | Minimum travel location |
|                      | Maximum        | pos00                             | Maximum travel location |
| Focus Stage          | Neutral        | pos00                             | Center of travel        |
|                      | Minimum        | pos01                             | Minimum travel location |
|                      | Maximum        | pos02                             | Maximum travel location |

a. “(comp)” means that the positions are mapped into component positions, and not sent directly to a mechanism.

## 2–3.4 Detailed Command Description

A general table of the commands accepted by the NIRI/GNIRS OIWFS, together with their arguments, is given in Table 1 on page 7. This section contains a detailed description of each of the commands that will be sent between the A&G and the NIRI/GNIRS control software. It is based on the required behavior of the commands specified in [21] and [32].

If a directive is accepted it will cause the NIRI/GNIRS OIWFS to change its configuration. The “applyC” CAR record will go “BUSY” while the OIWFS control software is changing its configuration and go to “IDLE” if completed successfully or to “ERR” if the configuration change fails.

### 2–3.4.1 Sequence Commands

#### 2–3.4.1.1 INIT

On receipt of this command, the OIWFS control software will execute its initialization sequence, in which it will: read its hardware setup files and lookup tables, and reset itself to the start-up state. The “initC” CAR record will go “BUSY” while the initialization is being carried out and go to “IDLE” if completed successfully or to “ERR” if the initialization fails. No mechanisms will be moved.

#### 2–3.4.1.2 DATUM

The OIWFS will locate the datum (i.e. index, home or reference point) for all its mechanisms. The “datumC” CAR record will go “BUSY” while the datum search is being carried out and go to “IDLE” if completed successfully or to “ERR” if the datum searched fails.

#### 2-3.4.1.3 PARK

On receipt of this command, the OIWFS will move its mechanisms to a configuration in which it can safely be switched off. It will utilize all the low-level CAD records dedicated to parking the individual mechanisms. The “parkC” CAR record will go “BUSY” while the mechanisms are being parked and go to “IDLE” if completed successfully or to “ERR” if any mechanism fails to park successfully.

#### 2-3.4.1.4 DIAGNOSE

On receipt of this command, the OIWFS will perform self tests to ensure that its subsystems are healthy. The “diagnoseC” CAR record will go “BUSY” while the tests are being carried out and go to “IDLE” if completed successfully or to “ERR” if the tests fail.

*Note: unlike the TEST command, the DIAGNOSE command will move mechanisms, as necessary to diagnose problems.*

### **2-3.4.2 General Engineering Commands**

#### 2-3.4.2.1 REBOOT

On receipt of this command, the OIWFS control software will reboot its IOC and go through the same start-up procedure it would normally do when first switched on. The working system will reboot into the simulation mode NONE and debug mode NONE. The simulator will reboot into simulation mode FULL and debug mode NONE. There is no CAR record associated with this command because connection with the IOC will be lost while it reboots.

*This command will also reboot the NIRI CC and IS. It will not reboot the GNIRS CC and IS.*

#### 2-3.4.2.2 DEBUG

On receipt of this command, the OIWFS control software will change to the given debug mode (NONE, MIN or FULL). This mode determines the amount and frequency of information logged by the NIRI control software, as described in the “Debugging Modes” section of ICD 1a, [24]. The “debugC” CAR record will go briefly “BUSY” and then back to “IDLE”.<sup>4</sup>

### **2-3.4.3 OIWFS Standard Mechanisms**

The OIWFS will have a single filter wheel, a detector focusing stage, and a two-axis gimbal mirror. The filter wheel is considered a “discrete” mechanisms, because it normally is moved only to one of a small number of locations. The focus stage and both axes of the gimbal mirror are considered continuous mechanisms, because they can move to a wide range of locations.

The following commands will act on all of the above mentioned mechanisms; however, as noted below, the use of certain commands with certain mechanisms is discouraged.

#### 2-3.4.3.1 Sel—Select Position by Name

This is the principal interface for all continuous mechanisms. It is provided as a convenience for discrete mechanisms. Note that for multi-axis components (e.g., the steering mirrors), there are commands that act on the combined components and on the individual mechanisms.

4. Note: This was the required behavior for this system. This requirement was dropped, so the debug command is non-functional.

On receipt of this command, the OIWFS control software will:

Ensure an observation is not taking place.  
 Translate desired position name into an engineering position with the aid of a look-up table.  
 Look up the expected value for the Hall-effect sensors at the desired position location.  
 Check to insure that protective interlocks are not set.  
 Wait until the number of active mechanisms drops below the number of mechanisms that the power supply can support.  
 Move the mechanism to the engineering position corresponding to the desired position.  
 Verify that the expected sensor value matches the measured value.

#### 2-3.4.3.2 Move—Move to specified Position

This is the principal interface for all discrete components. It is available for continuous components, but is of limited use for normal use of the instrument<sup>5</sup>. Note that for multi-axis components (e.g., the steering mirrors), there are commands that act on the combined components and on the individual mechanisms.

On receipt of this command, the OIWFS control software will:

Ensure an observation is not taking place.  
 Check to insure that protective interlocks are not set.  
 Wait until the number of active mechanisms drops below the number of mechanisms that the power supply can support.  
 Move the focus mechanism to the engineering position corresponding to the desired focus.  
 For some mechanisms, compute the expected values for the Hall-effect sensors.  
 For some mechanisms, verify that the final hall effect sensors match the expected values.

#### 2-3.4.3.3 Park—Park Mechanism

This command is used to prepare a mechanism for instrument shutdown. Note that for multi-axis components (e.g., the steering mirrors), there are commands that act on the combined components and on the individual mechanisms.

On receipt of this command, the OIWFS control software will:

Ensure an observation is not taking place.  
 Check to insure that protective interlocks are not set.  
 Wait until the number of active mechanisms drops below the number of mechanisms that the power supply can support.  
 Move the mechanism to its parking position.  
 Verify that the expected sensor value matches the measured value.

The park command is not needed for most OIWFS components, and is provided for uniformity. The single exception is the focus stage. The focus stage should be parked while the instrument being cooled down or warmed up.

#### 2-3.4.3.4 Datm—Datum mechanism

This command is used to find the datum (home) position which defines the mechanism locations. Note that for multi-axis components (e.g., the steering mirrors), there are commands that act on the combined components and on the individual mechanisms.

5. For discrete mechanisms, the units are position numbers; i.e., the first position is at 1.0, the second position is at 2.0.

On receipt of this command, the OIWFS control software will:

Ensure an observation is not taking place.  
 Check to insure that protective interlocks are not set.  
 Wait until the number of active mechanisms drops below the number of mechanisms that the power supply can support.  
 Move the mechanism and search for its datum.  
 Set the hardware datum position.

#### 2-3.4.3.5 MoveEng—Set position in engineering units (motor microsteps)

This command is used to move the components to locations using units that are convenient for working directly with the hardware. Note that for multi-axis components (e.g., the steering mirrors), the command acts only on the individual mechanisms.

On receipt of this command, the OIWFS control software will:

Ensure an observation is not taking place.  
 Check to insure that protective interlocks are not set.  
 Wait until the number of active mechanisms drops below the number of mechanisms that the power supply can support.  
 Move the mechanism to the specified engineering position.

The Hall-effect sensors will not be used for confirmation.

#### 2-3.4.3.6 Rdtm—Redatum mechanism.

This command provides a quick test that will detect a loss of positioning. It will detect small positioning errors, but can underestimate large positioning errors. Note that for multi-axis components (e.g., the steering mirrors), the command acts only on the individual mechanisms. It returns an estimate of the difference between the actual datum position and the current datum position (which can occur, e.g., if a mechanism is sticking, being driven too quickly, or being driven with too high an acceleration).

On receipt of this command, the OIWFS control software will:

Ensure an observation is not taking place.  
 Check to insure that protective interlocks are not set.  
 Wait until the number of active mechanisms drops below the number of mechanisms that the power supply can support.  
 Move the mechanism and search for its datum.

The hardware datum position will be left unchanged. This command returns a value in the RdtmVal record for this mechanism.

*This command is not supported by the pupil viewer or the window cover, because those mechanisms use a home-finding algorithm which is incompatible with this command.*

#### 2-3.4.3.7 Diag—Diagnose mechanism

This is a quick test that will detect a loss of positioning. It will detect arbitrarily large positioning errors, but will fail to detect small positioning errors. Note that for multi-axis components (e.g., the steering mirrors), command acts only on the individual mechanisms.



On receipt of this command, the OIWFS control software will:

Ensure that an observation is not taking place.  
 Check to insure that protective interlocks are not set.  
 Wait until the number of active mechanisms drops below the number of mechanisms that the power supply can support.  
 Move the mechanism and attempt to verify that the motors and Hall-effect sensors are operating.

If an error is detected, the diagnose command will fail and will return an error code in its CAR record.

## 2-3.5 Detailed Status information

### 2-3.5.1 General records

#### 2-3.5.1.1 name

This record contains a string constant describing the name of the instrument. For NIRI this record will contain “NIRI WFS”; for GNIRS the record will contain “NIRS WFS”.

#### 2-3.5.1.2 state

This record will take the values BOOTING, INITIALISING, RUNNING and CONFIGURING and reflect the state of the OIWFS control software while it is initializing. The activities corresponding to these states are

- BOOTING — the state during IOC Init and while any database default values are being set.
- INITIALISING — the OIWFS control software will be checking its own hardware and reading its hardware configuration files during this phase.
- RUNNING — the OIWFS control software is ready to accept commands.
- CONFIGURING — the OIWFS control software is ready to accept commands but it is busy configuring and unable to make observations.

#### 2-3.5.1.3 health

This record will reflect the overall health of the OIWFS. Expected values are GOOD, WARNING or BAD. If the overall health is BAD then the OIWFS will be unusable. If the Health is set to WARNING then the OIWFS will be able to execute some functions but not perhaps to specification.

#### 2-3.5.1.4 historyLog

The record used to contain messages to be recorded in the history log.

### 2-3.5.2 Component Status Records

#### 2-3.5.2.1 Datumed

This record indicates that a component has been datumed. For multi-axis components, this flag will only be set if all mechanisms have been datumed. This record takes on the following values:

- 0 = Datumed
- 1 = Not Datumed

2-3.5.2.2 EngCyclic

This record indicates that a mechanism may be set to a location beyond its physical limits (convenient for circular mechanisms like filter wheels). Note that this only affects the higher level commands; the low-level software may choose to ignore any invalid requests. This record takes on the following values:

- 0 = Limits can be exceeded
- 1 = Limits may not be exceeded

*This record is for internal use by the OIWFS software. Its value should not be changed. It is documented here only for completeness.*

2-3.5.2.3 EngMax

This record contains the maximum position that the mechanism can be moved to, in engineering units (motor microsteps).

2-3.5.2.4 EngMin

This record contains the minimum position that the mechanism can be moved to, in engineering units (motor microsteps).

2-3.5.2.5 EngName

This record contains the engineering name of the current mechanism position. This represents a physical configuration of a component, and will never change (unlike the “Name” record which may change, if filters or other optical elements are replaced or rearranged). The names of the positions are:

- park;
- home;
- pos00, pos01, pos02, ... , up to the maximum number of positions;
- INVALID (if not in a standard location).

2-3.5.2.6 EngPos

This record contains the current position of a mechanism, in engineering units (motor microsteps).

2-3.5.2.7 Label

The FDSC field of this record contains a text string which describes a component. This is convenient in providing a title string for edd/dm screens.

2-3.5.2.8 Max

This record contains the maximum position that the mechanism can be moved to, in real-world units. A text string describing the real-world units may be found in the Units record. For mechanisms such as a filter wheel in which there are no appropriate units, the position will typically be given as a position number, i.e., position 0 is 0.0, position 1 is 1.0.

2-3.5.2.9 Min

This record contains the maximum position that the mechanism can be moved to, in real-world units. A text string describing the real-world units may be found in the Units record. For mechanisms such as a filter wheel in which there are no appropriate units, the position will typically be given as a position number, i.e., position 0 is 0.0, position 1 is 1.0.

2-3.5.2.10 Name

This record contains the name of the current component position. This name typically describes a filter or other optical element and will be redefined if filters or other optical elements are replaced or rearranged (unlike the “EngName” record which describes a component’s physical configuration). The current names are listed in [36].

Note that for multi-axis components, there will be a Name record for each individual mechanism, as well as for the component as a whole.

2-3.5.2.11 Parked

This record indicates that a component is in a parked state. For multi-axis components, this record will be set only if all mechanisms are parked. This field takes the values:

- 0 = All mechanisms parked;
- 1 = Not all mechanisms are parked.

2-3.5.2.12 Pos

This record contains the current position of a mechanism, in real-world units. A text string describing the real-world units may be found in the Units record. For mechanisms such as a filter wheel in which there are no appropriate units, the position will typically be given as a position number, i.e., position 0 is 0.0, position 1 is 1.0.

2-3.5.2.13 RdtmVal

This record contains the result of the last redatum command, in engineering units (motor microsteps). (The redatum command returns an estimate of the how far the current mechanism datum position is from the actual datum position; it is useful in diagnosing hardware problems.)

2-3.5.2.14 Reject

This record contains a combination of the values of all CAD records for all mechanisms that make up a component. It contains the following values:

- -1 = At least one CAD record is in a reject state;
- 0 = No CAD records are in a reject state.

*This record exists as a convenience for the user interface. There is no reason for the OCS to use it. It is documented here only for completeness.*

2-3.5.2.15 State

This record contains a string which describes the current state of the mechanism-support code.

*This record is for debugging the OIWFS software. The possible values that this record may take may change, without notice. It is documented here only for completeness.*

2-3.5.2.16 Units

This record contains a text string which describes the real world units (e.g., “mm” or “arc sec”)

**2-3.5.3 Component Health Records**

All of these records return the standard health values: “GOOD”, “WARNING”, and “BAD”.

2-3.5.3.1 DatumHealth

This record reports “WARNING” if a mechanism (or if any mechanism of a multi-axis component) is not datumed. It is cleared by successful execution of a datum command.

2-3.5.3.2 HallHealth

This record reports “WARNING” if a Hall-effect sensor has been disabled. (The sensors can only be disabled from the engineering user interface, and are normally disabled only to deal with a failed sensor.)

This record reports “BAD” if both sensors of a Hall-effect sensor pair have been disabled. Note that a mechanism is still usable if both sensors are disabled, but cannot be datumed. (The sensors can only be disabled from the engineering user interface, and are normally disabled only to deal with a failed sensor.)

2-3.5.3.3 Health

This record reports the most severe error value of all other health records.

2-3.5.3.4 ModuleHealth

This record reports “BAD” if the stepper-motor driver module has entered a fault state. This normally indicates a serious hardware problem. It may be possible to clear the error state by issuing a driver-reset from the engineering user interface. If it is not possible to clear the error, then the stepper-motor driver module or other related hardware has failed, and must be repaired.

2-3.5.3.5 PosErrHealth

This record reports “WARNING” if a mechanism (or if any mechanism of a multi-axis component) has been moved in such a way that proper backlash correction has not been made. In this case, the reported mechanism position may not be the actual mechanism position. This situation can arise if a movement command has been canceled while a motion is in progress; if a movement command failed to complete correctly, if certain low-level engineering commands are executed, and immediately after the control software has been rebooted. It is cleared by successful execution of any motion command.

2-3.5.3.6 SnlHealth

This record reports “BAD” or “WARNING” if the underlying mechanism support code detects an error while a mechanism is in motion. The only situation which is likely to arise is “WARNING”, if an operation takes longer than expected.

**2-3.6 Alarm conditions**

The following situations will generate an alarm.

- Hall effect sensor disabled
- Motor-control module fault
- Mechanism not datumed
- Mechanism not backlash corrected
- Problem detected by SNL code (time-out)
- Temperature changing
- Temperature out of range

### 2-3.7 Debugging and Maintenance

This interface can be tested and debugged by means of an engineering user interface supplied as part of the OIWFS.

#### 2-3.7.1 Debugging Modes

The OIWFS control software will have the standard debugging modes described in ICD 1a, [24].

#### 2-3.7.2 Calibration

If a hall sensor fails or if a mechanism must be disassembled for maintenance, the sensors will need to be recalibrated. A separate program is be provided for this purpose. Calibration procedures are be described in a separate document [5].

### 2-3.8 Simulation

The Gemini simulation modes are described in ICD 1a, [24]. The OIWFS control software supports only the “NONE” mode. The simulator supports only the “FULL” and “FAST” modes.

The “FAST” simulation mode removes most of the simulated delays which are built into the “FULL” simulation mode, but does not fully meet the Gemini specification for the FAST simulation mode. The “VSM” mode is not available.

To switch from operating mode to simulation mode requires an alternate start-up script to be selected, and a reboot (see the release notes [7]). The command to switch between the “FAST” and “FULL” simulations modes is an engineering-only command and does not fall within the scope of this document (see [8]).

---

## 2-4.0 Safety Issues

It will be necessary to avoid moving OIWFS components while the mechanical components are not at thermal equilibrium. Records are be provided to provide a temperature interlock. Details are described in [6].



# Gemini Near Infra-Red Imager and Wavefront Sensor



# Near IR Imager (NIRI) and OIWFS — OIWFS Component Position Names

**Hubert Yamada**

niri\_h ty\_014 / 01 / 1999-11-23

This document describes the position names for the OIWFS mechanisms (i.e., filter names).

## 1 Introduction

### 1.1 Purpose

This document describes the position names for the OIWFS mechanisms. In general, information which logically belongs within an ICD but which is subject to change with relatively trivial hardware changes are presented here. In particular, the names of the various filters contained within the different OIWFS filter wheels are presented here.

The intended audience of this document is:

- The NIRI reviewers;
- The developers of the Gemini Observatory Control System (OCS);
- The maintainers of NIRI after it is delivered to Gemini;
- Any groups that will be using the NIRI OIWFS.

### 1.2 Definitions, Acronyms and Abbreviations

See “Near IR Imager (NIRI)—Definitions, Abbreviations, and References”.

### 1.3 Stylistic Conventions

References to documents in Section 1.5, “References,” on page 2 of “Near IR Imager (NIRI)—Definitions, Abbreviations, and References” are given like this; [1].

## 1.4 References

See “Near IR Imager (NIRI)—Definitions, Abbreviations, and References”.

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## 2 Tables

---

### 2.1 Filter Wheel

Table 1 describes the named positions for the filter wheel.

---

**TABLE 1.**

Named Positions for Filter Wheel

| Engineering Position Name | Position Name |
|---------------------------|---------------|
| pos00                     | open0         |
| pos01                     | open1         |
| pos02                     | J/53.8        |
| pos03                     | J/537         |
| pos04                     | H/26.9        |
| pos05                     | H/214.8       |
| pos06                     | K/107.4       |
| pos07                     | K/1074        |
| pos08                     | K'/107.4      |
| pos09                     | K'/214.8      |
| pos10                     | open/214.8    |
| pos11                     | open11        |



## ICD 1.9/1.10[ab]

### “NIRI/GNIRS Science Instrument to On Instrument Wavefront Sensor”

niri\_h ty\_005

Hubert Yamada  
2001-06-27

Version: 1.0

Issued By: Hubert Yamada

Approved By: \_\_\_\_\_ Group Manager Date: \_\_\_\_\_  
\_\_\_\_\_ Group Manager (if req) \_\_\_\_\_  
\_\_\_\_\_ Systems Engineering \_\_\_\_\_

## Revision Control

- |                             |                                                        |
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| 1. Revision Version #       | 1.00alpha                                              |
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# Chapter 1 Introduction

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## 1–1.0 Description

---

This document describes the NIRI and GNIRS instrument-to-wavefront-sensor-components interlock system (not to be confused with the Gemini Interlock Bus). These interlocks provide a way for the instrument to inform the wavefront sensor that it should not move its components at this time.

### 1–1.1 Purpose

This Interface Control Document (ICD) serves the following purposes:

- To describe the commands and parameters that will pass between the Near Infra-Red Imager (NIRI) and its on instrument wavefront sensor (OIWFS) and between the Gemini Near Infra-Red Spectrograph (GNIRS) and its OIWFS.

The intended audience for this document is:

- the NIRI reviewers;
- the GNIRS reviewers and developers.

### 1–1.2 Scope

This document describes commands which pass between NIRI/GNIRS and their respective OIWFS's.

This document does *not* describe the following topics:

- the NIRI detector and components, (see [18], [11], and [12], summarized in [3]);
- the GNIRS hardware and electronics, (see [18] and (???))
- the OIWFS hardware and electronics, (see [11] and [12], summarized in [3]);
- the NIRI/GNIRS OIWFS detector and components (see [11], [12], and [13], summarized in [3]);

The software design is not described here. For that, see [2] (currently under preparation), and [3] (currently under preparation).

### 1–1.3 Differences between NIRI and GNIRS

For GNIRS, the commands here affect only the OIWFS. For NIRI, the commands affect the entire instrument.

## 1–2.0 Related Documents and Drawings

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### 1–2.1 References

See “Near IR Imager (NIRI)—Definitions, Abbreviations, and References”.

**1–2.2 Definitions, Acronyms and Abbreviations**

See “Near IR Imager (NIRI)—Definitions, Abbreviations, and References”.

**1–2.3 Stylistic Conventions**

References to documents in Section 1.5, “References,” on page 2 of “Near IR Imager (NIRI)—Definitions, Abbreviations, and References” are given like this; [1].

## Chapter 2 Details of the NIRI/GNIRS to OIWFS Interface

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### 2–1.0 Introduction

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### 2–2.0 Physical System Interfaces

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#### 2–2.1 Mechanical Interface

*Not Applicable for a software ICD.*

#### 2–2.2 Optical Interface

*Not Applicable for a software ICD.*

#### 2–2.3 Electronic Interface

*Not Applicable for this ICD.*

#### 2–2.4 Mass/Balance

*Not Applicable for a software ICD.*

#### 2–2.5 Thermal Interface

*Not Applicable for a software ICD.*

### 2–3.0 Software/Control Function Interface (wavefront sensor)

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#### 2–3.1 Overview

##### 2–3.1.1 Communications infrastructure

The communications infrastructure is the same as described in [16].

#### 2–3.2 Implementation

This section contains tables summarizing the EPICS records used for the interface. The next sections (Section 2–3.2.1.1 on page 6) contain the details of what those records do.

It is assumed in all cases that the record names given in this document will be prefixed by the string “niri:” or “nirs:” when the database is implemented, except for the SAD database which might be prefixed by “niri::” or “nirs::”.

##### 2–3.2.1 Command (CAD/CAR) records

The columns of the command record tables are as follows:

**Description.** A brief description of the command.

**Attribute.** The name of the attribute which the Observatory Control System associates with a particular parameter of a particular command.

**CAD record/field.** The name of the EPICS CAD record (minus the “niri:” or “nirs:” prefix) together with the field in that CAD record associated with each parameter of the command.

**CAR record.** The CAR record associated with the command. Note that several CAD records may share the same CAR record.

These commands are executed by asserting the top level WFS apply record.

**TABLE 1.** NIRI/GNIRS OIWFS CAD/CAR records

| Description             | Attribute                                                           | CAD record:field (“niri:wfs:” +) | CAR record (“niri:wfs:” +) |
|-------------------------|---------------------------------------------------------------------|----------------------------------|----------------------------|
| Configuration interlock | Lock for configuration (string, “LOCK”   “UNLOCK”)                  | lockCfgSel.A                     | lockC                      |
| Observation interlock   | Lock for observation (string, “LOCK”   “UNLOCK”)                    | lockObsSel.A                     |                            |
| General interlock       | Lock (string, “LOCK”   “UNLOCK”)                                    | lockGenSel.A                     |                            |
| Temperature interlock   | Lock for cool down / warm up (string, “WARM”   “COLD”   “CHANGING”) | lockTmpSel.A                     |                            |
| Datum mechanism         | Datum mechanisms                                                    | datum                            | datumC                     |

#### 2–3.2.1.1 Detailed Command Description

##### 2–3.2.1.2 lockCfgSel

An instrument may have limitations on which mechanisms which may be moving simultaneously. This interlock is intended to prevent the wavefront sensor components from moving while the instrument is being reconfigured, if it is possible for wavefront sensor components to be damaged. There are two possible states:

**“LOCK”.** Hardware is prevented from moving. Any new commands which might cause the hardware to move will fail.

**“UNLOCK”.** Hardware is permitted to move (if no other locks are set).

The lock may be set at any time; however, it does not interrupt any currently executing commands. (See section 2–6.0, “Activity Completion,” p. 11, below.)

*This command should be used with caution. Excessive use of this lock will interfere with the proper functioning of the OIWFS (and NIRI) and will keep the OIWFS (and NIRI) from obeying commands that are issued by other systems (e.g., the Acquisition and Guiding sub-system).*

*This command is not needed by NIRI.*

2-3.2.1.3 lockObsSel

This command is provided to provide a low noise environment for sensitive observations. There are two possible states.

**“LOCK”**. Hardware is prevented from moving. Any new commands which might cause the hardware to move will fail.

**“UNLOCK”**. Hardware is permitted to move (if no other locks are set).

The lock may be set at any time; however, it does not interrupt any currently executing commands. (See section 2-6.0, “Activity Completion,” p. 11, below.)

*This command should be used with caution. Use of this lock will interfere with the proper functioning of the OIWFS (and NIRI) and will keep the OIWFS (and NIRI) from obeying commands that are issued by other systems (e.g., the Acquisition and Guiding subsystem).*

*This command is not needed by NIRI.*

2-3.2.1.4 lockGenSel

This command is provided as a “general” command to lock all hardware for any unspecified reason.

There are two possible states:

**“LOCK”**. Hardware is prevented from moving. Any new commands which might cause the hardware to move will fail.

**“UNLOCK”**. Hardware is permitted to move (if no other locks are set).

The lock may be set at any time; however, it does not interrupt any currently executing commands. (See section 2-6.0, “Activity Completion,” p. 11, below.)

*This command should be used with caution. Use of this lock will interfere with the proper functioning of the OIWFS (and NIRI) and will keep the OIWFS (and NIRI) from obeying commands that are issued by other systems (e.g., the Acquisition and Guiding subsystem).*

*This command is not needed by NIRI.*

2-3.2.1.5 lockTmpSel

The OIWFS motors (and, possibly, other components) can be damaged if a move command is issued before the system has reached a uniform temperature throughout. Furthermore, the physical parameters of the system (e.g., the datum position) may depend on the temperature. The lockTmpSel command informs the OIWFS about the current temperature of the system. There are three possible states:

**“WARM”**. Hardware is permitted to move (if no other locks are set). The system is at or near room temperature.

**“COLD”**. Hardware is permitted to move (if no other locks are set). The system is at or near its operating temperature.

**“CHANGING”.** Hardware is not permitted to move. The system is currently being cooled down or warmed up.

When the temperature setting is changed, the OIWFS will enter the not-datumed state. (The datum shifts slightly, depending on the temperature of the system.) Upon reboot, the system always begins in a “CHANGING” state, until explicitly set to another state.

The lock may be set at any time; however, it does not interrupt any currently executing commands. (See section 2–6.0, “Activity Completion,” p. 11, below.)

It is always better to be conservative and leave the state of the system “CHANGING”, until it can be guaranteed that the OIWFS mechanisms have reached their steady-state temperature.

This command also sets the value of various internal temperature-dependent parameters within the OIWFS. If this parameter is set incorrectly, mechanisms will move, but may incorrectly report that there is a hardware problem.

*This command should be used with caution. Use of this lock will interfere with the proper functioning of the OIWFS (and NIRI) and will keep the OIWFS (and NIRI) from obeying commands that are issued by other systems (e.g., the Acquisition and Guiding subsystem).*

#### 2–3.2.1.6 DATUM

This command will cause all mechanisms to be datumed. Note that the OIWFS does *not* automatically datum itself when a move command is issued.

*This command should be used with caution. Use of this lock may interfere with the proper functioning of the OIWFS and will keep the OIWFS from obeying commands that are issued by other systems (e.g., the Acquisition and Guiding subsystem).*

### **2–3.2.2 Status Information Records**

The following tables show the status information provided by the NIRI/GNIRS OIWFS components controller which are of importance to the instrument as a whole.

Unless otherwise noted, the SIR records described here are to be used for status information, and not to be used for real-time control purposes. It is possible that rapid changes in the underlying system may not be reflected in the SIR records.

The tables have the following columns:

**SIR record.** The part of the name of the EPICS SIR record which follows the “niri:” prefix<sup>1</sup>.

**Units.** The units in which the quantity is stored

**Comments.** A description of the item. The first paragraph of this will go into the SIR record description and also the comment field in the FITS header.

---

1. Note that the actual system has been designed so the Status Alarm Database can easily be given a different prefix from the rest of the NIRI database (for example “niri:”). It is assumed in this document that the Status Alarm Database prefix will be “niri:”.



Note that only the type, units and comments are actually contained in the SIR records. The FITS keyword is provided by the DHS.

**TABLE 2.** NIRI/GNIRS OIWFS status Information

| SIR record<br>("niri:wfs:" +) | FITS<br>keyword | FITS<br>includ<br>ed? | Type   | Units | Comments                                                                                                                                                                                                                                                                                                                               |
|-------------------------------|-----------------|-----------------------|--------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| lockHealth                    |                 | no                    | string |       | Health of subsystem ("GOOD"   "WARNING"   "BAD"). Intended to be combined with overall instrument Health. This can also be set if there is a hardware or configuration problem with the database. Diagnosis requires the use of the Engineering user interface, and is not considered a function of the Instrument to OIWFS interface. |
| lockCfg                       |                 | no                    | long   |       | Locked for configuration (non-zero = locked). If this is set, the system will be put into a "WARNING" state.                                                                                                                                                                                                                           |
| lockObs                       |                 | no                    | long   |       | Locked for observation (non-zero = locked). If this is set, the system will be put into a "WARNING" state.                                                                                                                                                                                                                             |
| lockGen <sup>a</sup>          |                 | no                    | long   |       | Locked (generic) (non-zero = locked). If this is set, the system will be put into a "WARNING" state.                                                                                                                                                                                                                                   |
| lockTmp                       |                 | no                    | long   |       | Locked for cool down / warm up (non-zero = locked. I.e., lockTmpVal is "CHANGING". If this is set, the system will be put into a "WARNING" state.                                                                                                                                                                                      |
| lockInit                      |                 | no                    | long   |       | Locked for initialization (non-zero = locked)? If this is set, the system will be put into a "WARNING" state.                                                                                                                                                                                                                          |
| lockLock                      |                 | no                    | long   |       | Combined value ( <i>logical or</i> ) of lockCfg, lockObs, lockGen, lockTmp, and lockInit.                                                                                                                                                                                                                                              |
| lockTmpVal                    |                 | no                    | string |       | Current temperature state ("COLD"   "WARM"   "CHANGING <sup>b</sup> ")                                                                                                                                                                                                                                                                 |
| lockMaxMove                   |                 | no                    | long   |       | How many motors may be moved at the same time. <sup>c</sup>                                                                                                                                                                                                                                                                            |
| lockActive                    |                 | no                    | long   |       | How many motors have been commanded to move and are waiting to move or are currently moving <sup>c</sup> .                                                                                                                                                                                                                             |
| lockMoving                    |                 | no                    | long   |       | How many motors are currently moving <sup>c</sup> .                                                                                                                                                                                                                                                                                    |
| wfsBeam                       |                 | no                    | long   |       | Flag to indicate that the WFS components are allowing light to fall on the on-instrument wavefront sensor. <sup>d</sup>                                                                                                                                                                                                                |

- The instrument should not trigger any special actions based on the state of the lockGen record, because it is intended that the lockGen should be usable at any time, for any reason, without side effects.
- For safety, when NIRI or the standalone OIWFS is rebooted, the system always restarts in the "CHANGING".
- For NIRI, this number is the total of the NIRI CC motors and NIRI OIWFS motors, *excluding the gimbal (prbx, prby) motors, which may be moved at any time*. For GNIRS, and any other non-NIRI systems, this number only represents the OIWFS motors. If the system attempts to move a motor when the maximum number of motors is already in motion, then the motion of that motor will silently be delayed until the requested action may be carried out; no error or warning will be issued.
- It is the responsibility of the instrument to combine the value of the wfs:wfsBeam with information from the components controller to produce the value of the global wfsBeam SIR record. This field indicates only that there is no obstruction, not that light is actually getting to the detector. Furthermore, it does not consider the obscuration caused by the pick-off mirror.

### 2-3.2.3 Detailed SIR record description

### 2-3.2.3.1 The lockMaxMove, lockActive, and lockMoving records

The lockMaxMove, lockActive, and lockMoving records provide a mechanism to restrict the number of mechanisms which may move simultaneously (necessary, because of power supply restrictions). The lockMaxMove record contains the maximum number of mechanisms which may be moved simultaneously. The lockMoving record contains the number of mechanisms which are currently moving. The lockActive record contains the number of mechanisms which are moving or which are waiting to move.

If an attempt is made to move more mechanisms than are permitted, then the system will silently postpone the motion, so that no more than the maximum number of mechanisms are in motion.

This restriction does not apply to the gimbal (prbx, prby) mechanisms, which are permitted to move at any time.

## 2-4.0 Software/Control Function Interface (Components Controller)

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### 2-4.1 Overview

#### 2-4.1.1 Communications Infrastructure

The communications infrastructure is the same as described in [16].

### 2-4.2 Implementation

The instrument may control the temperature interlock system by writing to the lock records which are described above. Alternatively, it may provide a status and information record<sup>2</sup> which will be used to automatically update the lock records.

#### 2-4.2.1 Monitoring an external record.

The temperature interlock may be set by using a record in another IOC. This feature must be enabled as a configuration option (see [7], section 5.4, “Configuration,” p. 7 and section 6.4, “Configuration,” p. 10). *All three of the variables defined below must be set as described, or all three must be left unset.*

Set the configuration variable *NIRI\_TMPVAR* to the name of a field of an EPICS record which contains a numeric value. If the field contains a 0, the value of the temperature interlock will be “CHANGING.”; a 1, the value is “COLD”; and a 2, the value is “WARM”.

The configuration variable *NIRI\_TMPSCAN* should be set to the rate at which the field defined by *NIRI\_TMPVAR* should be monitored. This should be a valid capfast scan rate, e.g., “1 second”. There is no reason to scan at a rate faster than “1 second.”

The configuration variable *NIRI\_TMPHB* should be set to the name of a field which is changing at least as frequently as *NIRI\_TMPSCAN*, and which resides within the same IOC as *NIRI\_TMPVAR*. This is used to detect communications problems and IOC problems, and

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2. Gemini requires a SIR record to be the interface between two systems. The software will work correctly with any record that can return a numeric value.

will automatically force the system into a “CHANGING” (locked) state if there is a problem.

#### 2–4.2.2 Status Information Records

The following tables show the status information provided by the NIRI/GNIRS OIWFS components controller which are of importance to the instrument as a whole. The tables have the following columns:

**SIR record.** The part of the name of the EPICS SIR record which follows the “niri:” prefix<sup>3</sup>.

**Units.** The units in which the quantity is stored

**Comments.** A description of the item. The first paragraph of this will go into the SIR record description and also the comment field in the FITS header.

*Note that only the type, units and comments are actually contained in the SIR records. The FITS keyword is provided by the DHS.*

**TABLE 3.** NIRI/GNIRS OIWFS status Information

| SIR record<br>(“niri:wfs:” +) | FITS<br>keyword | FITS<br>includ<br>ed? | Type   | Units | Comments                                                                                                                              |
|-------------------------------|-----------------|-----------------------|--------|-------|---------------------------------------------------------------------------------------------------------------------------------------|
| lockTmpVal                    |                 | no                    | string |       | Current temperature state (“COLD”   “WARM”   “CHANG-<br>ING”)                                                                         |
| lockTmpHB                     |                 | no                    | long   |       | A non-zero value indicates a communications error or a<br>problem with the IOC containing $\{NIRI\_TMPVAR\}$ and<br>$\{NIRI\_TMPHB\}$ |

#### 2–5.0 Safety Issues

The instrument software must always call lockTmpSel when the system is expected to be at a non-equilibrium temperature. If the alternative interface is used (see section 2–4.2, “Implementation,” p. 10), the sir records which are being monitored must hold the appropriate temperature state. *Various components may not operate correctly (and could be damaged) if they are moved when there is a significant temperature gradient across the component (more than 50 degrees C).*

#### 2–6.0 Activity Completion

The interlocks only prevent new actions from starting. They do not interrupt currently executing commands.

3. Note that the actual system has been designed so the Status Alarm Database can easily be given a different prefix from the rest of the NIRI database (for example “niri:”). It is assumed in this document that the Status Alarm Database prefix will be “niri:”.

In order to be sure that the filter wheel and the focusing stage have stopped, it is necessary to monitor their CAR records. These records are standard CAR records; when the mechanism is idle, these records will have the value “IDLE”.

The Gimbal Mirror is unusual; if it is in “follow” mode, it is possible for the mechanism to be in motion, even if the CAR record shows “IDLE”. For this reason, the CAR record should not be used to determine mechanism activity. Instead, use the IsIdle SIR record. When this SIR record has a zero value, the gimbals are guaranteed to be inactive.

**TABLE 4.** NIRI/GNIRS OIWFS Activity Records

| Record<br>("niri:wfs:" +) | Record Type | Type                    | Units | Comments                                 |
|---------------------------|-------------|-------------------------|-------|------------------------------------------|
| IsIdle                    | SIR         | long<br>0=TRUE, 1=FALSE |       | Both axes of gimbal mirror are inactive. |
| filtC                     | CAR         | Standard CAR values     |       | State of the filter wheel.               |
| focC                      | CAR         | Standard CAR values     |       | State of the focusing stage.             |

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Infra-Red  
Imager  
and  
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Sensor**



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# Near IR Imager (NIRI) Software— Project History Document

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**Hubert Yamada**

niri\_h ty\_010 / 02 / 1999-10-23

**This report is a description of lessons learned while developing the NIRI software.**

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## 1 Introduction

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This report is a summary of lessons learned during the process of developing the NIRI software.

This report was heavily influenced by [34].

### 1.1 Purpose of the document

This report is a summary of useful information that was learned during the development of the NIRI software. It is provided here as a guide to future projects that may choose to use NIRI as a basis for project development.

The intended audience of this document is:

- the Gemini instrument development community;
- the Gemini Controls group.

### 1.2 Scope

No software project will be developed from this document.

### 1.3 Definitions, Acronyms and Abbreviations

See “Near IR Imager (NIRI)—Definitions, Abbreviations, and References”.

## **1.4 Stylistic Conventions**

References to documents in Section 1.5, “References,” on page 2 of “Near IR Imager (NIRI)—Definitions, Abbreviations, and References” are given like this; [1].

## **1.5 References**

See “Near IR Imager (NIRI)—Definitions, Abbreviations, and References”.

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## **2 Caveats**

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At this point in time, the NIRI software developers feel that NIRI, as implemented, is not a good choice as a starting point for other software projects. There are some severe flaws in the current design that should not be propagated to other instrument projects.

This document is not intended to be a criticism of the designers and implementers of the CICS and NIRI software. It is recognized that there were severe design constraints which were placed on them at an early stage, which forced some unwise design decisions. Although many of these constraints were later lifted, it was too late to correct the implementation of the software.

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## **3 History**

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The software design for NIRI began in the early part of 1997; NIRI is expected to be delivered in late 1999. During this time, the design of NIRI has changed significantly for many reasons.

### **3.1 Early attempts**

Originally, the NIRI developers were told that very little software development would be necessary. The CICS was a prototype system which would be provided for use as a template for the instrument, which could easily be customized to fit the NIRI mechanisms. It was to be the simple process of rearranging a few capfast diagrams, and adding links to existing records, to provide the actual mechanism control. At that time, the design for NIRI seemed quite straightforward. Very little time had been budgeted for software development, on the belief that the design would be trivial, based largely on an existing design. The implementation was also expected to be a relatively simple task, since it was expected to be a minor modification of an existing implementation.

At that early stage, the only task for the NIRI software developers was to learn EPICS, which appeared to be a relatively simple task. At that time, no attention was paid to the process of record development, indeed, to any of the mechanisms for embedding C code within an EPICS system. Simple EPICS databases were built. From this early experience, it seemed that building a mechanism-control database would be a fairly simple task.

A copy of the CICS was obtained fairly quickly. Unfortunately, this was not the final CICS release, nor would the final release be available for some time, because the CICS project was

experiencing unexpected delays. The CICS turned out to be quite a bit larger and more complicated than expected; however, at the time, this did not seem to be a major problem, because it was believed that it would not be necessary to deal with most of the complexity of the system. It was thought that the bulk of the implementation could be treated as “black boxes,” not requiring understanding of the details.

The first task seemed obvious: Customize the CICS to produce a working prototype for the NIRI system. It was felt that the working prototype would be updated when the final version of the CICS was provided, and it was believed that the prototype could trivially be modified to control actual motors. Although it was realized that this was an early release of the CICS, it was thought that it would be fairly easy to upgrade the system to a new CICS release, because the bulk of the implementation would be left untouched.

Unfortunately, it was found that the CICS was far more difficult to modify than expected. Weaknesses of capfast became obvious, and the extremely large number of connections required were found to be quite unwieldy. Furthermore, the existing diagrams had not been designed with room for expansion, requiring a significant amount of effort to increase the number of mechanisms from the small number that were present in the simulated database. Furthermore, the diagrams were extremely difficult to understand. It took considerable time to figure out precisely what each mechanism was doing. (In later releases, comments the CICS developers added comments, which reduced the amount of effort required to understand the diagram.) Ultimately, however, a working prototype for the NIRI mechanisms was completed, and the NIRI developers had developed a much better understanding of the way that the CICS worked.

Soon after, however, a new version of the CICS was released. This too, was not the final release. It was immediately obvious that there was a problem. Minor changes had been made throughout the entire system. A shortcoming of capfast suddenly became painfully visible: even the most trivial changes in a capfast diagram made significant changes in the underlying schematic files. Although the files were text-based, trying to compare the new version to the older version was fruitless. It was clear that the prototype would have to be re-implemented using the newer version of the CICS. Because this was not the final version of the CICS, it was felt that any time spent in developing a CICS-based prototype would be wasted, so work was temporarily suspended on the public interface to NIRI.

While the work on the prototype was taking place, early prototype mechanisms were being designed and fabricated, and motors were available for testing. Simple databases were created, and, in surprisingly short time, it was possible to create a database to control the motor using the stepper motor record. It was also quite easy to read out the values from the Hall effect sensors. This early success was to prove misleading, however, because further testing showed the stepper motor record to be quite unreliable. It took quite a bit of time to isolate the problems with the stepper motor record, and at the time the NIRI developers discovered far more about the inner workings of EPICS than was desired; though this knowledge would eventually turn out to be quite handy. The stepper motor record was never completely reliable, though.

At this point, an attempt was made to implement the proposed home finding algorithm. This required the Hall effect sensors to be read, while moving the motor. When the strongest signal was read from the sensor, that would be the motor home position. It was realized that a more sophisticated algorithm would be needed for the final system, but that this would be sufficient for an early prototype.

At this time, the NIRI developers were constrained by the requirement—verbally presented, but apparently mandatory, and not negotiable—that everything be done using capfast and the standard EPICS records. The subroutine record had explicitly been discouraged, though it might be allowed to be used. A significant amount of effort was expended in trying to create a search algorithm using these tools. Eventually, a diagram was created that was able to perform the necessary functions, to some extent. However, it was clear from the complexity of the system that implemented the simplest possible algorithm, that the effort required in implementing any refinements would be prohibitive.

At that time, the NIRI developers were still attempting to fulfil the all requirements, both written and verbal. Inspection of the CICS showed that the EPICS state notation language (SNL) was used extensively, so it was felt that SNL was a permissible development tool. Implementing a simple search algorithm proved to be relatively simple to implement. However any attempts to refine the algorithm quickly exposed weaknesses in SNL. In particular, the lack of simple ways to perform more than the most trivial loops was a constant source of aggravation, as was the poor support for code reuse. It was clear that implementing any sort of useful algorithm would require far more time and effort than was available, and that the resulting system would be completely inflexible and impossible to maintain.

### **3.2 The first working system.**

At this point, it was clear that, as designed, the NIRI software would never be built. The available tools were simply insufficient, and frustration was building. Fortunately, at this time the attitude at Gemini had been changing. Experience with actual systems, and lack of progress on the part of the software groups had led to a more liberal attitude. The arbitrary restrictions on the use of the full EPICS tool set were quietly dropped, and we were informed that we could use anything that we felt was necessary, though Gemini did not want to see the entire NIRI control system embedded within a single subroutine record.

At that point, embedding the entire NIRI control system within a single subroutine record was a very tempting prospect, but with a certain amount of regret, it was decided to take a more moderate approach. A single gensub (generalized subroutine) record would be used to embed most of the control algorithms. This gensub record would be connected to the motor record which would control the motor and to analog input records which would read the Hall effect sensor values. It would contain a subroutine which would embed the necessary control functions. For the first time, it was actually possible for a NIRI mechanism to find its home position, using code that was somewhat maintainable.

Soon, it was realized that the gensub record was insufficient. The NIRI control algorithms required a fair amount of control over the output links, and the gensub record was not intended to provide this. Furthermore, there seemed to be no natural way to deal with slight variations between the different mechanisms, while taking advantage of the similarity between the mechanisms.

To get around these problems, it was decided to create a new, custom EPICS record, the hall-step record, which was based on the gensub record. Modifying the code within the record provided direct control over the outputs of the record. Different device types were defined; each device type would control the idiosyncrasies of each class of mechanism. The hallstep record would provide a common interface to EPICS so that all mechanisms could be controlled in the same way. The hallstep record was extended so that it could do a variety of tasks, not just finding the home position. Eventually, the hallstep record became the control



point for the mechanism, containing all of the more complicated looping and testing functions. A prototype control system was created which was capable of controlling all of the different mechanisms that were within NIRI. A mechanism simulator was written, which made it possible to build a complete, functional NIRI control system which was capable of controlling two prototype mechanisms and a full compliment of simulated mechanisms.

The problems with the stepper motor record continued to be a constant source of frustration, and time was not available to fix its problems. Fortunately, at this time, we were provided with a copy of the motor record. Unfortunately, the motor record was not a simple drop-in replacement for the stepper motor record, but modifying the existing system to use the motor record instead of the stepper motor record was a relatively minor task.

The process of replacing the stepper motor record as well as the process of refining the main control diagram pointed out a serious weakness in the current system. There was a separate capfast diagram to control each mechanism type. Even the slightest change in a capfast diagram had to be made manually in each diagram. It was realized that proper use of capfast definitions would make it possible to replace multiple diagrams with a single diagram, which would simplify the task of maintenance and modification. Although some symbols would be unused for particular mechanisms, the gain in efficiency far outweighed the slight cost of an unused symbol.

### **3.3 Integration with the CICS**

The Core Instrument Control system (the CICS) was then integrated with the NIRI software, with the assistance of Steven Beard. This integration was a very smooth process, requiring only minor modifications, with the CICS and the NIRI software. This was a source of great comfort for the NIRI and CICS designers, and it was felt that the remaining work for NIRI was to refine the control algorithms to work with each actual working mechanisms. This early optimism was soon to be found to be unjustified.

It was very quickly realized that the NIRI/CICS system took an excessive amount of time to boot. This was a serious problem, and there was no way to overcome it, without completely rewriting the CICS or replacing it. In order to maintain a reasonable level of productivity, the NIRI developers chose to continue developing the NIRI software with minimal testing of the NIRI/CICS integrated system. This was, on the whole, a good decision, and significantly increased the productivity of the NIRI software. Unfortunately, it also caused several interface issues to be postponed, which caused a significant underestimate of the time required to finish the system.

### **3.4 The interlock system**

At this point, working mechanisms were becoming available, and a great deal of time and effort were devoted to getting the mechanism control refined and to verify correct operation of the mechanisms. At this point, it was felt that the decision to postpone development of the overall system was a sound one. Unfortunately, this meant that several user interface elements had to be built into the NIRI low-level support system. Various interlocks to prevent operation of the motors at critical times were added (partly at the request of the GNIRS developers). It was felt that the interlocks should be implemented in the lower-level code, so that the implementation would be simple to verify by inspection.

Getting the interlocks to work with the CICS code turned out to be an unexpected challenge. Since there was no intrinsic support for state variables in the CICS, it was felt that the easiest approach was to treat the interlock variables as a simple two-position mechanism. Since the interlocks were all interrelated, it was felt that the interlocks could easily be treated as a four axis mechanism. It was realized that this approach would lead to a more complicated implementation, but would permit large amounts of the CICS code to be used unmodified. In retrospect, this proved to be a very bad decision, possibly the worst decision that the NIRI developers made on the project.

Expanding the existing two-axis discreet mechanism to support a four-axis mechanism was found to be rather challenging. It required a surprising amount of rearrangement of the existing diagrams, to add the new axes. Furthermore, in order to support all of the fields that were required by the SNL code, it was necessary to add a large number of dummy field, including a completion field which would toggle immediately (since the flags could be set instantly).

Ultimately, when it became necessary to make major modifications to the mechanism-control (described below) it was decided that it was a waste of effort to make the same changes to the pseudo mechanisms. At that point, the code was copied (unfortunately, no simple way to share common code was found), and the supporting code for the interlocks was simplified.

The apparent simplifications that arose from making a flag look like a mechanism turned out to be illusory, and actually ended up wasting far more time and effort than it saved.

### **3.5 The NIRI temperature control system**

The low level control system was made to function by writing a custom record to interface to the off-the-shelf temperature controllers. The mechanisms were supported using a custom record, which communicated with the built-in serial ports on the CPU and a XYCOM serial I/O interface board, which was chosen because of previous experience with the board. The existing EPICS serial control software was not used, because the control task was quite simple. In retrospect, some time might have been saved by using the existing software, and choosing a board with existing EPICS support.

The NIRI temperature control system had many of the problems in interfacing to the CICS that the interlock system had. The initial attempt at controlling the system was made by treating the temperature control as a three axis mechanism. The difficulties which plagued the interlock system also plagued the temperature control system, and ultimately, the temperature control system was unable to share much code with the mechanism control system; most of the capfast diagrams had to be replicated and customized.

### **3.6 The NIRI diagnostic functions**

### **3.7 The NIRI verification functions**

## **4 CICS**

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### **4.1 Successful elements of the CICS**

The CICS provides all of the necessary ICD compliant external interfaces.

### **4.2 Problems with the CICS**

The CICS is very slow to reboot, and difficult to modify. There are a large number of nearly-identical diagrams, all of which must be modified in order to make any changes to the system. It assumes that all mechanisms are handled in an identical fashion, and requires a large number of diagrams to be duplicated if one mechanism must be handled differently.

## **5 The NIRI software**

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### **5.1 Successful elements of NIRI**

The strong separation of the higher-level interface from the low-level mechanism-control database has turned out to be very useful. This has made it possible to rapidly develop customized mechanism controls, without worries about meeting interface specifications. It has streamlined the development cycle, by reducing the time wasted waiting for the system to reboot. It made it possible to replace the mechanism implementation with a simulation, so that the public interfaces could be tested and debugged long before the actual mechanisms were available.

Using identical capfast diagrams for all mechanisms has turned out to be a tremendous time saver.

Using the same custom record for all mechanisms, with custom device support for the different device types has been quite useful in reducing the amount of work involved. Being able to replace the drivers with simulated mechanisms has insured that the simulation is able to reproduce the working system with a high degree of accuracy. This has allowed much development and debugging to be carried out before the mechanisms were available.

## **5.2 Problems with NIRI**

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### **6 Lessons learned**

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- If a template is to be provided, the initial implementation should be completed before any of the software which is to be derived from it is complete.
- Time should be allocated to re-implement the template after experience with working systems.
- Time should be allocated to re-implement the working systems, after the template has been modified.
- If the template is complicated, then the working systems will be very complicated.
- If the template is difficult to maintain, then the working systems will be impossible to maintain.
- Arbitrarily restricting developers to a subset of the available tools can lead to overly complicated designs.
- Be careful about scaling problems. Designs that work for simple test systems can become unmanageable for larger systems.
- It is better to have unnecessary symbols in a diagram than to have multiple nearly-identical diagrams. Proper use of definitions can combine nearly-identical diagrams.

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### **7 Suggestions**

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### **8 Conclusion**

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At this point, the NIRI software should be considered a qualified success, in that it is capable of controlling the instrument and providing the required functions. However, the implementation is considerably more complicated than it needs to be, and is quite difficult to modify. The time that the system takes to reboot is a concern, because it makes it quite time consuming to fix even the smallest bugs. Insufficient time was available to refine many of the algorithms, particularly the algorithms which are used to find the home position.

**Gemini  
Near  
Infra-Red  
Imager  
and  
Wavefront  
Sensor**



# Near IR Imager (NIRI) — Hardware Notes

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**Hubert Yamada**

niri\_h ty\_012 / 04 / 2000-05-22

This report contains a few notes that may be useful for configuring NIRI hardware.

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## 1 Introduction

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This report contains a few notes that are useful in configuring the NIRI hardware.

### 1.1 Purpose of the document

This report contains a few informal notes which may be useful to anyone who has to configure or maintain NIRI hardware.

The intended audience of this document is:

- the maintainers of the NIRI software and hardware.

## 2 The Motor Driver Modules

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The following table lists the current settings for all motor driver modules at the time that NIRI was initially shipped. Note that if necessary, the current may be raised to overcome sticking in the mechanisms; however, this is frequently a sign of a mechanical problem which should be resolved. The currents listed have been found to be sufficient for reliable operation at room temperature and at cryogenic temperatures. Excessive current settings should be avoided because of the added heat load to the system and the possibility of significant thermal gradients across mechanisms.

**TABLE 1.** Motor Driver Modules: Current settings

|       |                              | Column 1                       | Column 2                    | Column 3       | Column 4               | Column 5               |
|-------|------------------------------|--------------------------------|-----------------------------|----------------|------------------------|------------------------|
| Row 1 | Mechanism                    | Pupil Mask /<br>Filter Wheel 3 | Filter Wheel 1              | Filter Wheel 2 | Steering Mir-<br>ror 1 | Steering Mir-<br>ror 2 |
|       | Current Setting <sup>a</sup> | 2                              | 2                           | 2              | 2                      | 3.75                   |
| Row 2 | Mechanism                    | Empty                          | Science Detec-<br>tor Focus | Pupil Viewer   | Gimbal x               | Focal Plane<br>Mask    |
|       | Current Setting <sup>a</sup> |                                | 3.75                        | 1              | 1.25                   | 2.5                    |
| Row 3 | Mechanism                    | Gimbal y                       | Beam Splitter               | WFS Focus      | WFS Filter             | Window Cover           |
|       | Current Setting <sup>a</sup> | 1                              | 1.25                        | 1              | 2                      | 2                      |

- a. The current settings on the Phytron driver modules go from slightly less than 0 to slightly more than 4, in unspecified units. The actual range of current values is 0.6 A to 2.6 A.

# Gemini Near Infra-Red Imager and Wavefront Sensor



## OIWFS — Mechanism Testing

Hubert Yamada

niri\_h ty\_015 / 1.00 / 2001-07-02

This document describes the procedure for verifying that the OIWFS mechanical components are properly working.

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### 1 Introduction

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#### 1.1 Purpose

This document describes the procedure for insuring that the OIWFS mechanical components are working properly. It assumes that all wiring has already been verified for correctness, and that the default channels are used for all input and output boards. It assumes that all of the boards have been correctly configured<sup>1</sup>.

Note that although the Digital Input / Output board is intended to permit the OIWFS to interface to the interlock bus, there is no requirement that it actually must connect to the interlock bus. Support for the interlock bus is minimal, and is provided so that the user may define instrument-specific support, as required.

The current release of the OIWFS software requires a Bancomm board to interface to the A&G system. It can, however, be tested without the Bancomm board. The only operations that will fail are those that relate to the “follow” command.

Note that the OIWFS was initially designed to work with NIRI. Since delivery of NIRI, changes have been made to the NIRI software (including the OIWFS software). Later changes have not been propagated into the NIRI developers’ software, so some changes may need to be performed by the Gemini staff in order to integrate the OIWFS with the Gemini control system.

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1. Note: For GNIRS, all boards except the Digital Input/Output board were correctly configured at the time that the system was shipped. The Digital Input/Output board should be configured according to Gemini standards, as described in [30].

## **1.2 Definitions, Acronyms and Abbreviations**

See “Near IR Imager (NIRI)—Definitions, Abbreviations, and References”.

## **1.3 Stylistic Conventions**

References to documents in Section 1.5, “References,” on page 2 of “Near IR Imager (NIRI)—Definitions, Abbreviations, and References” are given like this; [1].

## **1.4 References**

See “Near IR Imager (NIRI)—Definitions, Abbreviations, and References”.

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## **2 Precautions**

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### **2.1 Temperature**

The mechanisms should never be operated while the instrument is being cooled or warmed. This is particularly important if the mechanism is being tested in a small liquid-nitrogen cooled dewar. Severe temperature gradients may occur, which can cause the mechanisms to bind. Operating under those circumstances can break the mechanism.

It is strongly recommended that a temperature sensor be placed on a part of the mechanism which is as far away from the source of heating or cooling, in order to verify that the entire mechanism has reached thermal equilibrium before operation. Note that for the focus stage, the moving part of the mechanism has very poor thermal contact with the rest of the mechanism. Also, the motor shaft has very poor thermal contact with the motor housing.

### **2.2 Motor Heating**

The mechanisms are not intended to be operated continuously for extended periods of time (i.e., longer than five minutes.) During prolonged, continuous operation, the motors can become hot. This should be avoided. It is recommended that any tests which require continuous motion of the mechanisms should be performed intermittently, with adequate time to permit the motor to cool.

### **2.3 Wear**

The mechanisms are not intended to be operated continuously. Avoid day-long tests, as this will reduce the lifetime of the mechanisms.

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## **3 Procedure for Testing Mechanisms**

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The procedure described here assumes that the OIWFS control software (at least the Engineering subsystem) has already been configured and installed, as described in the release



notes. It assumes that the Engineering subsystem has been verified to work correctly in simulation mode.

### **3.1 Preparation**

1. Source your local configuration file
2. Run niriEngStart. This will bring up the top level engineering screen.
3. Press the Interlocks button. This will bring up the interlocks screen.
4. Press the “WARM” button, to set the system temperature. (For cold tests, the “COLD” button should be used instead.) Wait for one minute. You must set the system temperature each time that the IOC is rebooted. You must set the temperature each time that the system completes warming or cooling. It is strongly recommended that the system temperature be set to “CHANGING” while the system is being warmed or cooled; this will set an interlock that will protect the mechanisms by keeping them from moving.
5. Close the Interlocks screen.
6. Return to the top level engineering screen.
7. Press the “Wavefront Sensor CC” button. This will bring up the top-level Wavefront Sensor Components screen.

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## **4 Filter Wheel**

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1. From the top-level Wavefront Sensor Components screen, press the “Filter wheel” button. This will bring up the filter wheel engineering screen.
2. Identify the motor driver module which will be used to drive the filter wheel.
3. Enter a number in the new position (“New Pos”) field that is 40000 more than the current position (“Cur Pos”).
4. You should see all of the following things happen. If any of these does not, then there is a problem with the mechanism.
5. The LED on the motor driver module should turn yellow.  
If it is off, then the motor is not enabled  
If it is green, then the motor is enabled, but not receiving pulses  
If it is red, then the driver is in a fault state. Turn the motor driver box off for 20 seconds, then turn it back on again. If the LED again turns red, then there is a problem with the electronics, motor, or wiring.
6. The motor should immediately start moving.
7. The value of the sensors marked 1P and 1B should change slightly, but should be less than 0.100 V most of the time. Occasionally, the sensor values will rise up or drop down to a much larger value, and then immediately return back to a low value. It will take constant inspection of the meters to see this happen.  
If all of these happen, then the motor is turning, the system is working at a minimal level, but is not necessarily performing to specification.
8. Press the Datum/Diag button. This will find the home position for the mechanism and immediately perform a quick mechanism check. Check the Status and Error/Info boxes. If all goes well, the system should return to the Idle state. There should be no messages in the Error/Info box at this time. If there are any messages, then the system has failed the test. See the calibration procedure.

9. If all goes well, press the “cycle” button. This will perform a continuous mechanism test. Twenty or thirty cycles should be sufficient to verify correct operation. End the cycle test by pressing the stop button.

#### **4.1 Focus Stage**

1. From the top-level Wavefront Sensor Components screen, press the “Focus stage” button. This will bring up the filter wheel engineering screen.
2. Locate the motor which is to drive the focus stage
3. By hand, with motor off, turn the main gear of the mechanism near the center of travel
4. Enter a number in the new position (“New Pos”) field that is 500 more than the current position (“Cur Pos”).
5. Verify that the motor turns and then stops.
6. Verify that the hall-effect sensor meters (labelled 2P and 2B) move.
7. Press the Datum/Diag button.
8. Watch and listen to the mechanism very carefully. If you can see or hear the mechanism hit a limit, stop it immediately. If this happens, there is a serious problem with the mechanism or its sensors. Low-level calibration may be required.
9. Press the Datum/Diag button. This will find the home position for the mechanism and immediately perform a quick mechanism check. Check the Status and Error/Info boxes. If all goes well, the system should return to the Idle state. There should be no messages in the Error/Info box at this time. If there are any messages, then the system has failed the test. See the calibration procedure.
10. If all goes well, press the “cycle” button. This will perform a continuous mechanism test. Twenty or thirty cycles should be sufficient to verify correct operation. End the cycle test by pressing the stop button.

#### **4.2 Gimbal**

Only the x-axis will be described here. The y-axis is tested identically.

1. From the top-level Wavefront Sensor Components screen, press the “Gimbal X” button. This will bring up the gimbal x-axis engineering screen.
2. Locate the motor which is to drive the filter wheel
3. Move the mechanism away from the limits of travel. (May require mechanical adjustment).
4. Enter a number in the new position (“New Pos”) field that is 70000 more than the current position (“Cur Pos”).
5. Verify that the motor turns and then stops.
6. Verify that the mirror support moves. Note that because of the high gear ratio, this motion is very difficult to see.
7. Verify that the two right hall-effect sensor meters (marked 2P and 2B) move.
8. Verify that the two left hall-effect sensor meters (marked 1P and 1B) move. Note that these meters will normally have a value that is close to 0. They will move off-zero very briefly. It takes careful observation to see this motion. If no motion is seen, repeat from step 4, with a value that is 70000 less than the starting position. If motion is still not seen, then the hall sensors are not being read properly or the entire gear train is not turning.

9. Press the Datum/Diag button. This will find the home position for the mechanism and immediately perform a quick mechanism check. Check the Status and Error/Info boxes. If all goes well, the system should return to the Idle state. There should be no messages in the Error/Info box at this time. If there are any messages, then the system has failed the test. See the calibration procedure.
10. If all goes well, press the “cycle” button. This will perform a continuous mechanism test. Because of the very large range of travel of this mechanism, two or three cycles should be sufficient to verify correct operation. End the cycle test by pressing the stop button.

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## Filter Wheel

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**Gemini  
Near  
Infra-Red  
Imager  
and  
Wavefront  
Sensor**



# Summary of the GNIRS OIWFS Mechanism Testing

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**This document describes the tests which were performed to insure that the GNIRS OIWFS mechanisms were working correctly.**

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## **1 Introduction**

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The GNIRS OIWFS mechanisms were built using the same designs as the NIRI OIWFS mechanisms. The NIRI mechanisms had received extensive testing, both in the lab and on the telescope, so it was felt that the design was sound. For this reason, it was deemed unnecessary to perform extensive and time-consuming mechanism tests. It appeared sufficient to perform a basic test of all mechanism functions, with particular attention being paid to those elements which had proven to be problematic with the NIRI OIWFS.

### **1.1 Definitions, Acronyms and Abbreviations**

See “Near IR Imager (NIRI)—Definitions, Abbreviations, and References”.

### **1.2 Stylistic Conventions**

References to documents in Section 1.5, “References,” on page 2 of “Near IR Imager (NIRI)—Definitions, Abbreviations, and References” are given like this; [1].

### **1.3 References**

See “Near IR Imager (NIRI)—Definitions, Abbreviations, and References”.

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## **2 Filter Wheel**

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The filter wheel built from a design that has proven to be very reliable. For this reason, it was felt that it was acceptable to test the mechanism at room temperature, without cooling it to its operating temperature.

Testing was done using the actual interface boards and stepper-motor driver modules that will be used in GNIRS. Custom interface cables were used in place of the cables which will be used in the dewar; the other cables, which will be external to the dewar, were the cables which are to be used with GNIRS. The mechanism was configured as it was expected to be used, including the Hall-effect sensors.

The standard OIWFS datum / diagnose function was then used, with the NIRI calibration constants. It was expected that the datum function would work correctly, but that the diagnose function would fail until the system had been calibrated. It was found that the direction of rotation of the motor was the reverse of NIRI. It is believed that the NIRI motors were inconsistently wired. Since all of the GNIRS motors were wired identically, it was felt that it was preferable not to reverse the direction of motor motion to match NIRI. Instead, all calibration constants were reversed.

With the corrected calibration constants, the filter wheel was able to datum properly, but failed the diagnosis phase. This was expected, as the normal calibration constants are sensitive, and need to be regenerated for each mechanism. The low-level calibration constants are well-defined by the mechanism. They were not changed, and were not expected to need to be changed.

The standard mechanism calibration procedure was then carried out, and the new constants were verified to be reasonable, and then were loaded into the database. The datum / diagnose function was again performed, and was completed with no problems. The datum / diagnose function was then executed a few times, with no failures detected.

The mechanism was then moved to each filter position in sequence. The mechanism reached each position and the Hall-effect sensor reading confirmed that the mechanism was in the correct position.

Finally, the OIWFS cycle test was started. The mechanism was run through 30 test cycles. No errors were detected during the cycle test.

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## **3 Focus Stage**

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The filter wheel built from a design that has proven to be reliable. For this reason, it was felt that extensive testing was unnecessary, but that cryogenic operation needed to be verified.

Testing began with the mechanism sitting in an upright position. The mechanism was moved to the center of travel. Some slight misalignment of the mechanism were discovered and corrected at this time. The home position was found to be slightly closer to the center of travel of the mechanism than NIRI. This has the possibility of introducing slightly more inconsistency to the home position, because the mechanism is not pre-loaded at the home position. It

was felt that the home position was sufficiently distant from the central position to insure a repeatable home position.

The mechanism was centered, and moved back and forth from limit to limit several times. It was found that the mechanism moved freely, with no apparent problems. No evidence of motor stalling — a common problem with this type of mechanism, particularly near the limits of travel — was detected.

The low-level calibration constants were then adjusted to match this mechanism. This was expected to be necessary, since the constants are dependent on very fine positioning of the Hall-effect sensors. The limits of travel were adjusted, also expected, as the limits depend on the orientation of the main screw. The mechanism was datumed, and performed as expected. The normal calibration was then carried out. The datum / diagnose test was then performed several times, with no failure. Finally, a cycle test was carried out for about 30 cycles. The mechanism passed with no problems.

The mechanism was then cooled to its operating temperature. The mechanism was unable to datum itself. Further inspection showed that the home position had moved slightly as the system had cooled. This was not a problem with Niri, though it was a problem with a similar Niri mechanism. At that time, it was decided to resolve the problem by changing some of the low-level calibration constants, depending on the temperature. This was found to be necessary with the Niri science-channel focus stage, which has a similar design. There is no explanation of why this is necessary with the GNIRS OIWFS focus stage, but not with the Niri OIWFS focus stage. With this change to the software, the mechanism passed the datum / diagnose test several times. Finally, a cycle test was performed for about 30 cycles. No problems were discovered during the cycle test.

Finally, to verify that nothing had changed with the mechanism was warmed again. It was able to pass the datum / diagnose test and a cycle test of approximately 30 cycles.

At this time, it is possible to say that the GNIRS OIWFS mechanism passes its tests as well as the corresponding Niri mechanism did. However, optical testing is still required to insure that it is behaving as well as the Niri mechanism.

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## **4 Gimbal Mirror**

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The gimbal mirror was built from a working design, so it was felt that it was unnecessary to perform the detailed testing that was required for the Niri mechanism. This mechanism is designed to very high precision, so optical testing was required.

The mirror was first calibrated while it was warm. The mechanism was found to move smoothly. The mechanism was datumed a few times, and found to return to the same home position (as determined using the Hall-effect sensors). The mechanism was then calibrated and passed a small number of cycle tests. Slight mechanical adjustments were necessary in order to provide the full range of travel. The calibration and cycle tests were then repeated.

The system was then analyzed using an alignment telescope. The image of a cross-hair was projected onto the gimbal mirror, and the reflected image was examined as the gimbal was tipped. The reticle was able to be read to an accuracy of about 0.1 reticle unit, which corresponded to a mirror move of approximately 150 steps. (The exact scale was different for both axes, because of geometric factors.)

The mirror was offset by 100000 steps and then returned to its home position. This was repeated 20 times. There appeared to be a slight drift in the position, that was approximately 0.1 or 0.2 reticle units.

The mechanism was tested at cryogenic temperatures. One axis failed to move at all at cryogenic temperatures. The other axis failed also after further cold testing. The problem was eventually isolated, and repaired.

The previous calibration and test procedure was repeated, but the slight drift in position was not observed.

The mechanism was then cooled. The mirror was offset by 100000 steps and then returned to the home position. The cross-hair returned exactly to the zero position to the accuracy of observation.

The mechanism was calibrated, and the datum / diagnose function was repeated three times for each axis. Each axis was cycle tested for five cycles.

The mechanism was then moved to a series of points, and the reticle position of the cross hair was then read. The following data was recorded. Note that no corrections were made for any slight distortions which might have been caused by the optics of the test system. The data was found to be linear, with any discrepancies easily explained by the limitations of the test setup.

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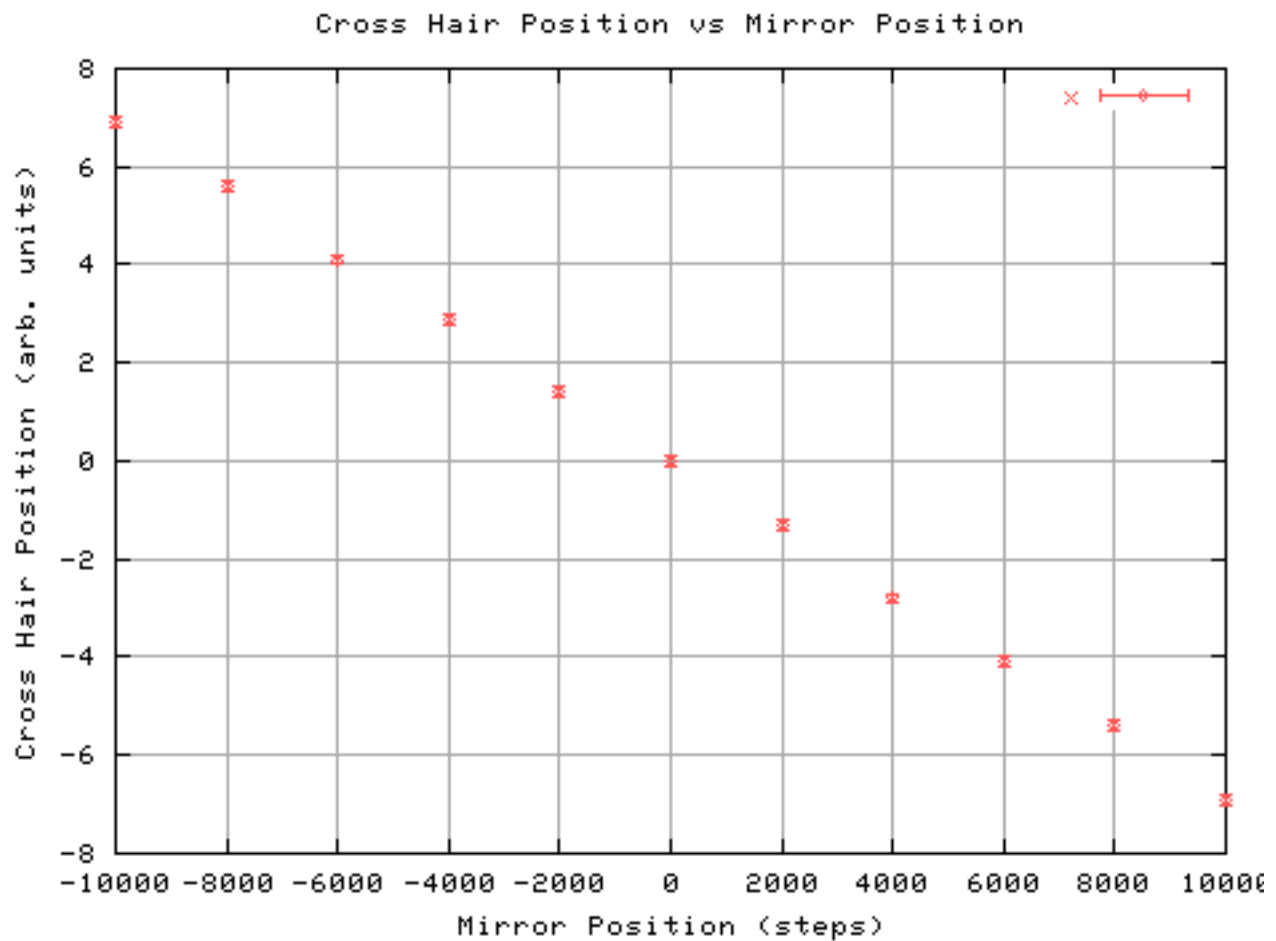
**TABLE 1.**

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**x Axis Cross Hair Position as a Function of Mirror Position**

| Mirror position<br>(ministeps) | Cross hair position<br>(Reticle units) | Cross hair uncertainty<br>(Reticle units) |
|--------------------------------|----------------------------------------|-------------------------------------------|
| -10000                         | 6.9                                    | 0.1                                       |
| -8000                          | 5.6                                    | 0.1                                       |
| -6000                          | 4.1                                    | 0.1                                       |
| -4000                          | 2.9                                    | 0.1                                       |
| -2000                          | 1.4                                    | 0.1                                       |
| 0                              | 0.0                                    | 0.1                                       |
| 2000                           | -1.3                                   | 0.1                                       |
| 4000                           | -2.8                                   | 0.1                                       |
| 6000                           | -4.1                                   | 0.1                                       |
| 8000                           | -5.4                                   | 0.1                                       |
| 10000                          | -6.9                                   | 0.1                                       |



**FIGURE 1.** x Axis Cross Hair Position vs Function of Mirror Position**TABLE 2.** y Axis Cross Hair Position as a Function of Mirror Position

| Mirror position (ministeps) | Cross hair position (Reticle units) | Cross hair uncertainty (Reticle units) |
|-----------------------------|-------------------------------------|----------------------------------------|
| -10000                      | 7.8                                 | 0.1                                    |
| -8000                       | 6.0                                 | 0.1                                    |
| -6000                       | 4.7                                 | 0.1                                    |
| -4000                       | 3.0                                 | 0.1                                    |
| -2000                       | 1.5                                 | 0.1                                    |
| 0                           | 0.0                                 | 0.1                                    |
| 2000                        | -1.5                                | 0.1                                    |
| 4000                        | -3.1                                | 0.1                                    |
| 6000                        | -4.5                                | 0.1                                    |

| Mirror position<br>(ministeps) | Cross hair position<br>(Reticle units) | Cross hair uncertainty<br>(Reticle units) |
|--------------------------------|----------------------------------------|-------------------------------------------|
| 8000                           | -6.0                                   | 0.1                                       |
| 10000                          | -7.5                                   | 0.1                                       |

FIGURE 2. y Axis Cross Hair Position as a Function of Mirror Position

